

TECHNICAL REPORT

Multi-Sample Comparison Report

Hydraulic Conductivity Analysis - 25 Samples

PROJECT

Project

PROJECT NO.

test

CLIENT

RASDASS

PREPARED BY

213123123

REPORT DATE

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Multi-Sample Comparison Report

Project: Project
Client: RASDASS
Analyst: 213123123
Report Date: July 21, 2026 at 09:21

Number of Samples: 25
Temperature: 20 °C
Porosity: Varies by sample (0.35 to 0.45)

Executive Summary

Comparison Analysis: This report compares 25 soil samples based on grain size distribution and hydraulic conductivity estimates.

Key Findings:

- Highest permeability: NBAL-06 [English] (8.28e-04 m/s)
- Lowest permeability: NBAL-24 [English] (6.90e-05 m/s)
- Permeability range: 12.0x difference

Methodology

Comparative Analysis Approach

This comparison report presents a side-by-side analysis of multiple soil samples to identify patterns, variations, and relationships between grain size characteristics and hydraulic conductivity estimates.

Analysis Components

Grain Size Comparison: Overlapping distribution curves allow visual assessment of particle size variations between samples.

K-Value Comparison: Box plots and statistical summaries reveal the range and reliability of hydraulic conductivity estimates across samples.

Method Reliability: A reliability matrix shows which empirical methods are applicable for each sample, helping identify the most suitable estimation approaches.

Results & Analysis

Sample Overview

Sample	Temp (°C)	Porosity	D10 (mm)	D50 (mm)	D60 (mm)	Cu
NBAL-01 [English]	20.00	0.435	0.171	0.295	0.319	1.86
NBAL-02 [English]	20.00	0.399	0.255	0.655	0.785	3.08
NBAL-03 [English]	20.00	0.405	0.122	0.305	0.350	2.86
NBAL-04 [English]	20.00	0.35	N/A	0.173	0.205	N/A
NBAL-05 [English]	20.00	0.419	0.098	0.192	0.233	2.38
NBAL-06 [English]	20.00	0.421	0.257	0.526	0.592	2.30
NBAL-07 [English]	20.00	0.425	0.219	0.425	0.473	2.16
NBAL-08 [English]	20.00	0.423	0.125	0.244	0.281	2.25
NBAL-09 [English]	20.00	0.432	0.164	0.297	0.325	1.97
NBAL-10 [English]	20.00	0.394	0.178	0.516	0.583	3.27
NBAL-11 [English]	20.00	0.439	0.090	0.149	0.158	1.74
NBAL-12 [English]	20.00	0.426	0.075	0.152	0.161	2.15
NBAL-13 [English]	20.00	0.35	N/A	0.155	0.169	N/A
NBAL-14 [English]	20.00	0.35	N/A	0.064	0.123	N/A
NBAL-15 [English]	20.00	0.45	0.108	0.149	0.156	1.44
NBAL-16 [English]	20.00	0.437	0.086	0.149	0.155	1.80
NBAL-17 [English]	20.00	0.356	0.093	0.365	0.463	4.99
NBAL-18 [English]	20.00	0.35	N/A	0.152	0.177	N/A
NBAL-19 [English]	20.00	0.429	0.170	0.321	0.349	2.06
NBAL-20 [English]	20.00	0.439	0.188	0.305	0.328	1.74
NBAL-21 [English]	20.00	0.434	0.167	0.292	0.316	1.90
NBAL-22 [English]	20.00	0.435	0.171	0.294	0.319	1.86
NBAL-23 [English]	20.00	0.431	0.134	0.237	0.268	1.99
NBAL-24 [English]	20.00	0.35	N/A	0.144	0.152	N/A
NBAL-25 [English]	20.00	0.414	0.146	0.324	0.369	2.52

Sample	Soil type	Descriptor	K geometric mean (m/s)
NBAL-01 [English]	Poorly-graded sand	Uniform sand low in fines	3.64e-04
NBAL-02 [English]	Poorly-graded sand	Moderately well sorted sand low in fines	7.45e-04

Sample	Soil type	Descriptor	K geometric mean (m/s)
NBAL-03 [English]	Poorly-graded sand	Moderately well sorted sand with fines	1.94e-04
NBAL-04 [English]	Sand	Clayey sand with fines	8.50e-05
NBAL-05 [English]	Poorly-graded sand	Moderately well sorted sand low in fines	1.21e-04
NBAL-06 [English]	Poorly-graded sand	Moderately well sorted sand low in fines	8.28e-04
NBAL-07 [English]	Poorly-graded sand	Moderately well sorted sand low in fines	6.05e-04
NBAL-08 [English]	Poorly-graded sand	Moderately well sorted sand low in fines	1.94e-04
NBAL-09 [English]	Poorly-graded sand	Uniform sand low in fines	3.38e-04
NBAL-10 [English]	Poorly-graded sand	Moderately well sorted sand low in fines	4.18e-04
NBAL-11 [English]	Poorly-graded sand	Uniform sand with fines	1.09e-04
NBAL-12 [English]	Poorly-graded sand	Moderately well sorted sand with fines	8.93e-05
NBAL-13 [English]	Sand	Clayey sand with fines	8.57e-05
NBAL-14 [English]	Sand	Clayey sand with fines	1.24e-04
NBAL-15 [English]	Poorly-graded gravel	Uniform gravelly sand low in fines	1.45e-04
NBAL-16 [English]	Poorly-graded sand	Uniform sand with fines	1.05e-04
NBAL-17 [English]	Poorly-graded sand	Moderately well sorted sand with fines	8.21e-05
NBAL-18 [English]	Sand	Clayey sand with fines	8.67e-05
NBAL-19 [English]	Poorly-graded sand	Moderately well sorted sand low in fines	3.78e-04
NBAL-20 [English]	Poorly-graded sand	Uniform sand low in fines	4.35e-04
NBAL-21 [English]	Poorly-graded sand	Uniform sand low in fines	3.47e-04
NBAL-22 [English]	Poorly-graded gravel	Uniform gravelly sand low in fines	3.63e-04
NBAL-23 [English]	Poorly-graded sand	Uniform sand low in fines	2.23e-04
NBAL-24 [English]	Sand	Clayey sand with fines	6.90e-05
NBAL-25 [English]	Poorly-graded sand	Moderately well sorted sand low in fines	2.73e-04

Grain Parameters Comparison

Large table moved to companion Excel appendix: Grain Parameters Comparison.

K-Value Aggregate Summary

Scope	Datasets	Geometric mean (m/s)	Arithmetic mean (m/s)	Median (m/s)	ln(K) std dev
Overall	25	2.31e-04	3.68e-04	2.54e-04	1.014
g1	5	2.47e-04	4.09e-04	2.77e-04	1.062
g2	6	3.53e-04	5.25e-04	3.63e-04	0.902

Scope	Datasets	Geometric mean (m/s)	Arithmetic mean (m/s)	Median (m/s)	ln(K) std dev
g3	6	1.03e-04	1.49e-04	1.11e-04	0.919
g4	8	2.62e-04	3.51e-04	2.77e-04	0.814

Scope	Included K cells	Warning cells	Permeability class
Overall	298 / 400	102	High (Clean Sand)
g1	61 / 80	19	High (Clean Sand)
g2	80 / 96	16	High (Clean Sand)
g3	62 / 96	34	High (Clean Sand)
g4	95 / 128	33	High (Clean Sand)

K-Value Calculations by Dataset and Method

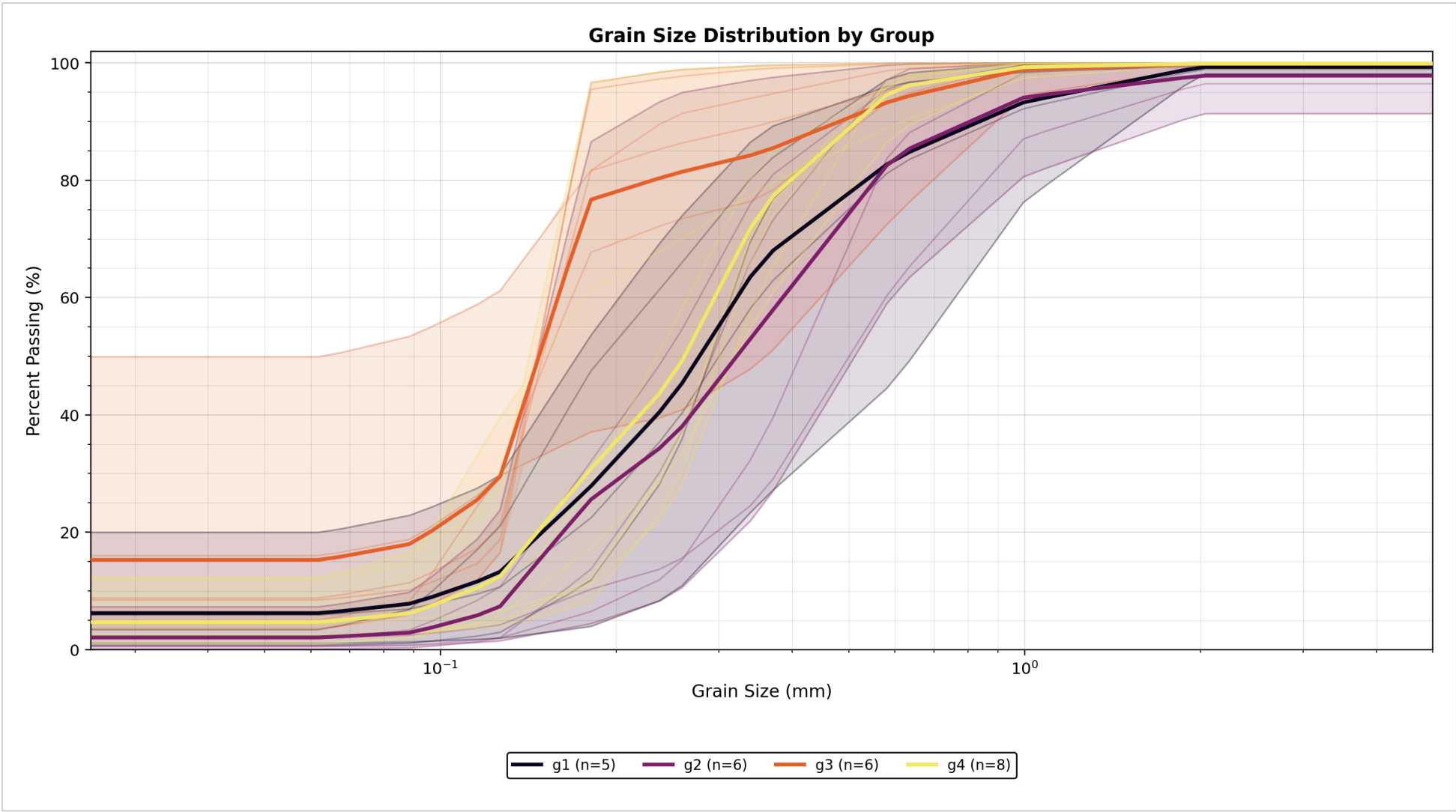
Large table moved to companion Excel appendix: K-Value Calculations by Dataset and Method.

Permeability Classification Summary

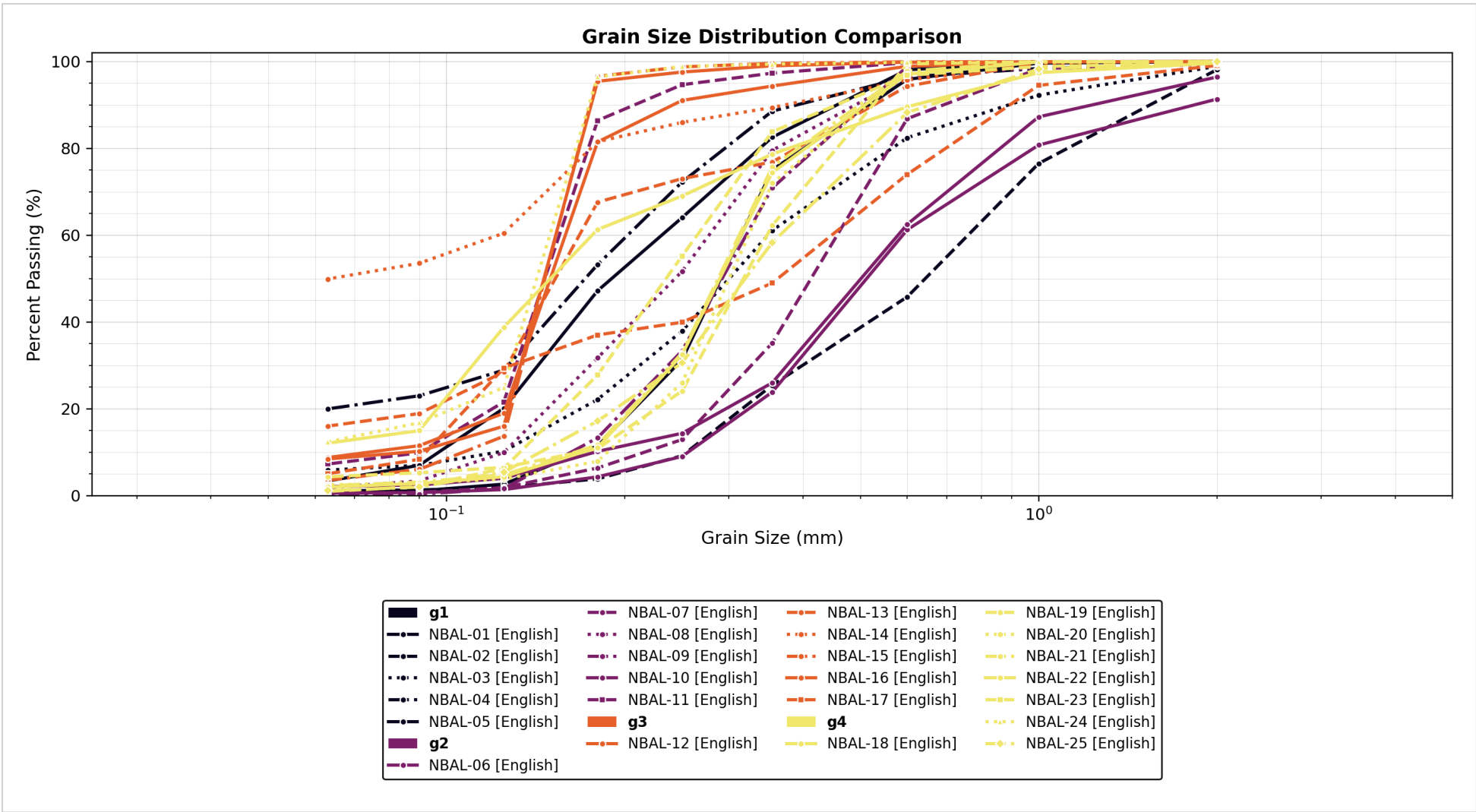
Sample Name	K geometric mean (m/s)	Classification
NBAL-01 [English]	3.64e-04	High (Clean Sand)
NBAL-02 [English]	7.45e-04	High (Clean Sand)
NBAL-03 [English]	1.94e-04	High (Clean Sand)
NBAL-04 [English]	8.50e-05	Moderate (Fine Sand)
NBAL-05 [English]	1.21e-04	High (Clean Sand)
NBAL-06 [English]	8.28e-04	High (Clean Sand)
NBAL-07 [English]	6.05e-04	High (Clean Sand)
NBAL-08 [English]	1.94e-04	High (Clean Sand)
NBAL-09 [English]	3.38e-04	High (Clean Sand)
NBAL-10 [English]	4.18e-04	High (Clean Sand)
NBAL-11 [English]	1.09e-04	High (Clean Sand)
NBAL-12 [English]	8.93e-05	Moderate (Fine Sand)
NBAL-13 [English]	8.57e-05	Moderate (Fine Sand)
NBAL-14 [English]	1.24e-04	High (Clean Sand)
NBAL-15 [English]	1.45e-04	High (Clean Sand)
NBAL-16 [English]	1.05e-04	High (Clean Sand)
NBAL-17 [English]	8.21e-05	Moderate (Fine Sand)

Sample Name	K geometric mean (m/s)	Classification
NBAL-18 [English]	8.67e-05	Moderate (Fine Sand)
NBAL-19 [English]	3.78e-04	High (Clean Sand)
NBAL-20 [English]	4.35e-04	High (Clean Sand)
NBAL-21 [English]	3.47e-04	High (Clean Sand)
NBAL-22 [English]	3.63e-04	High (Clean Sand)
NBAL-23 [English]	2.23e-04	High (Clean Sand)
NBAL-24 [English]	6.90e-05	Moderate (Fine Sand)
NBAL-25 [English]	2.73e-04	High (Clean Sand)

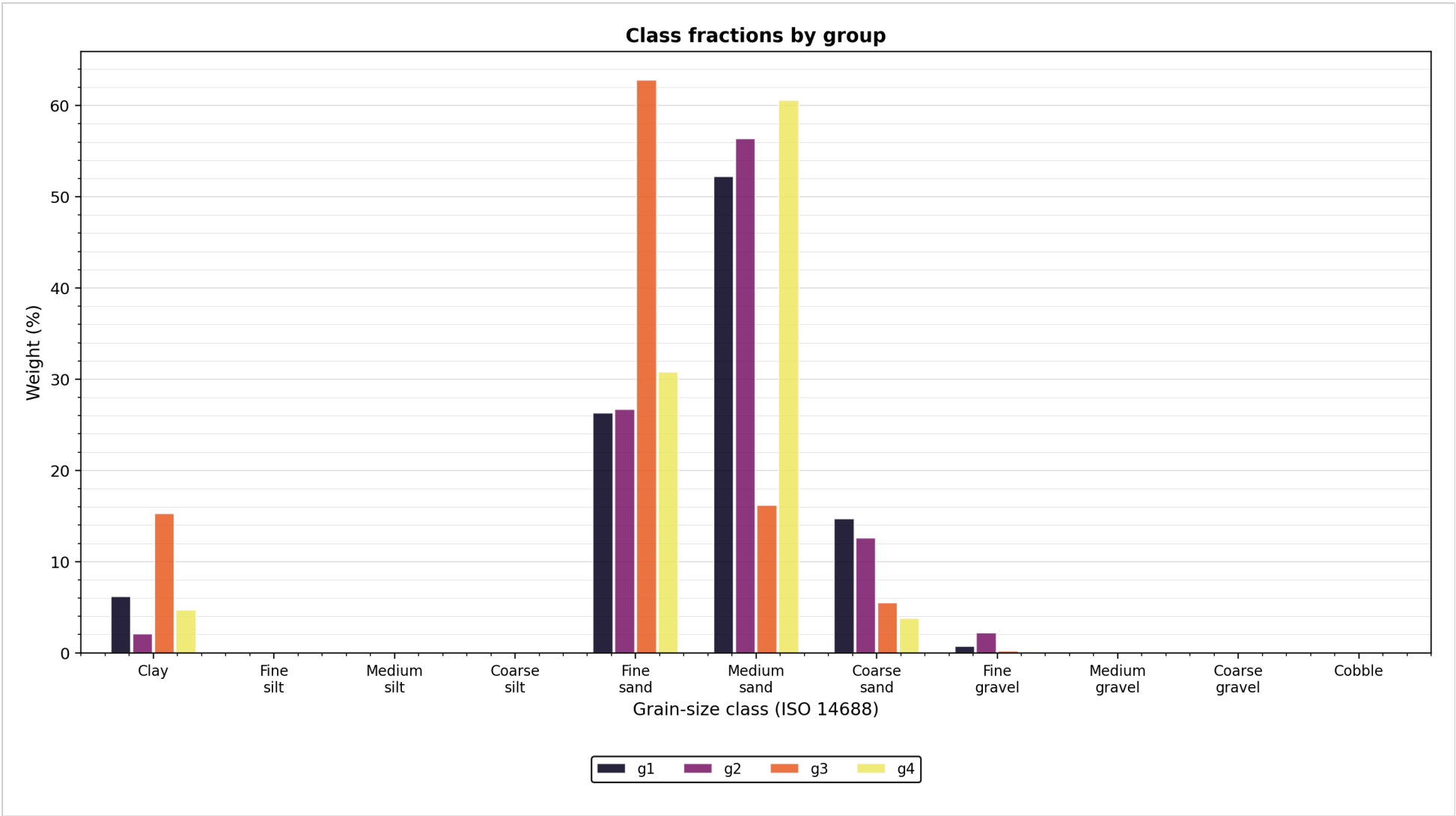
Grain Size Distribution Comparison — per group



Grain Size Distribution Comparison — per dataset



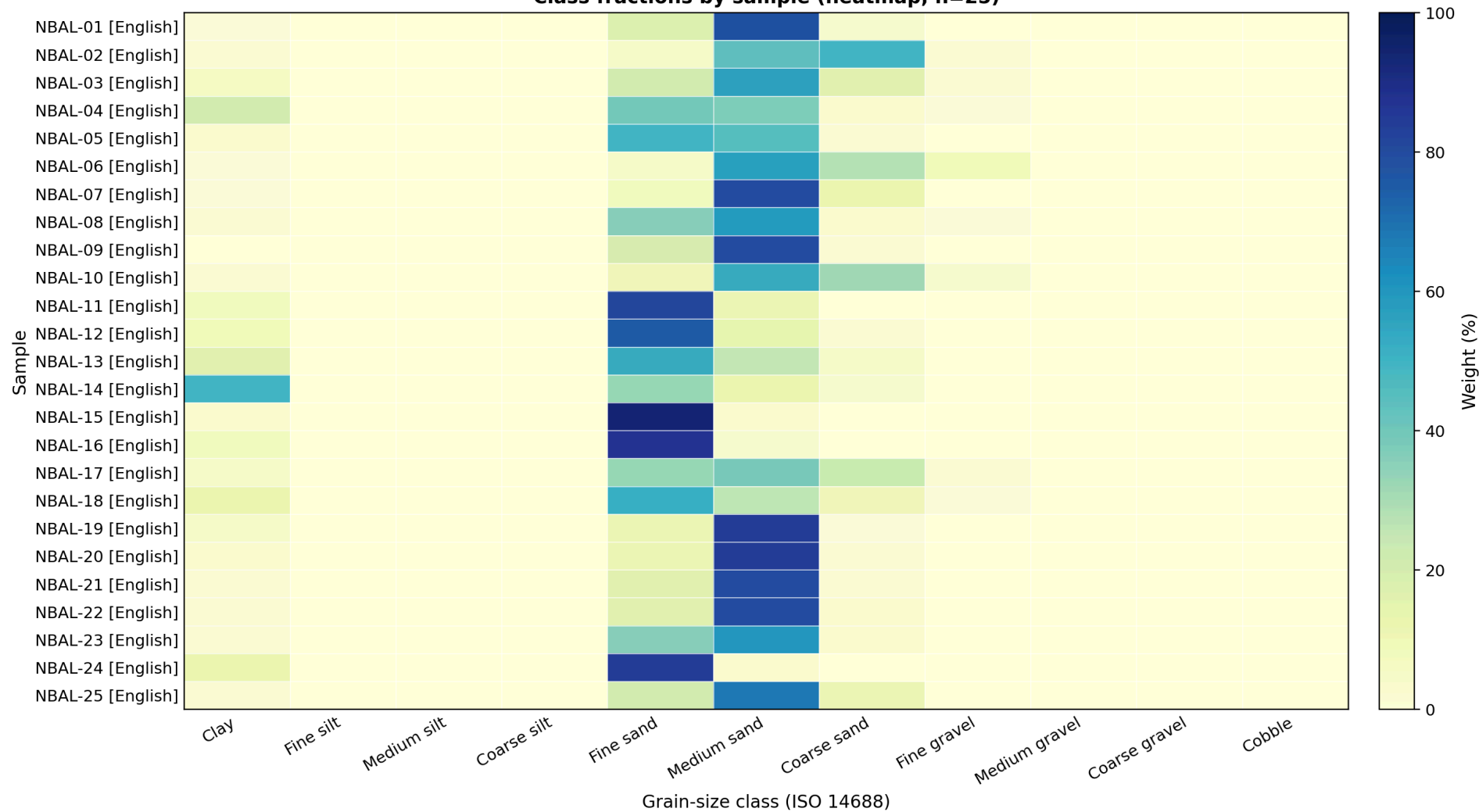
Grain-size Class Fractions — per group



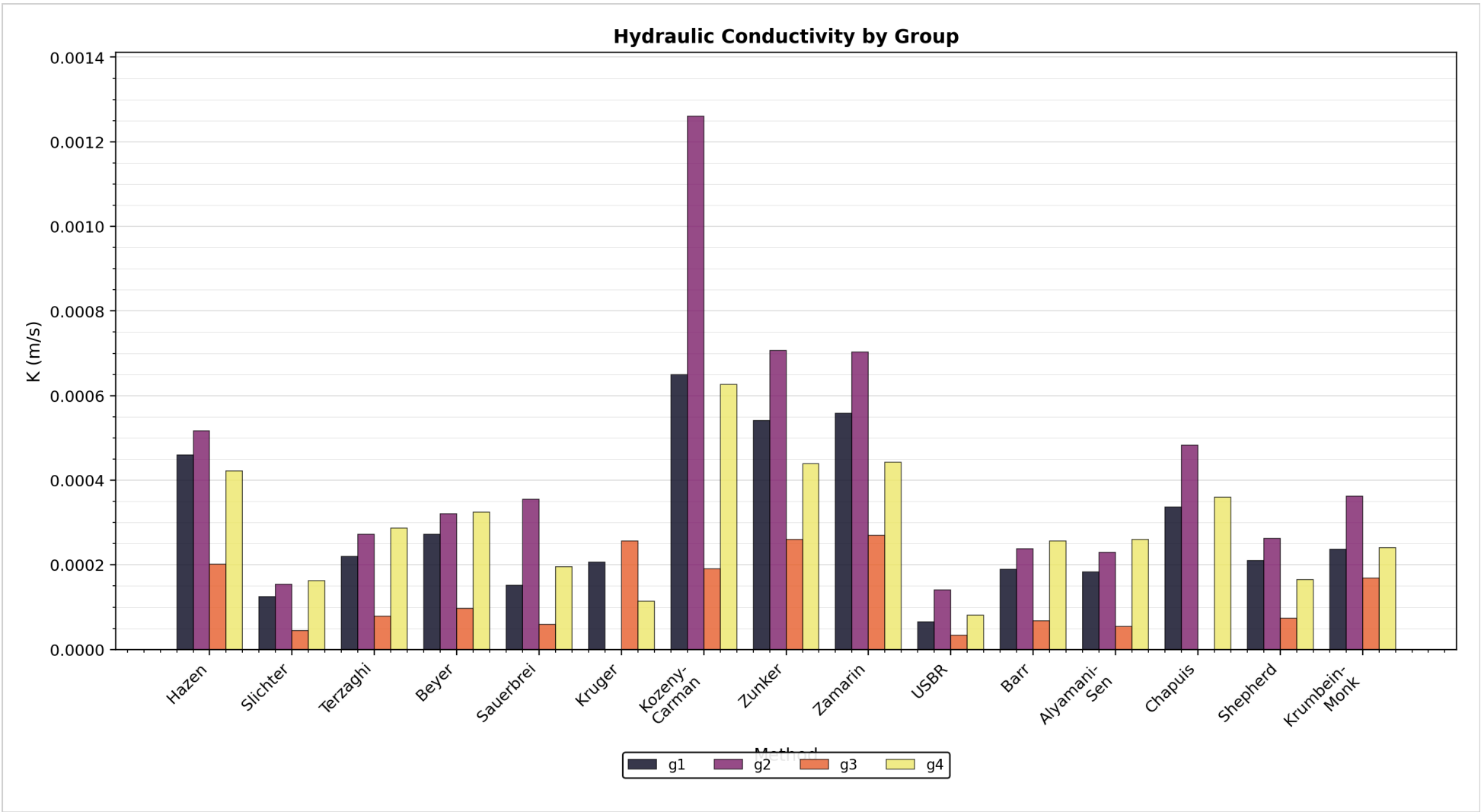
Grain-size Class Fractions — per dataset

Large-batch layout: Heatmap shown for 25 samples; rows are samples, and color shows weight percent (0-100).

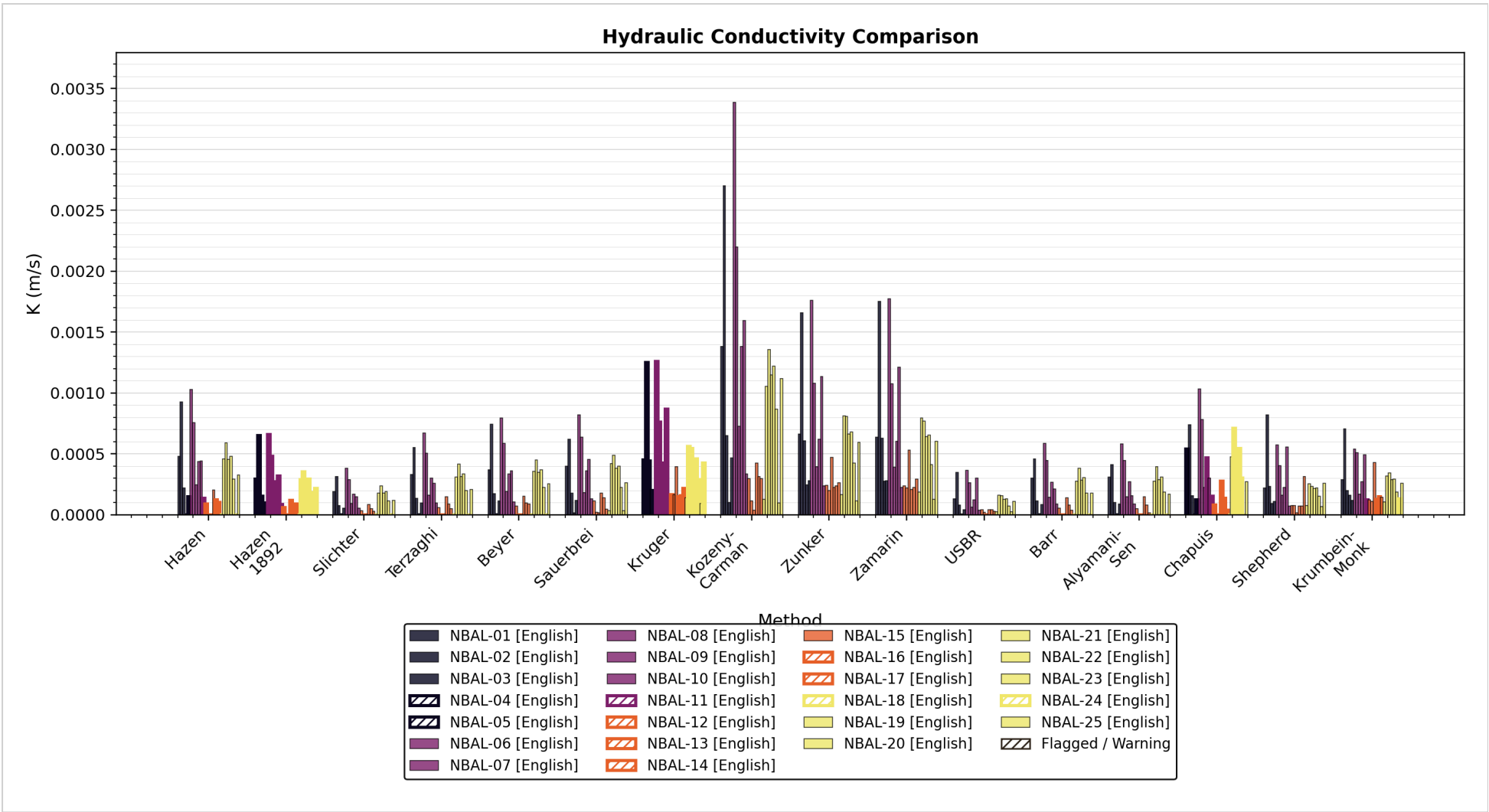
Class fractions by sample (heatmap, n=25)



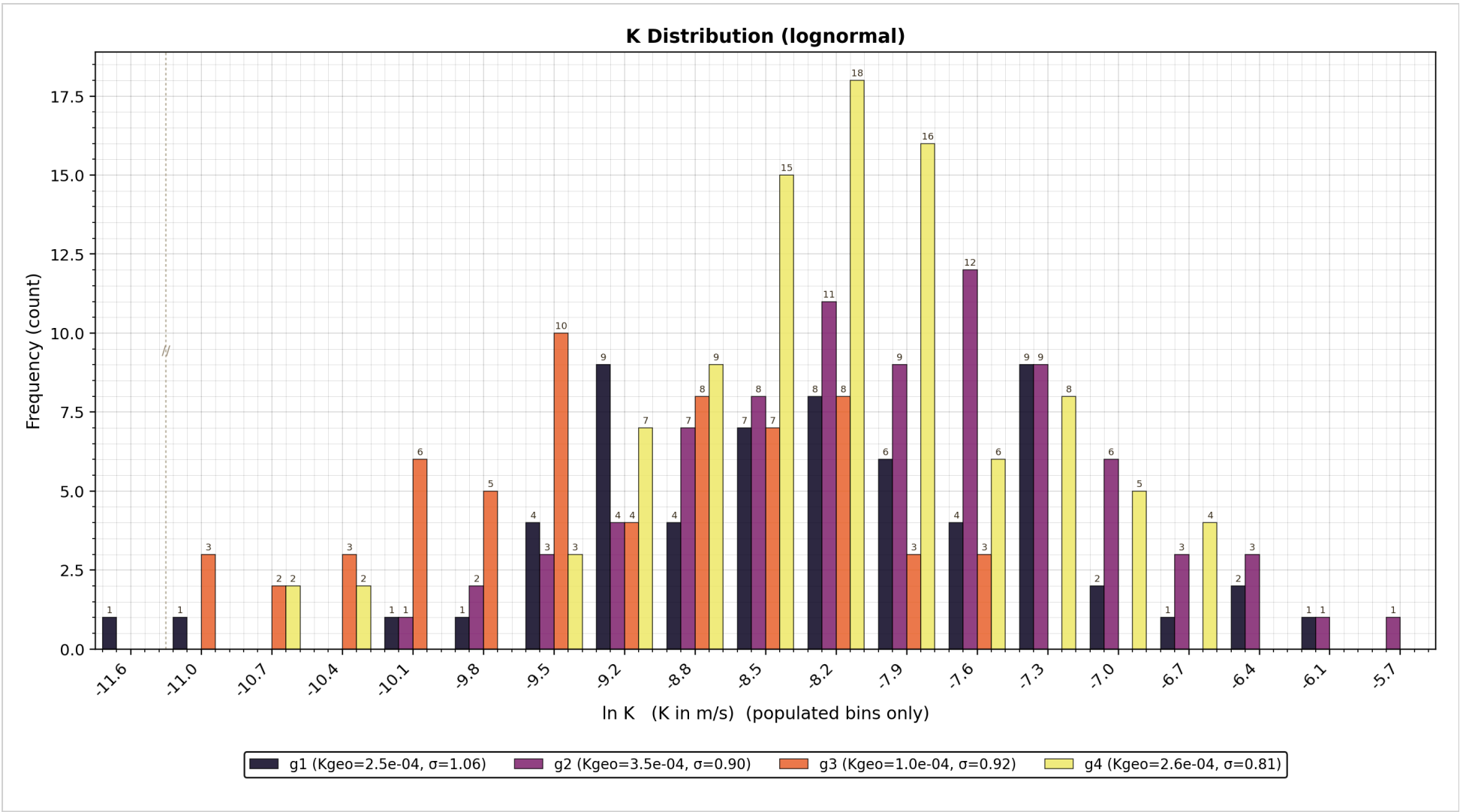
Hydraulic Conductivity by Method — per group



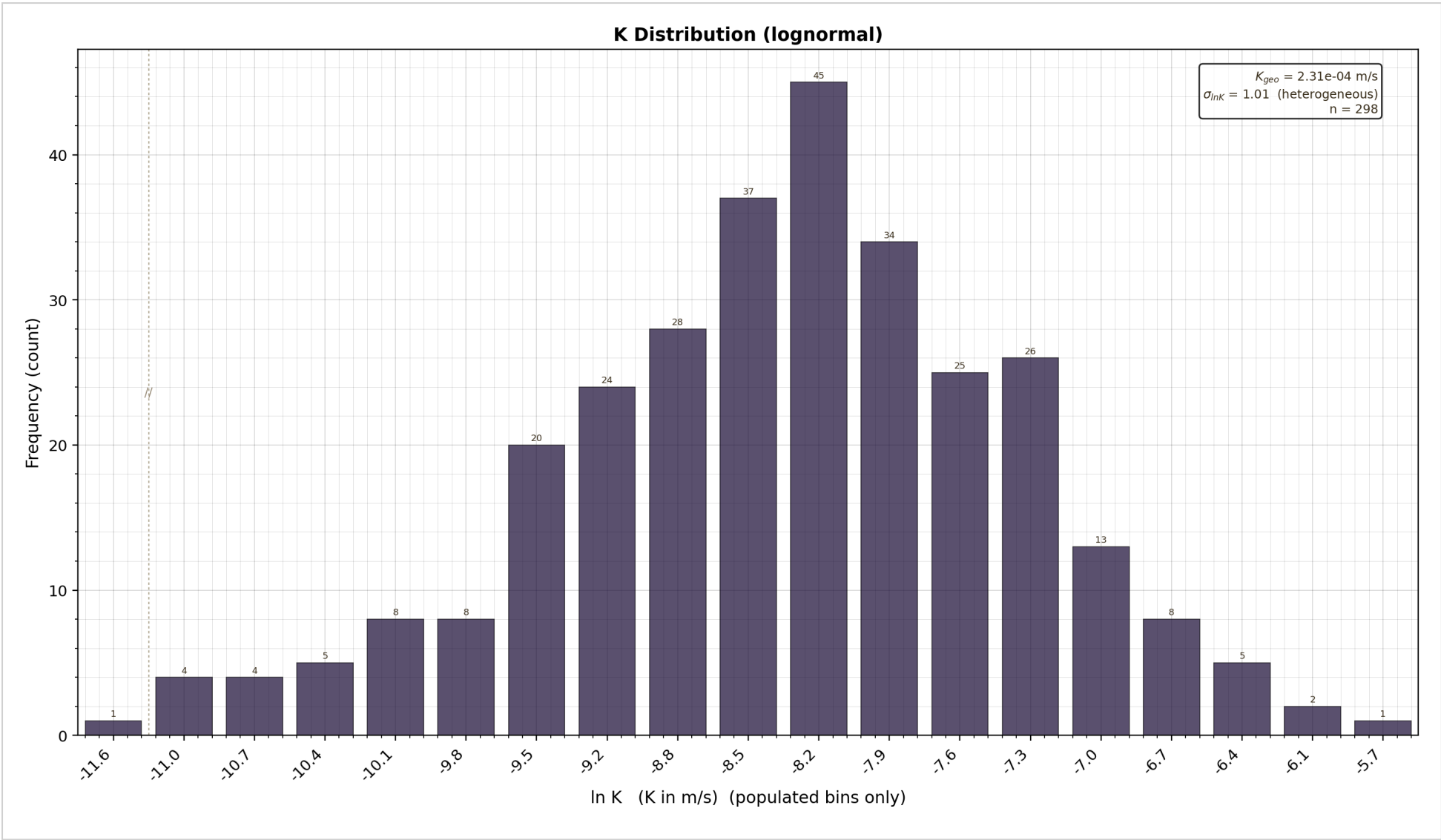
Hydraulic Conductivity by Method — per dataset



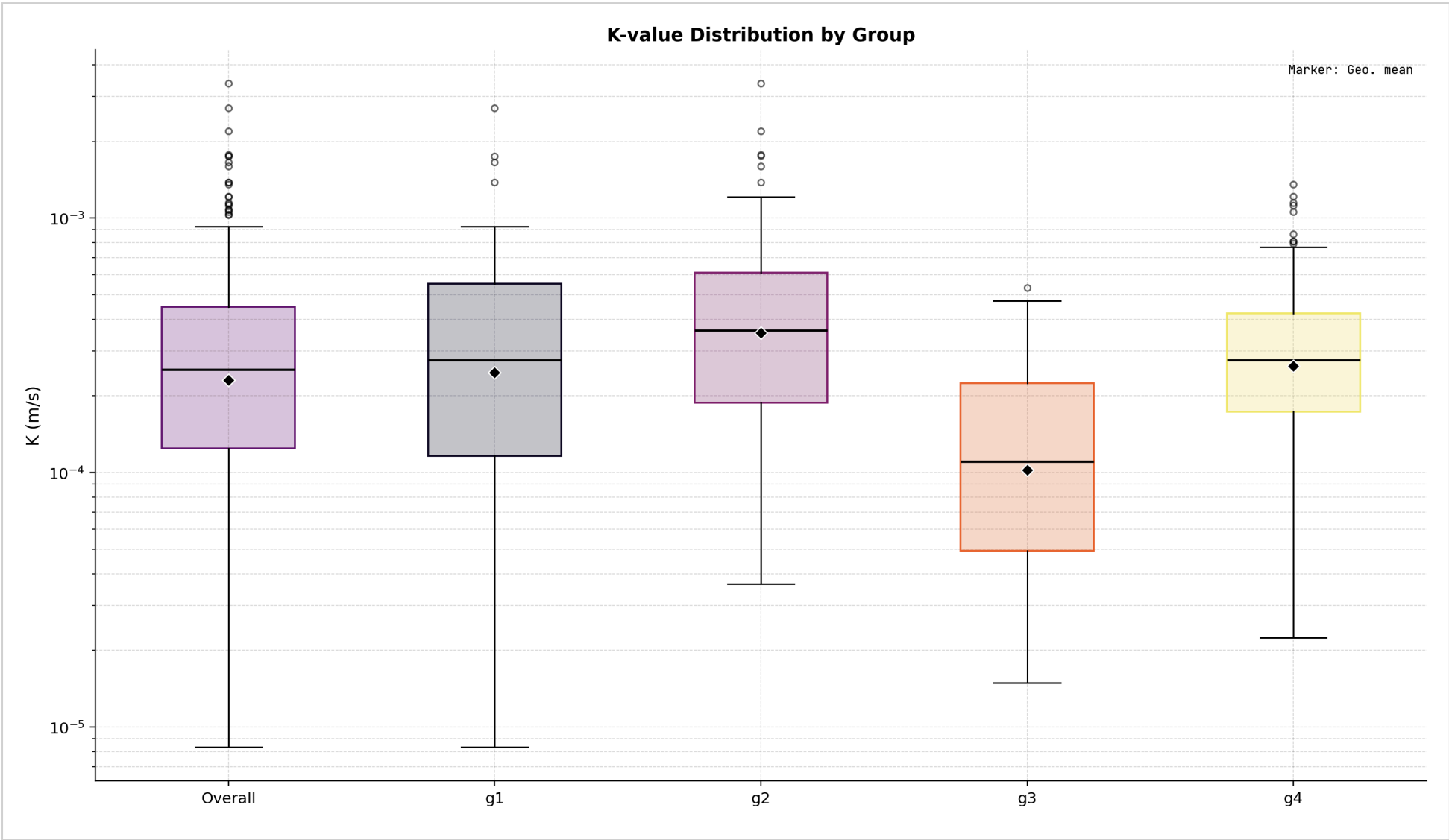
Hydraulic Conductivity Distribution (Lognormal) — per group



Hydraulic Conductivity Distribution (Lognormal) — per dataset



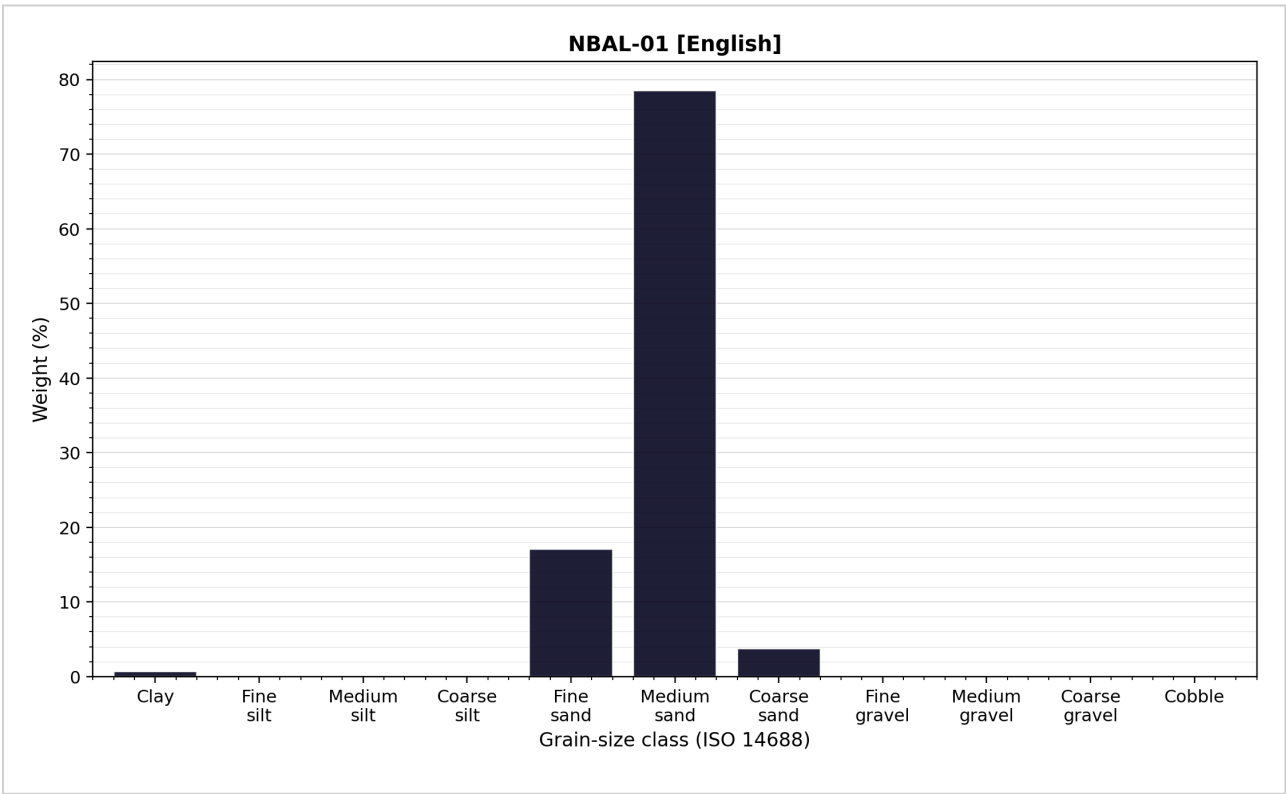
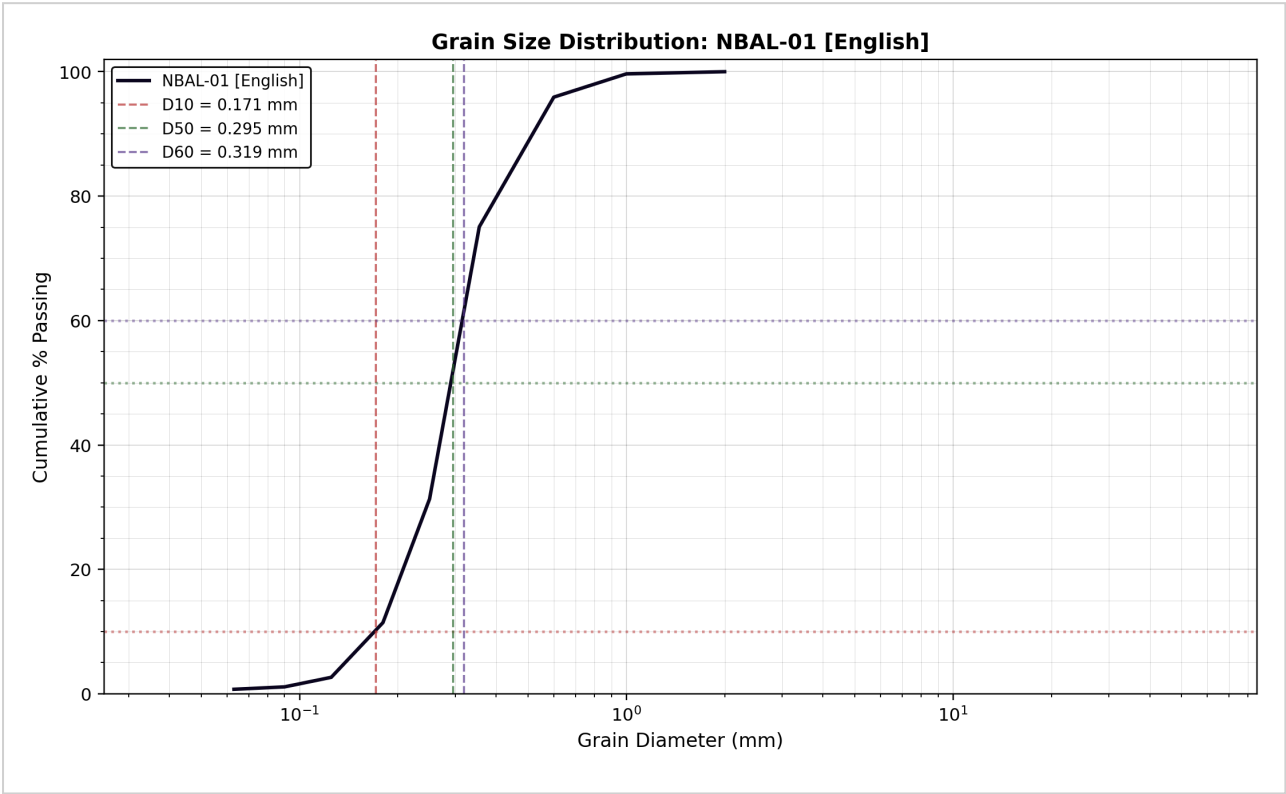
Hydraulic Conductivity Distribution

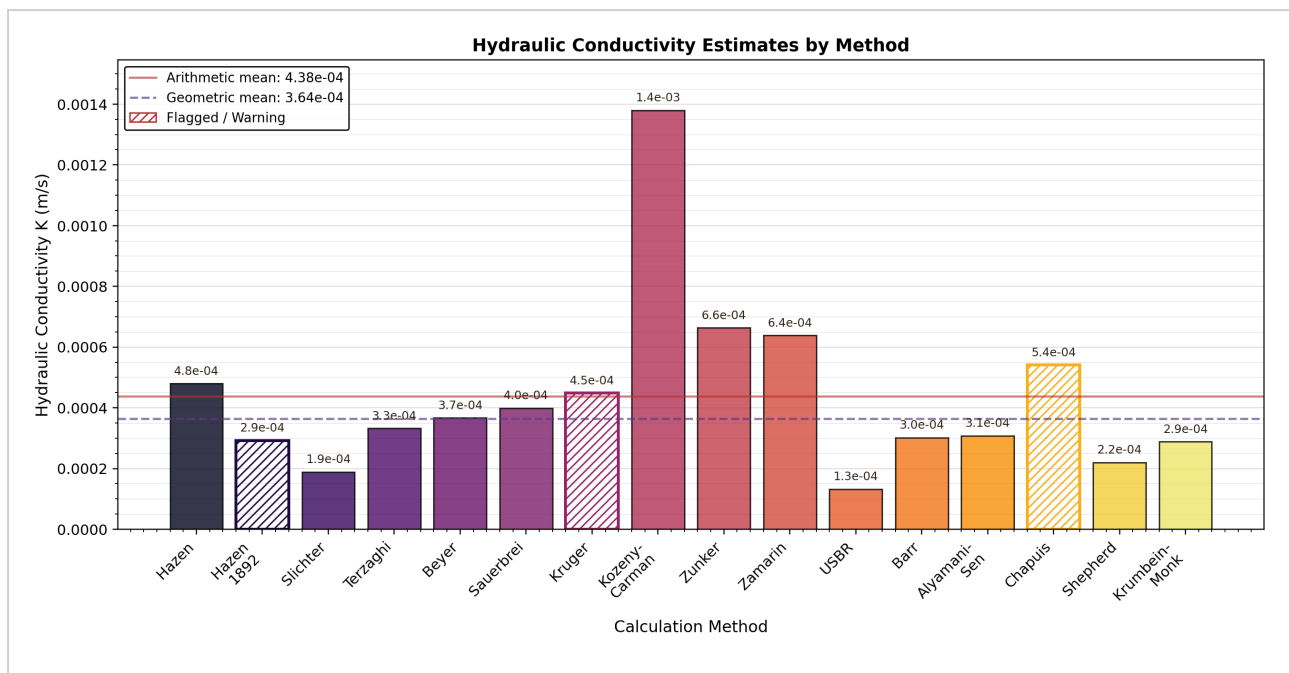


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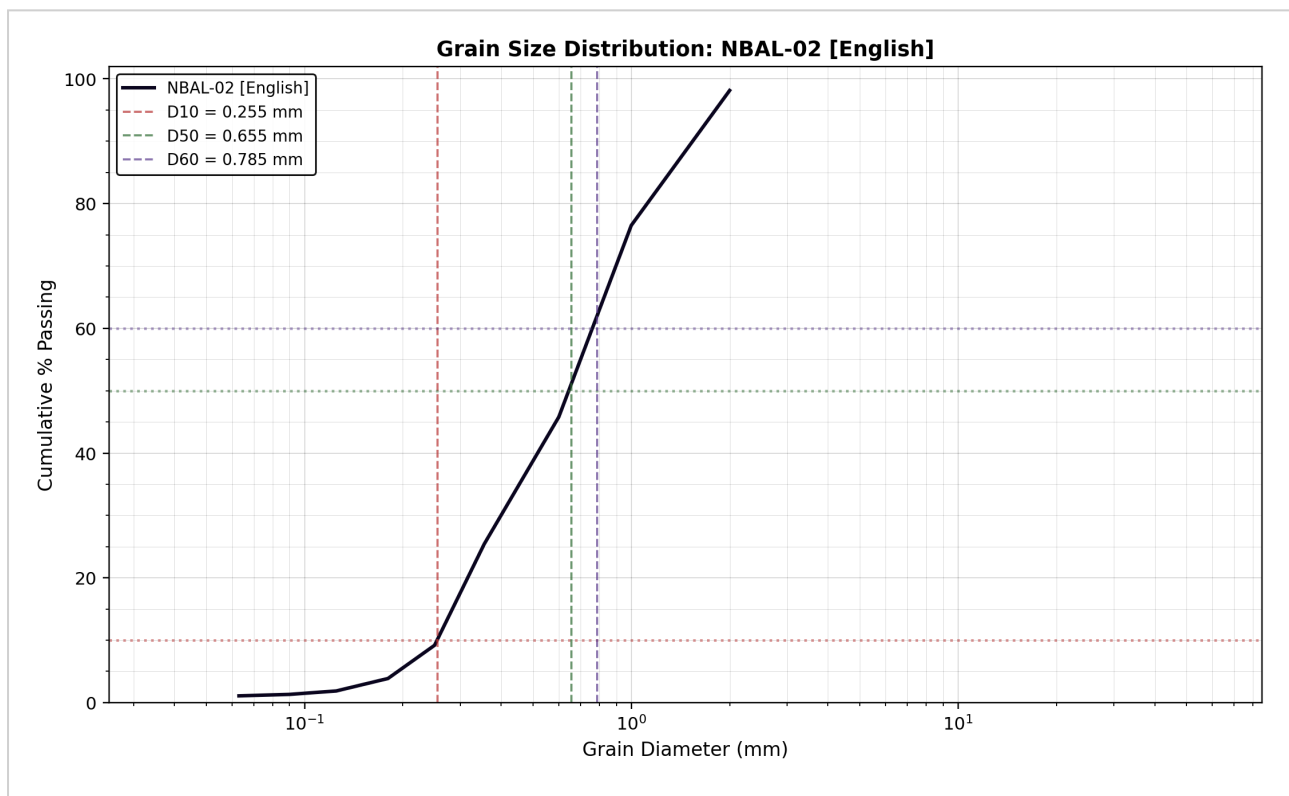
Individual Sample Plots

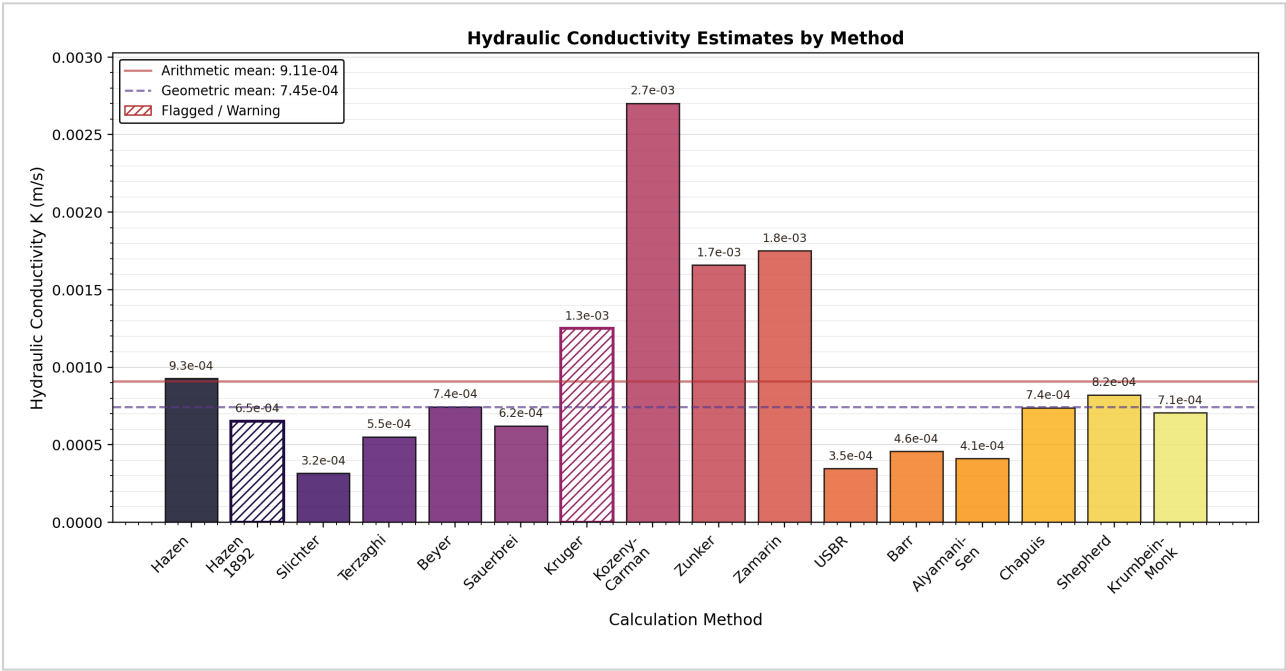
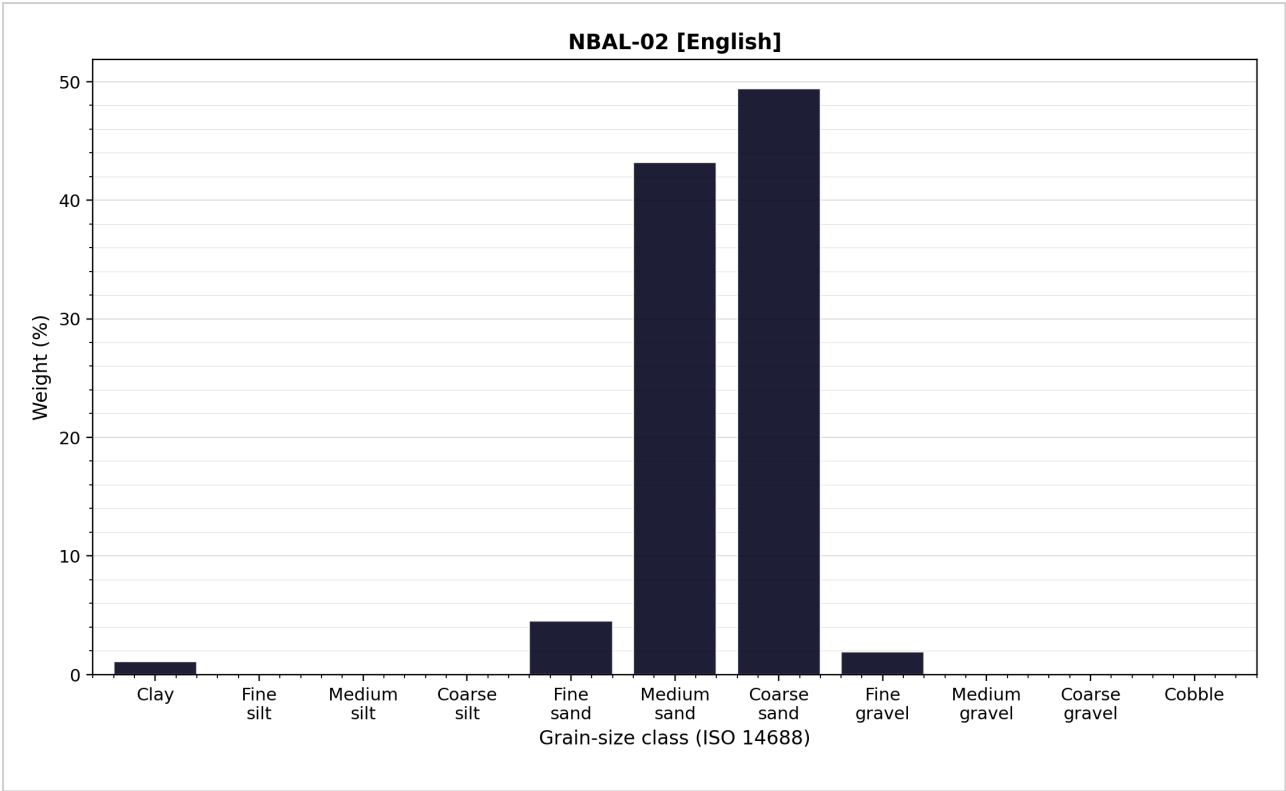
NBAL-01 [English]



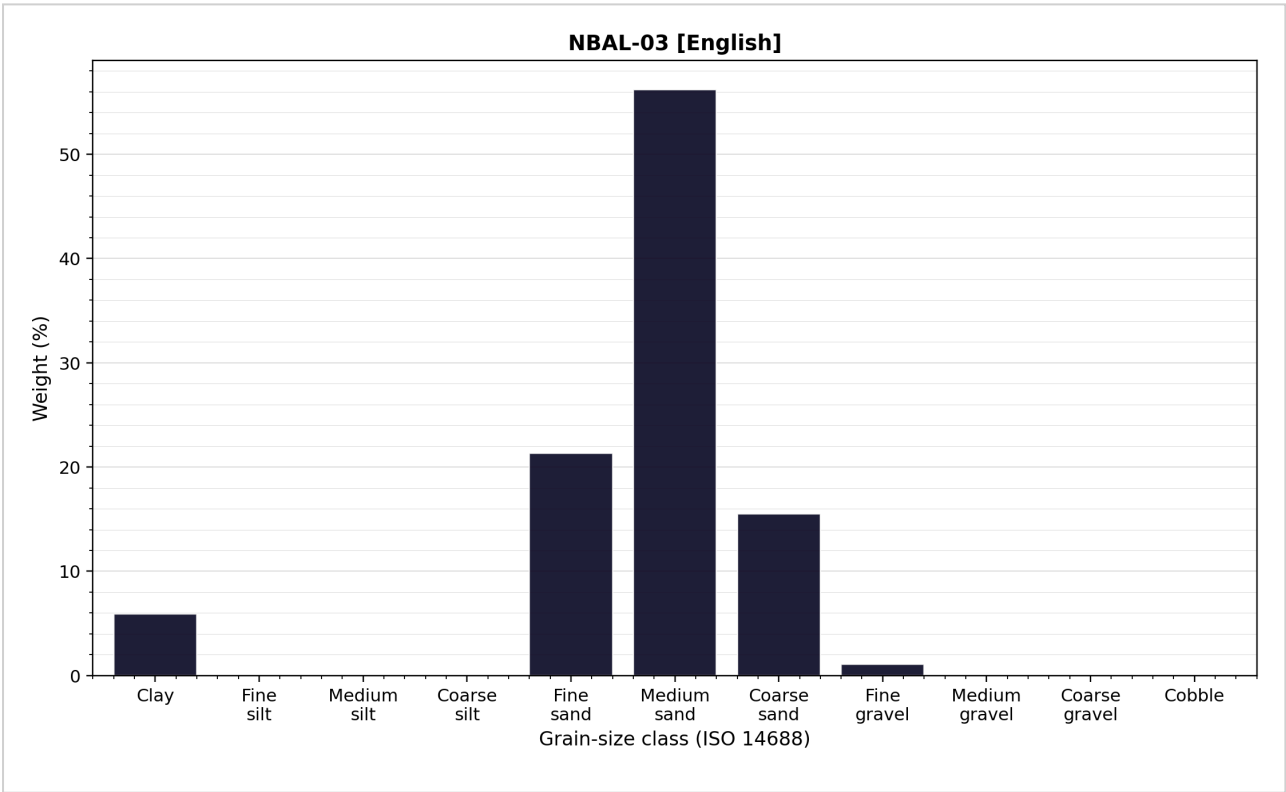
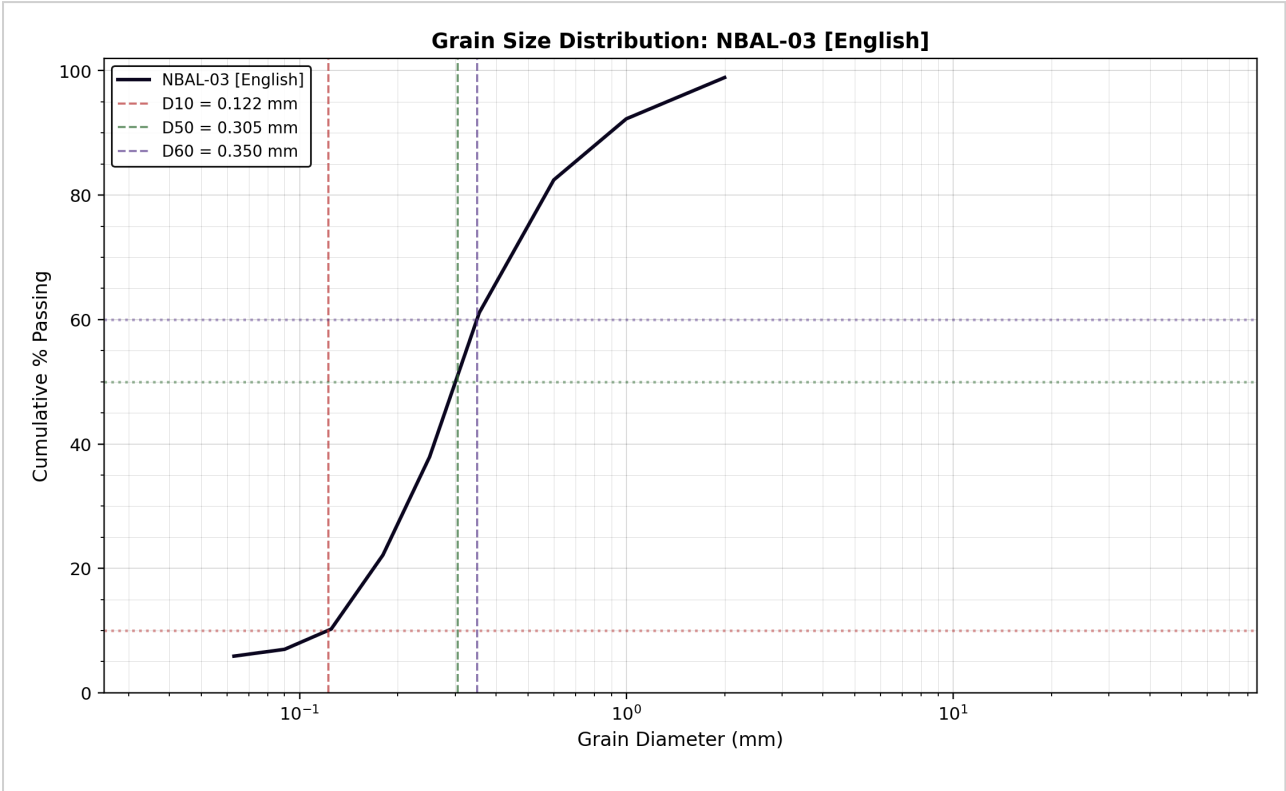


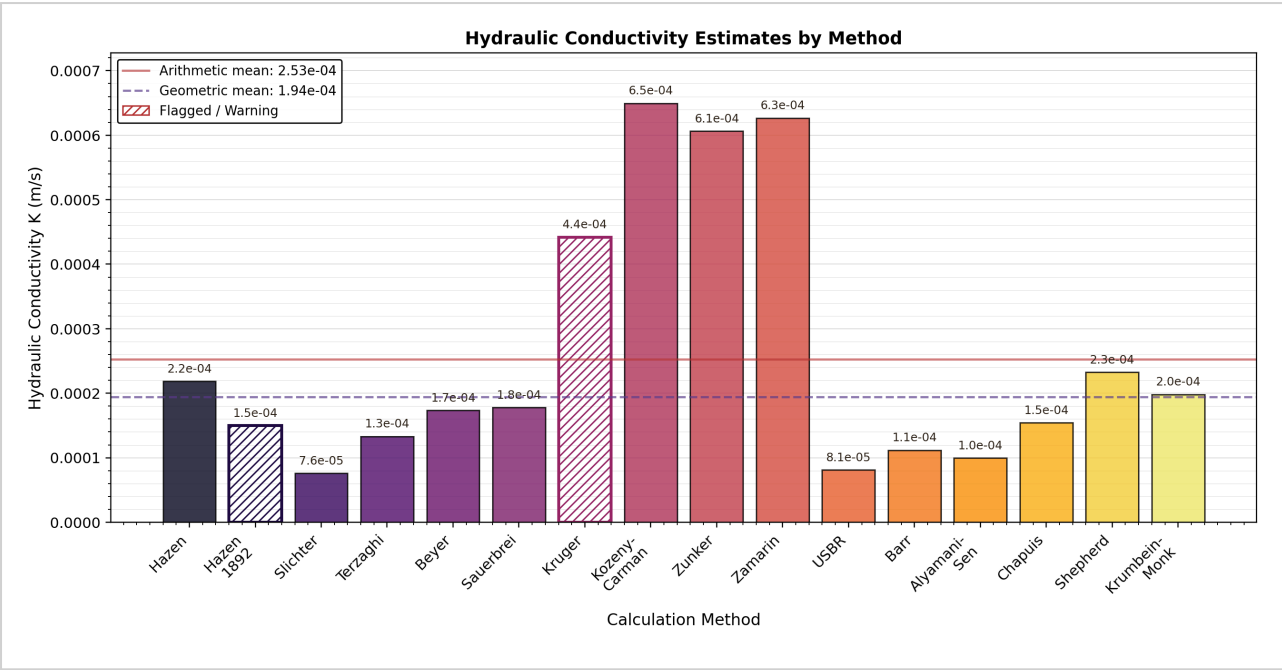
NBAL-02 [English]



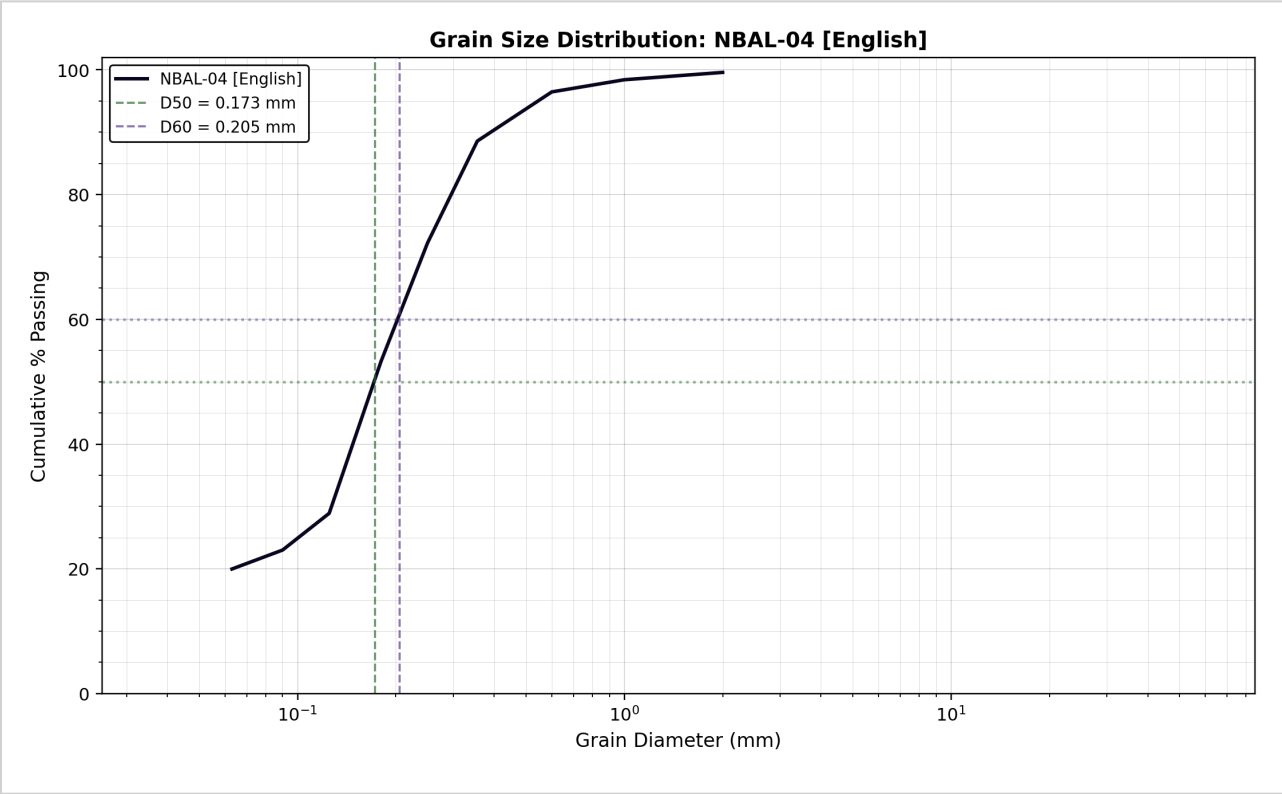


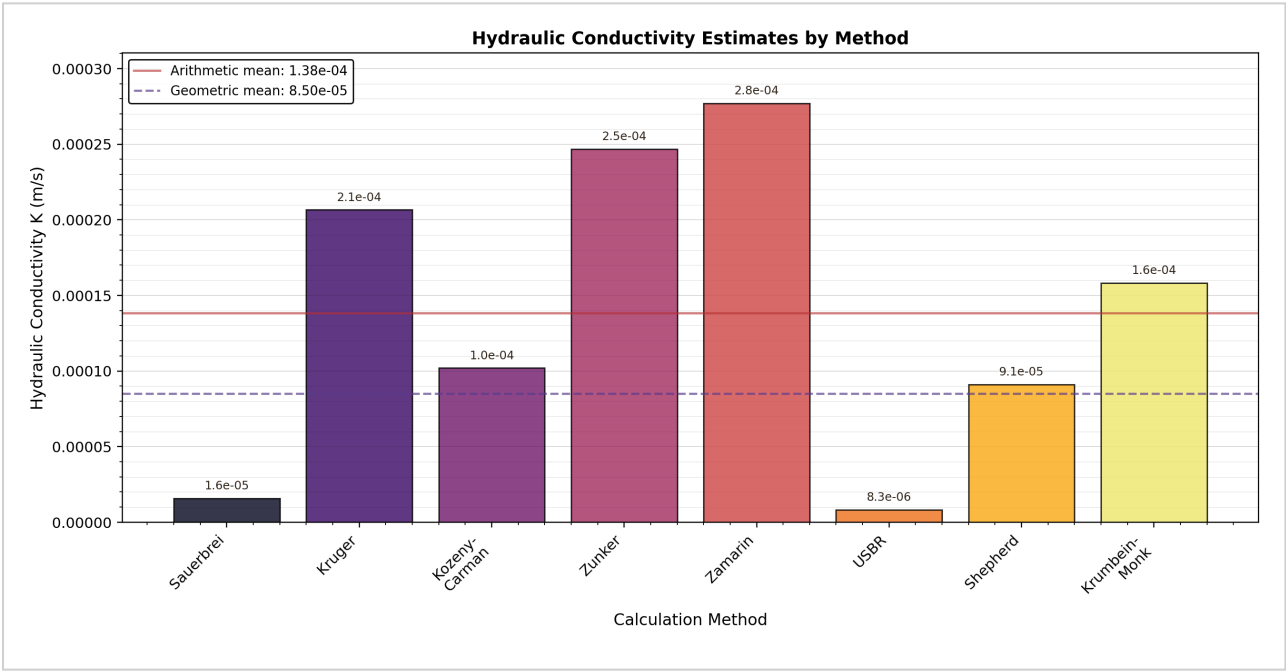
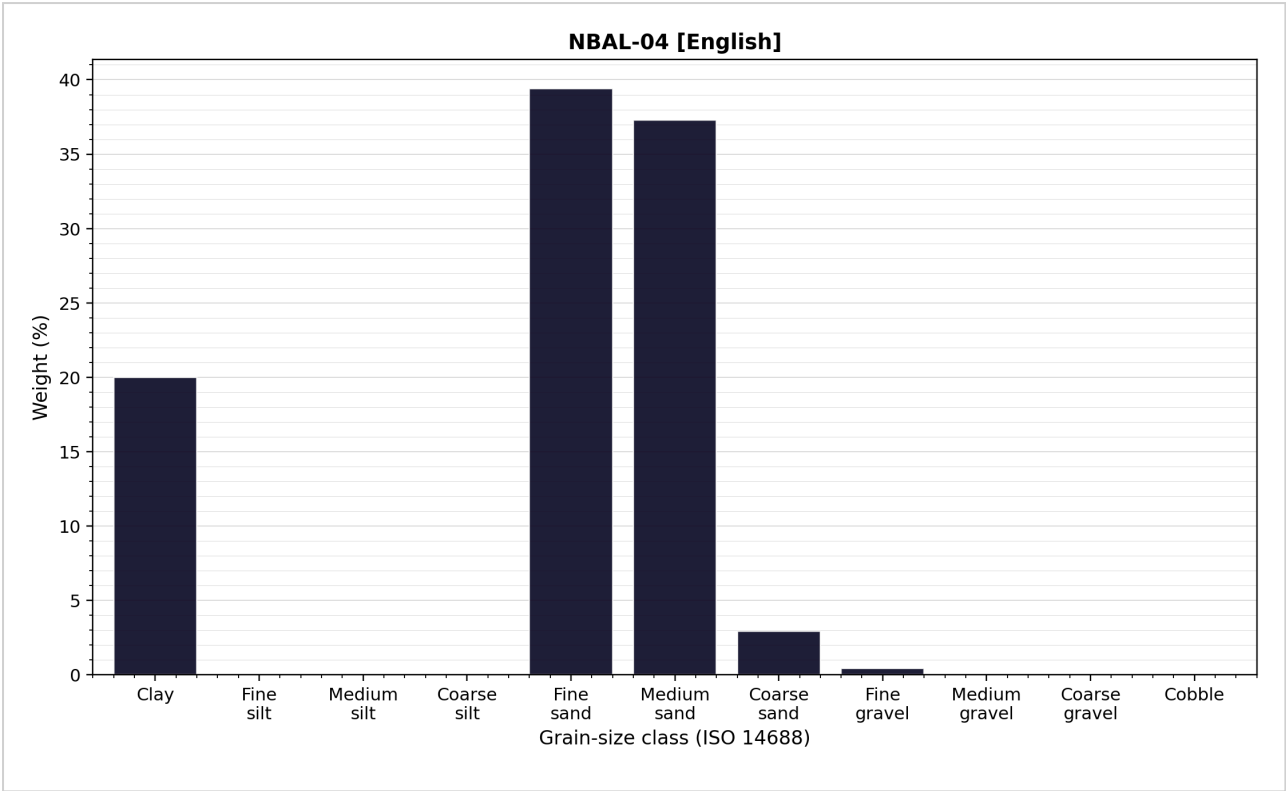
NBAL-03 [English]



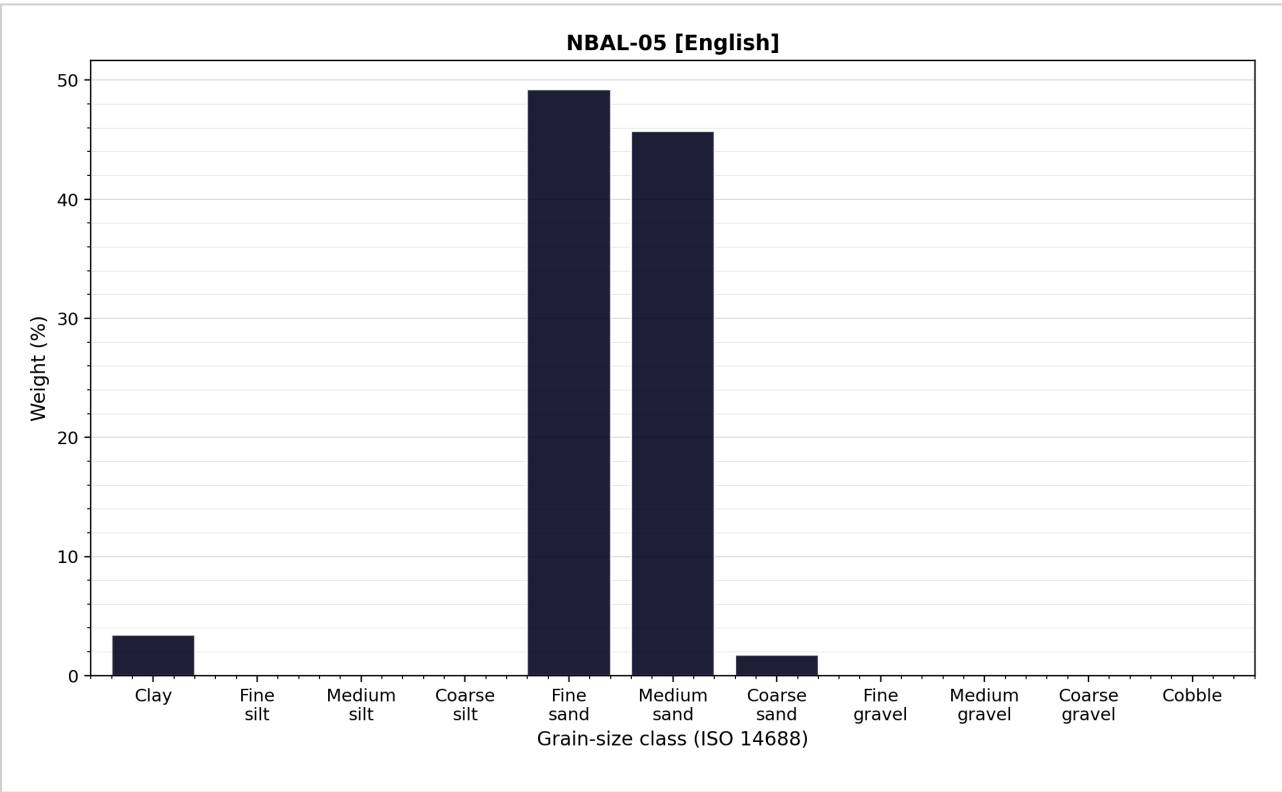
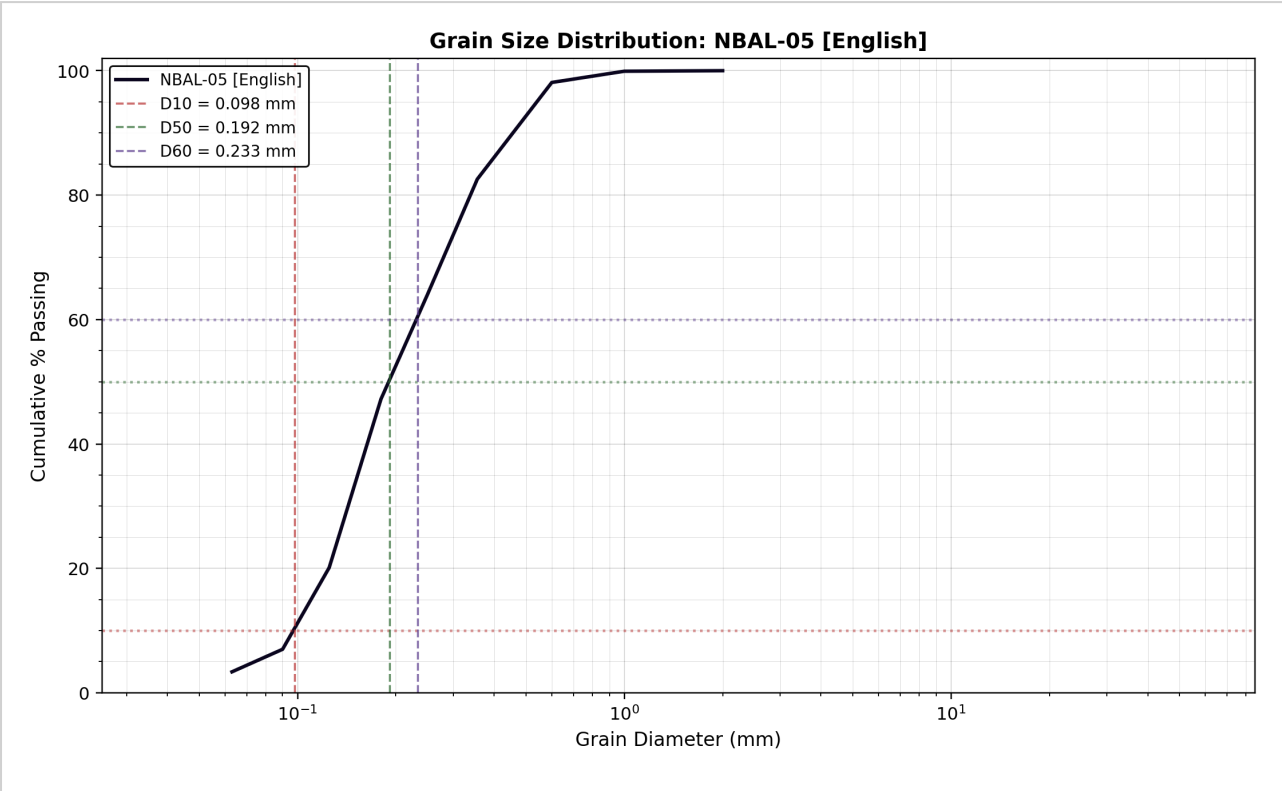


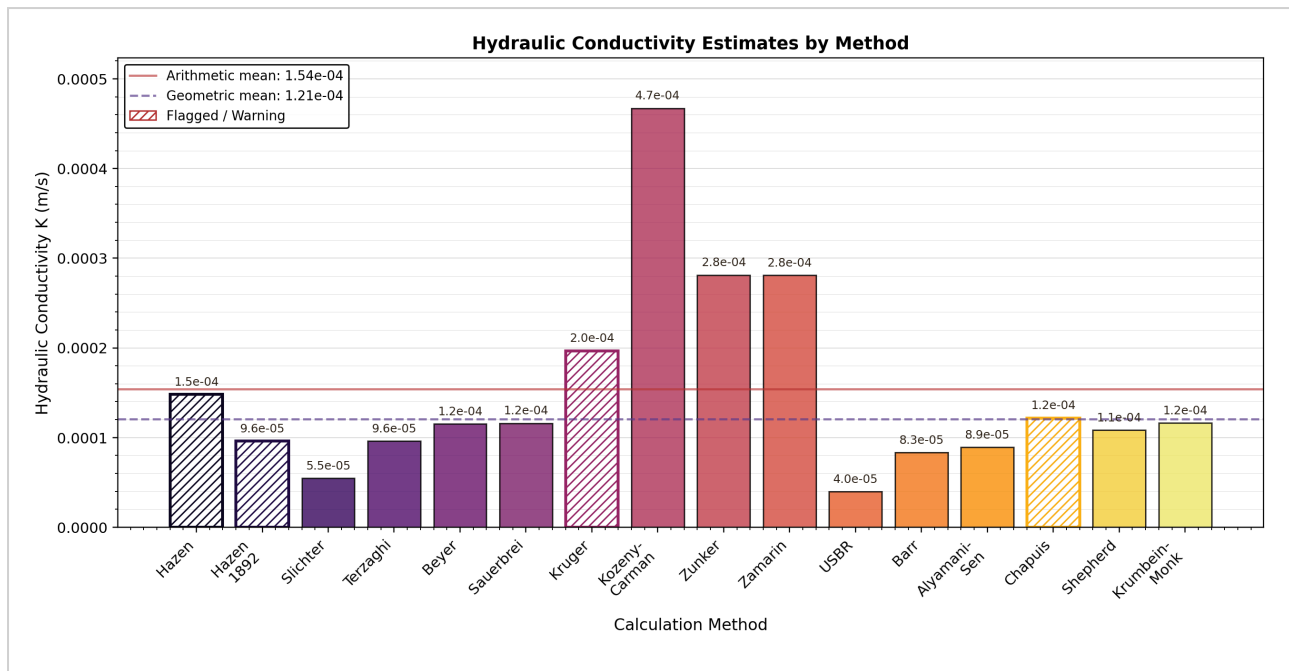
NBAL-04 [English]



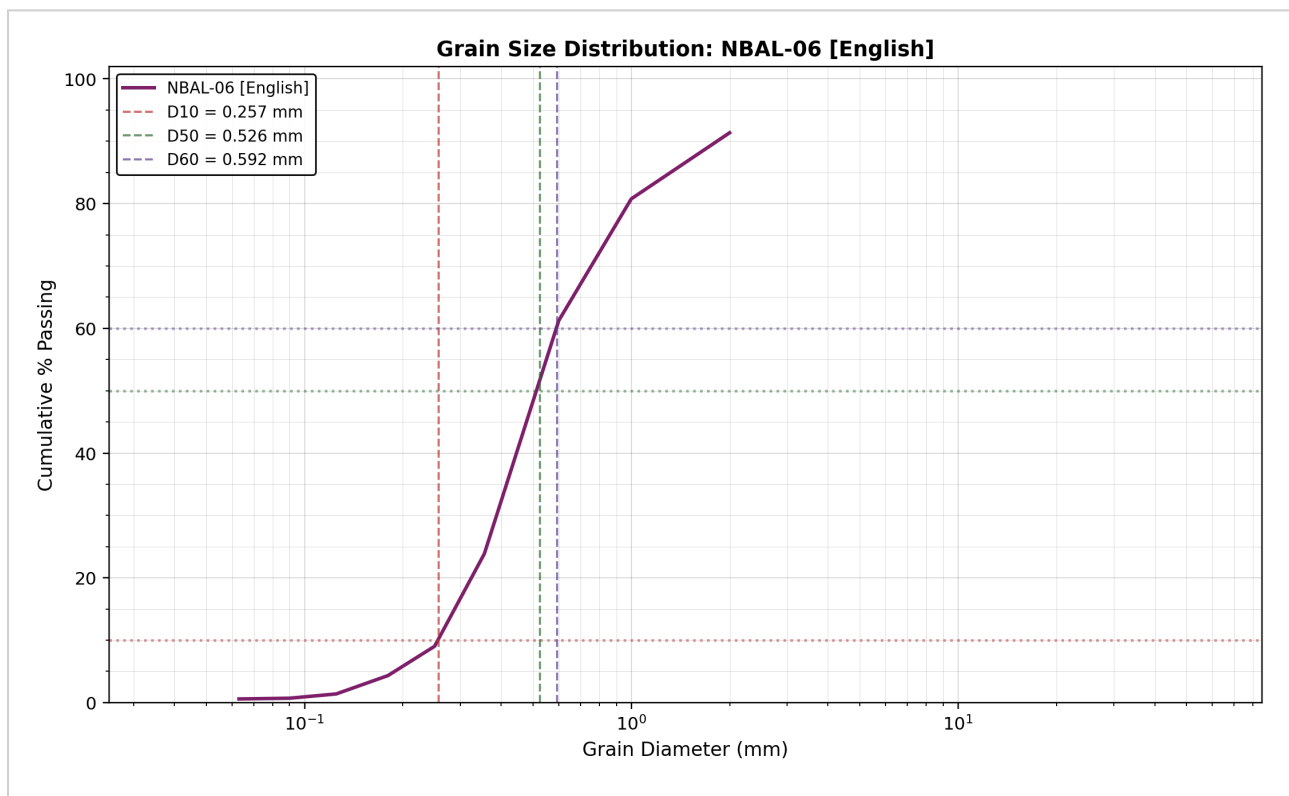


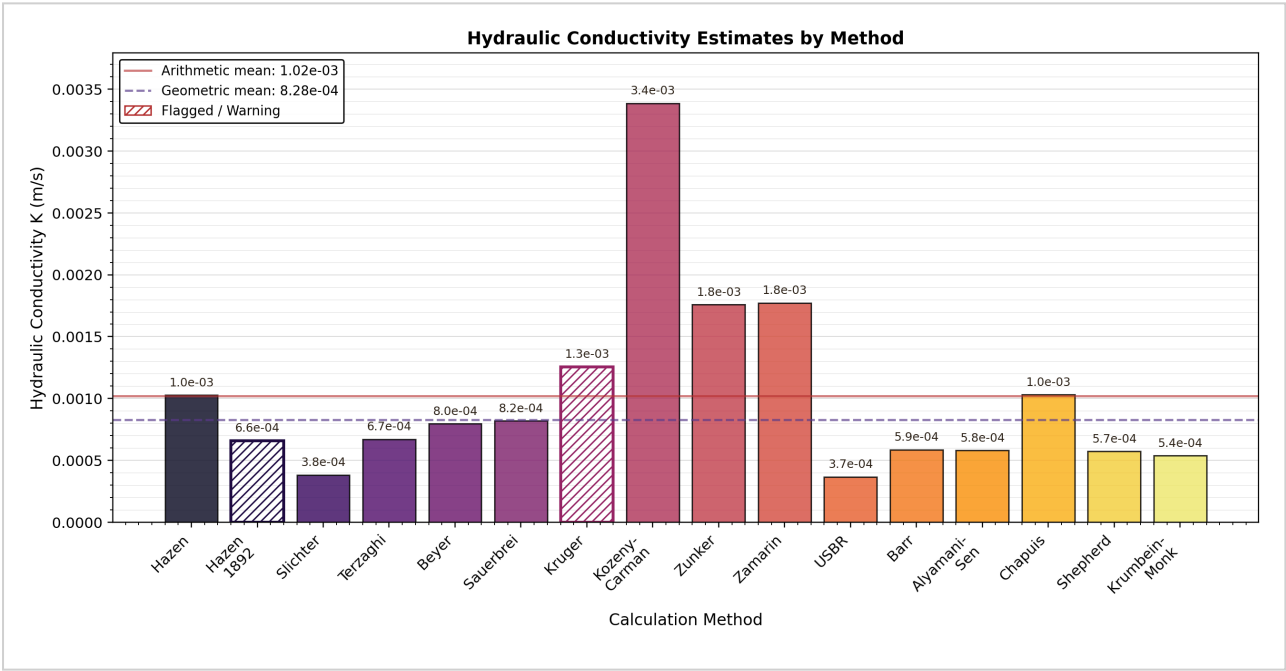
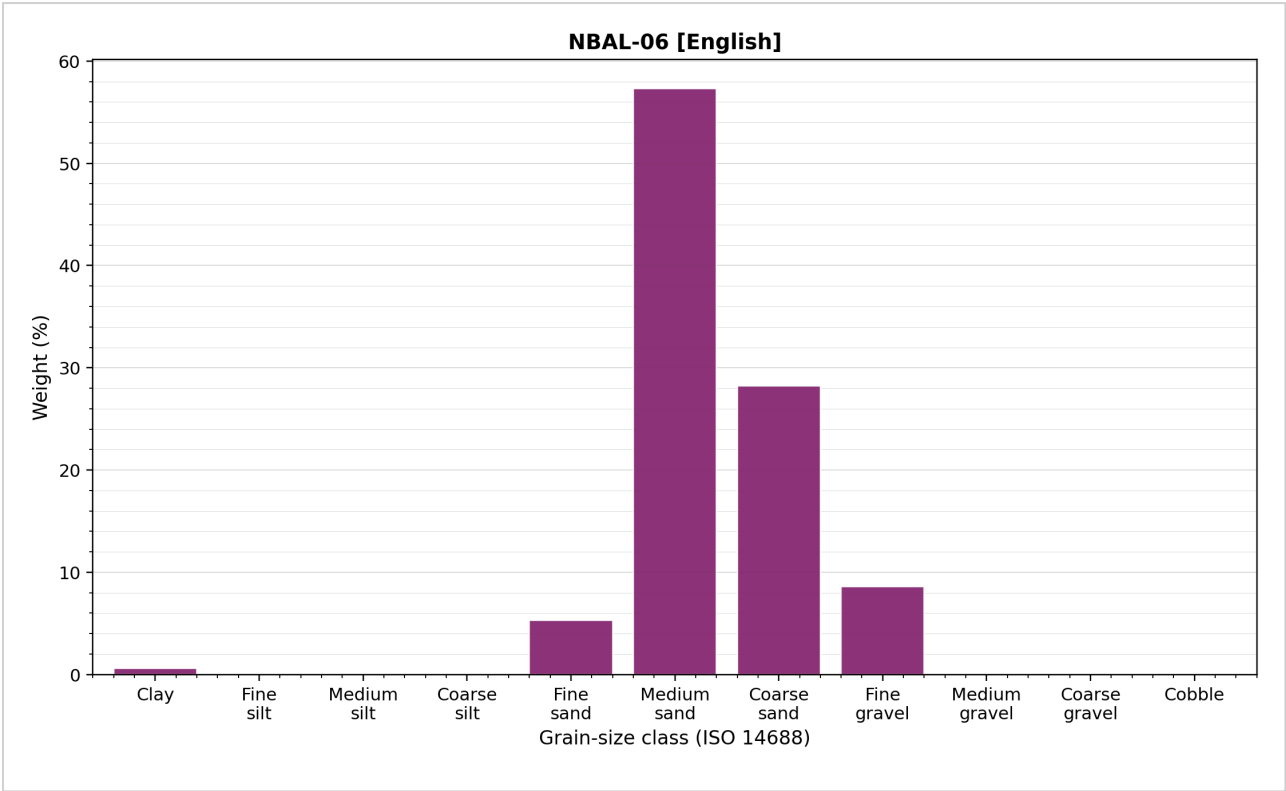
NBAL-05 [English]



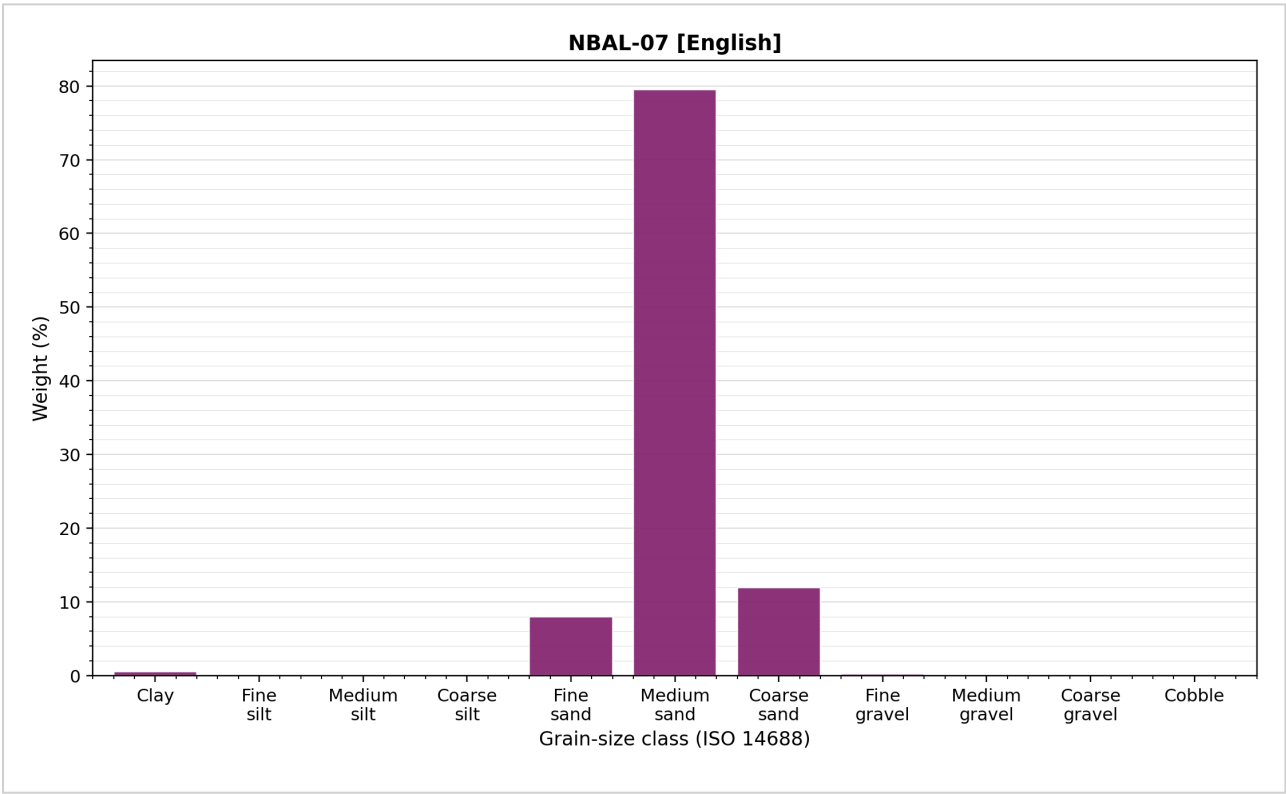
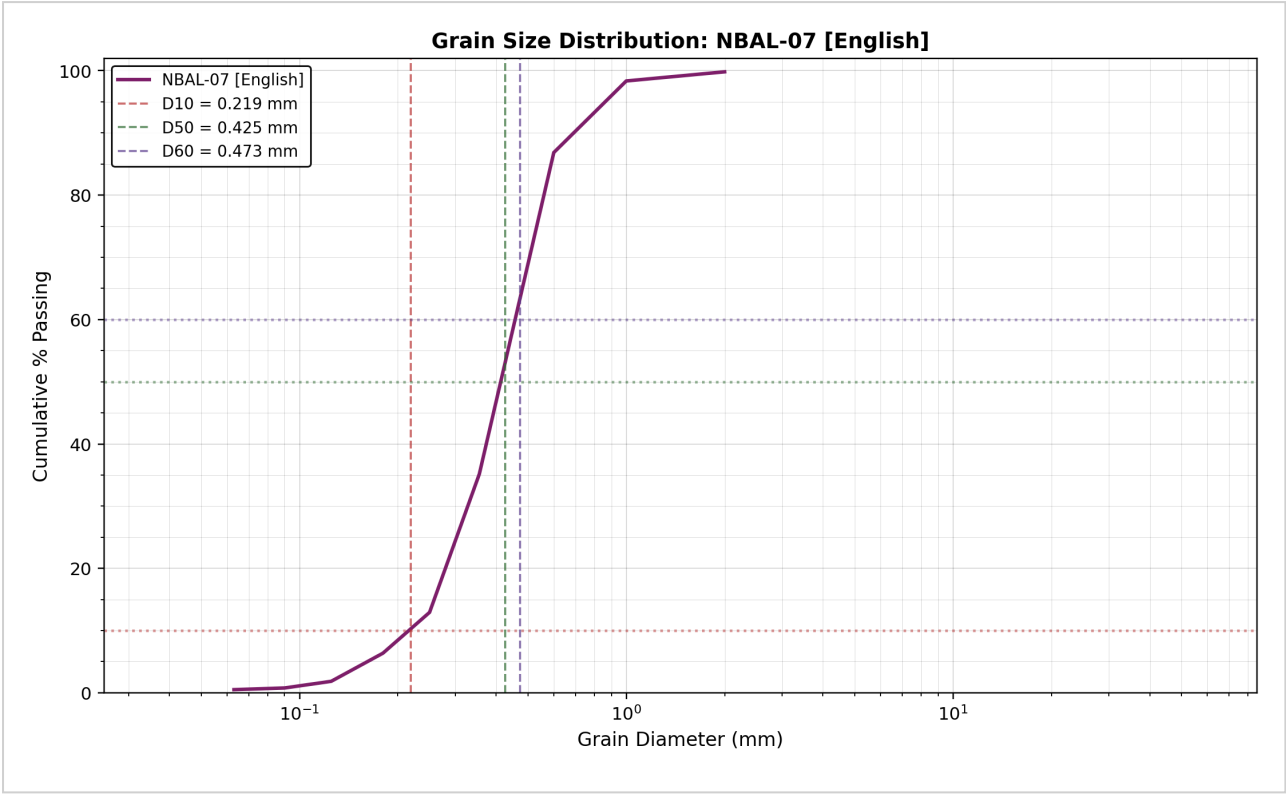


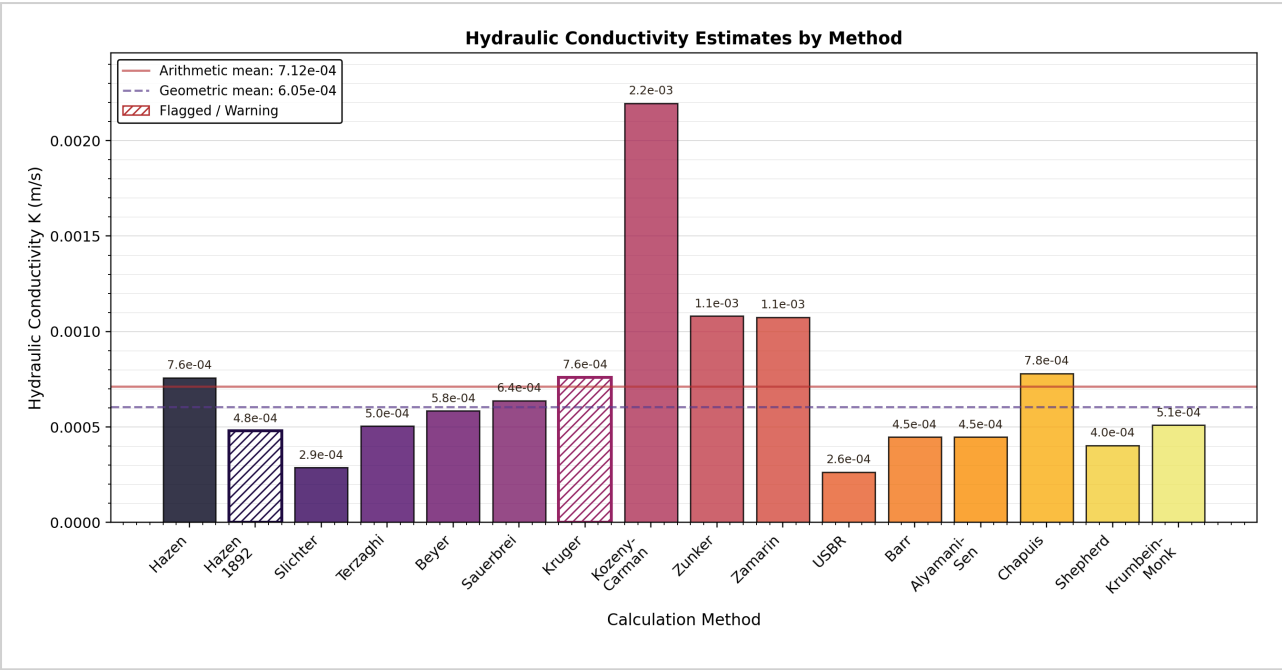
NBAL-06 [English]



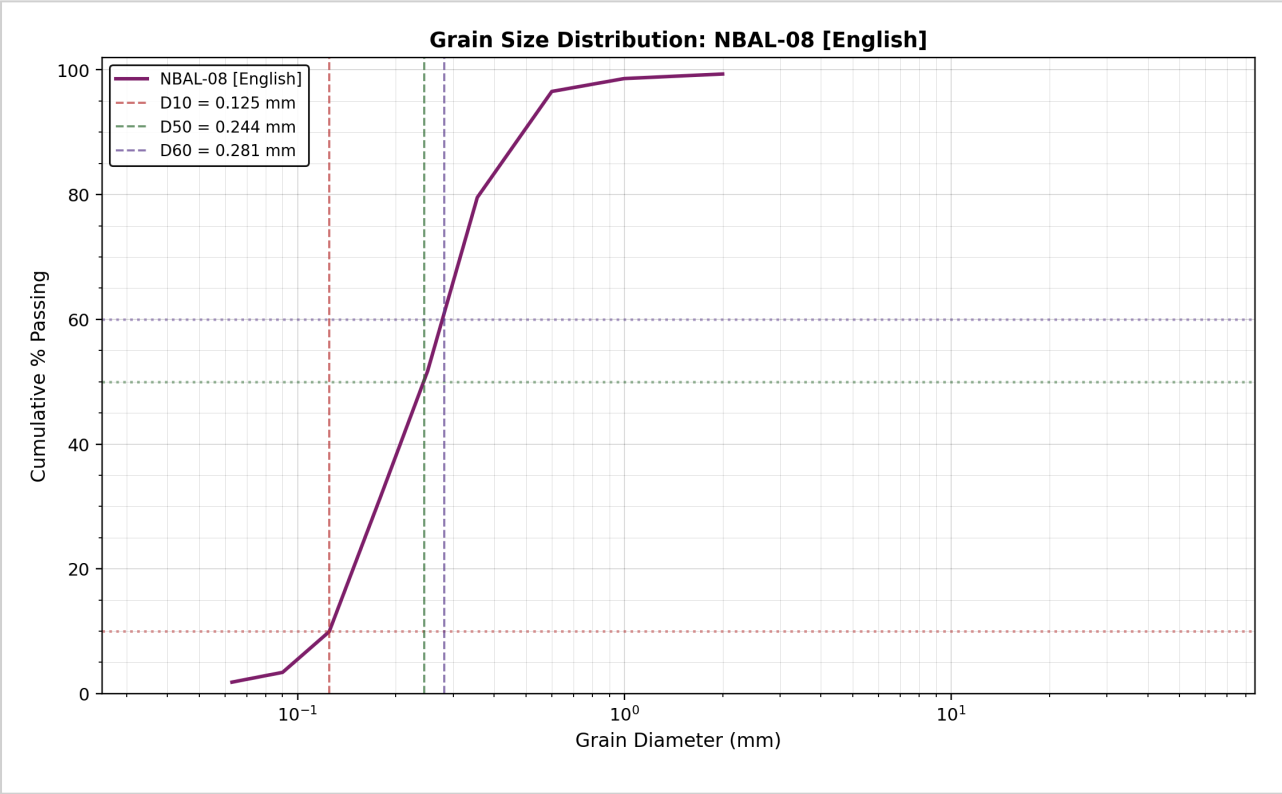


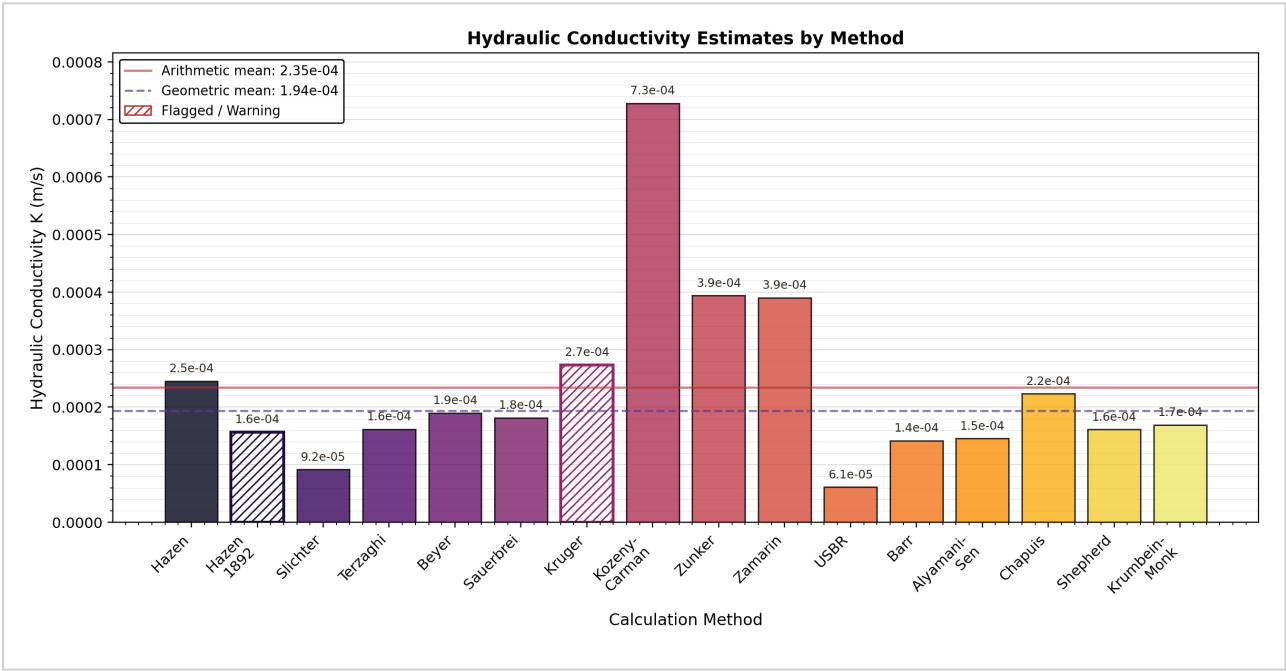
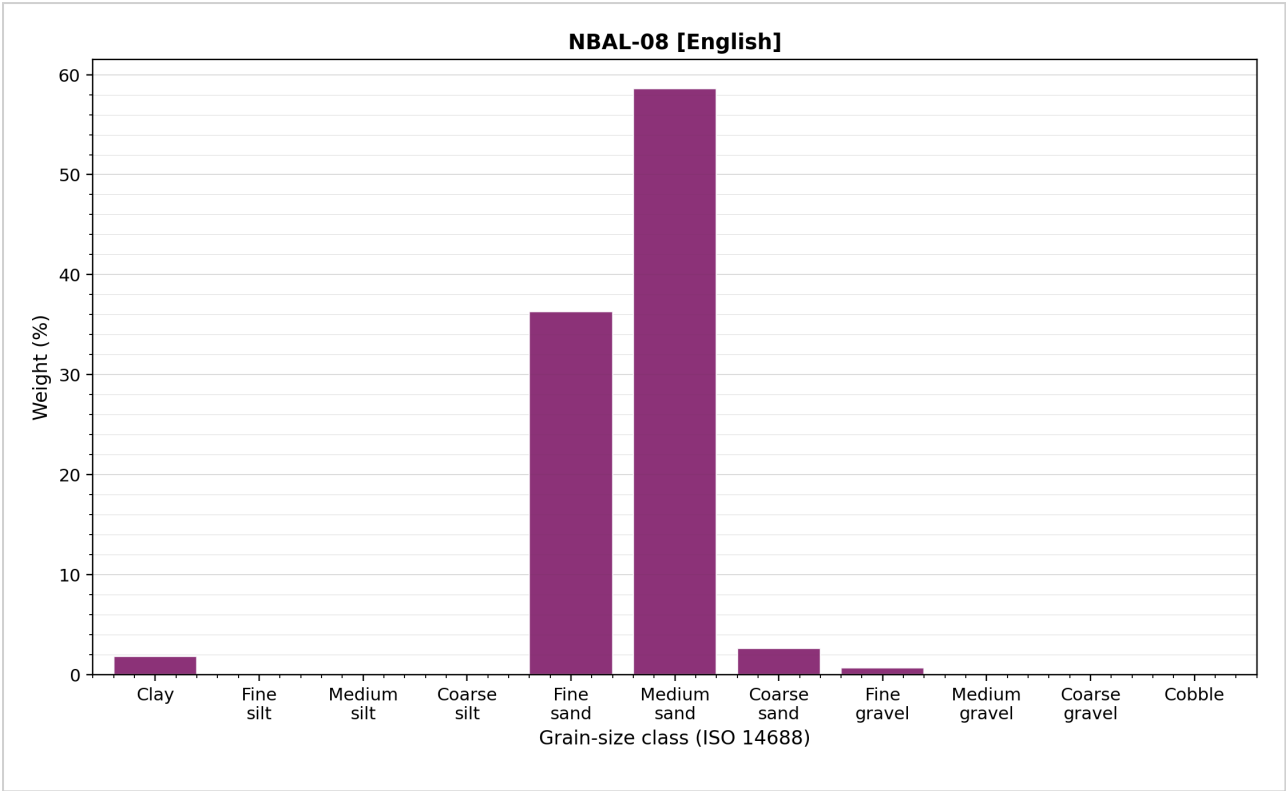
NBAL-07 [English]

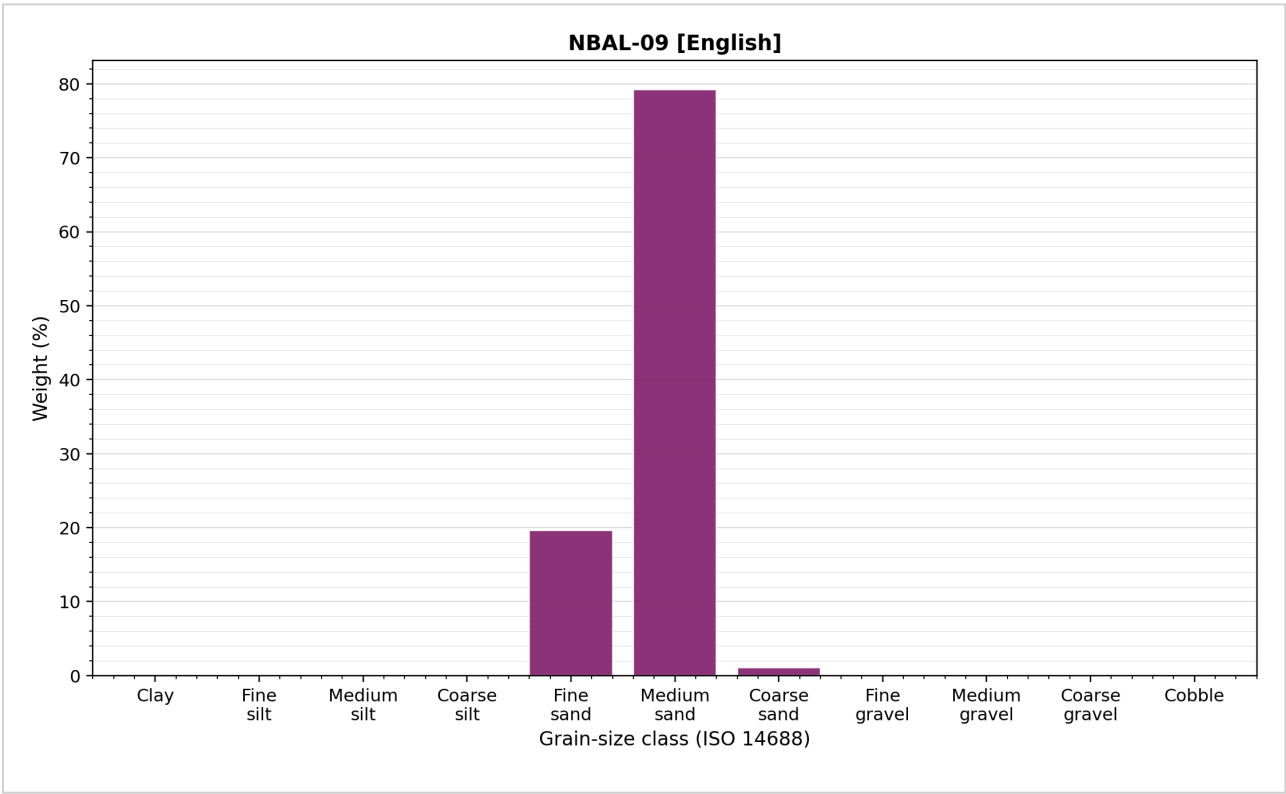
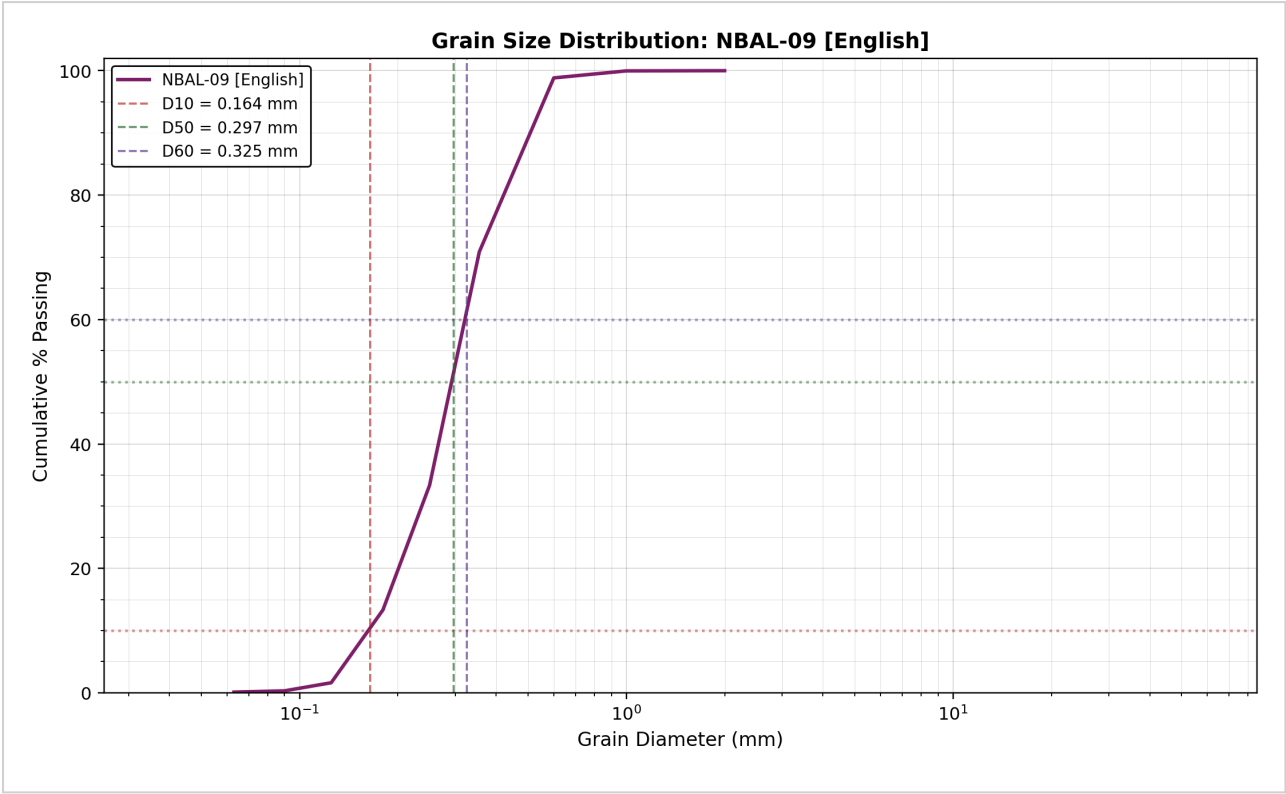


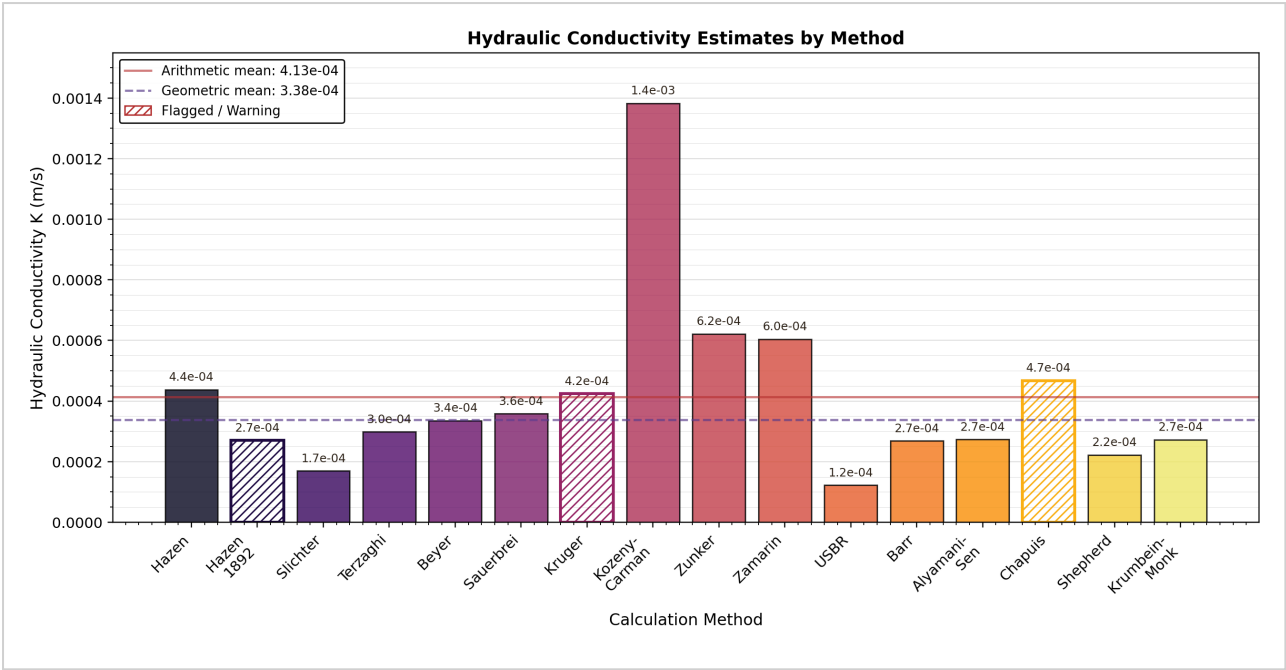


NBAL-08 [English]

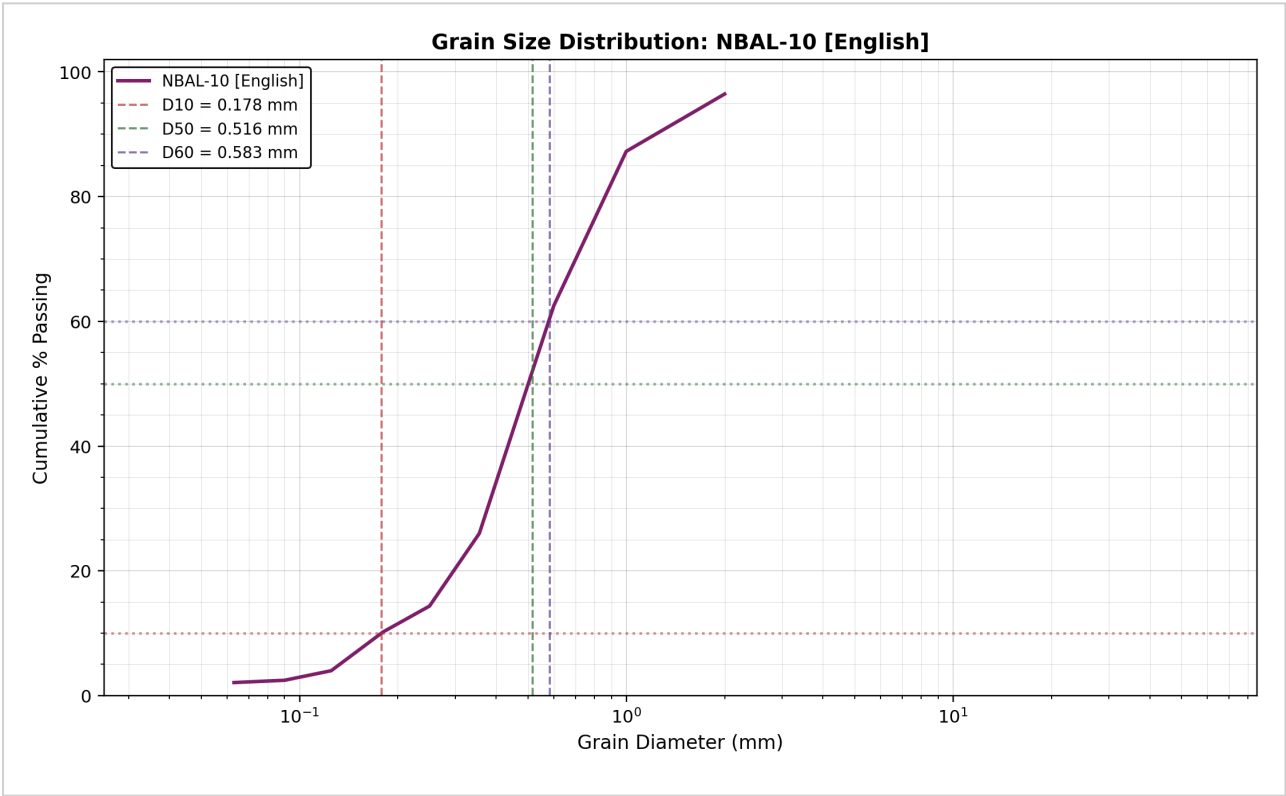


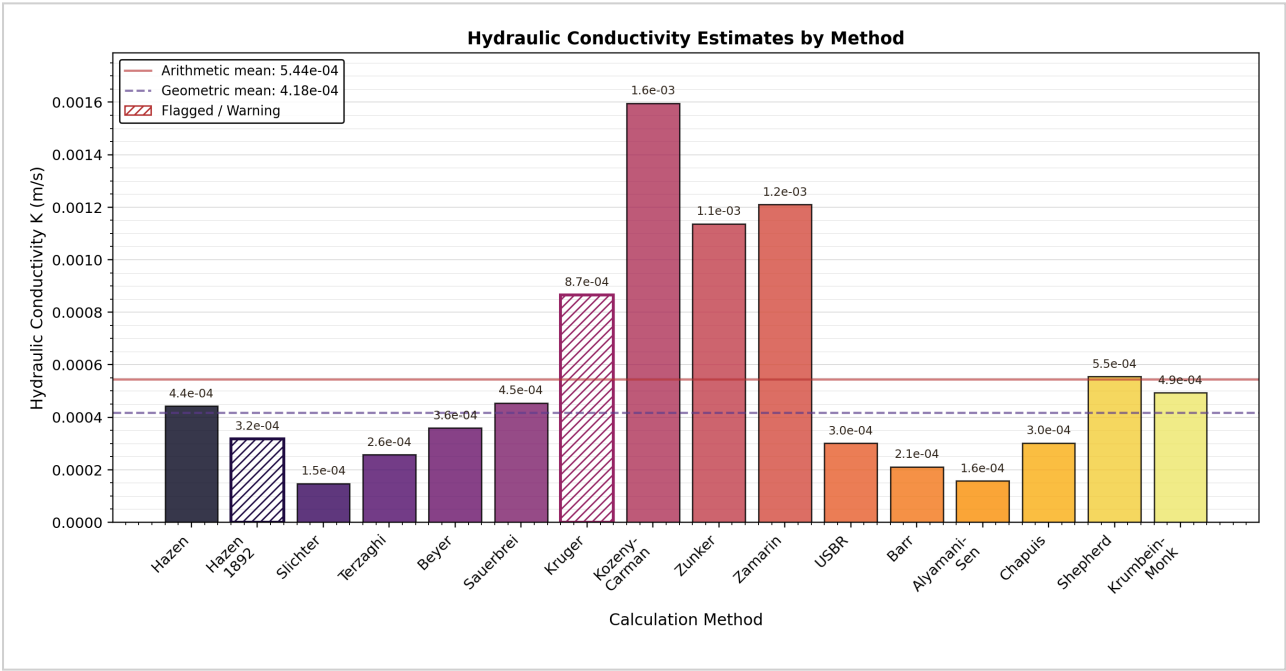
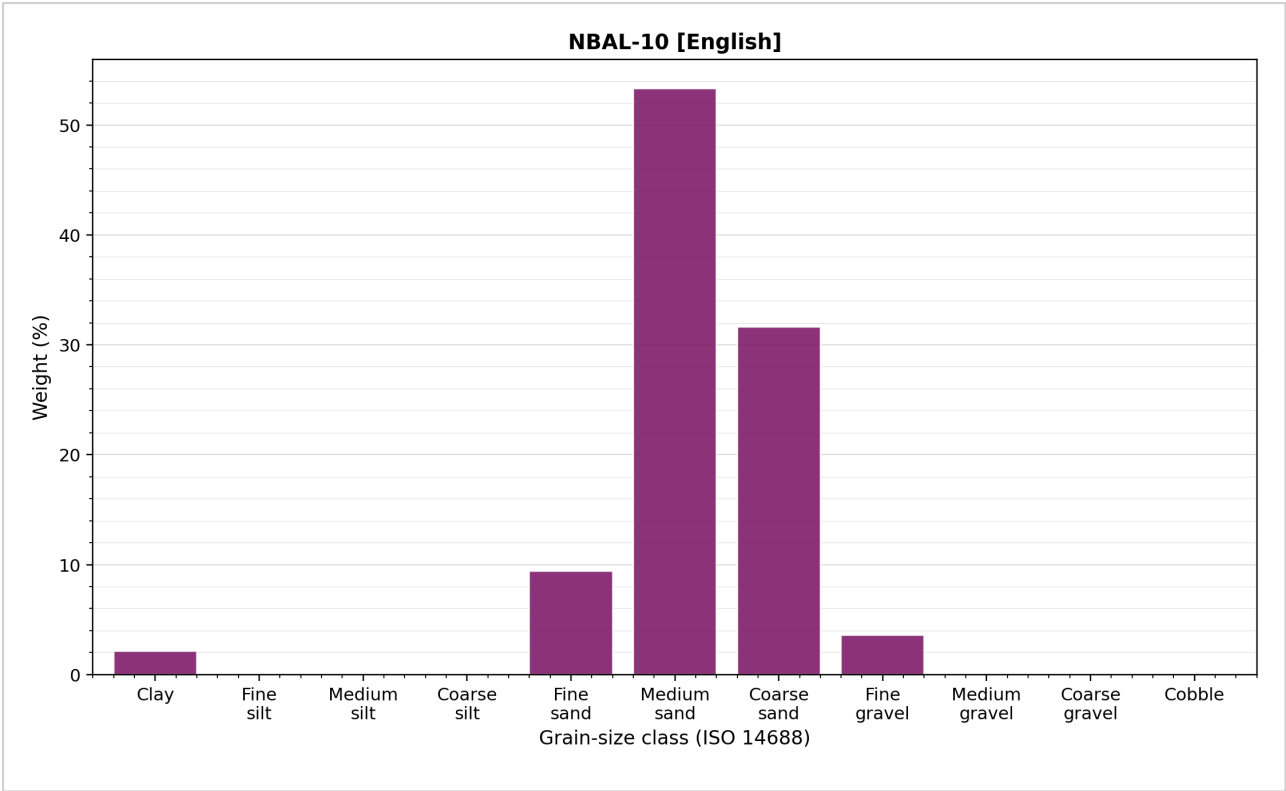




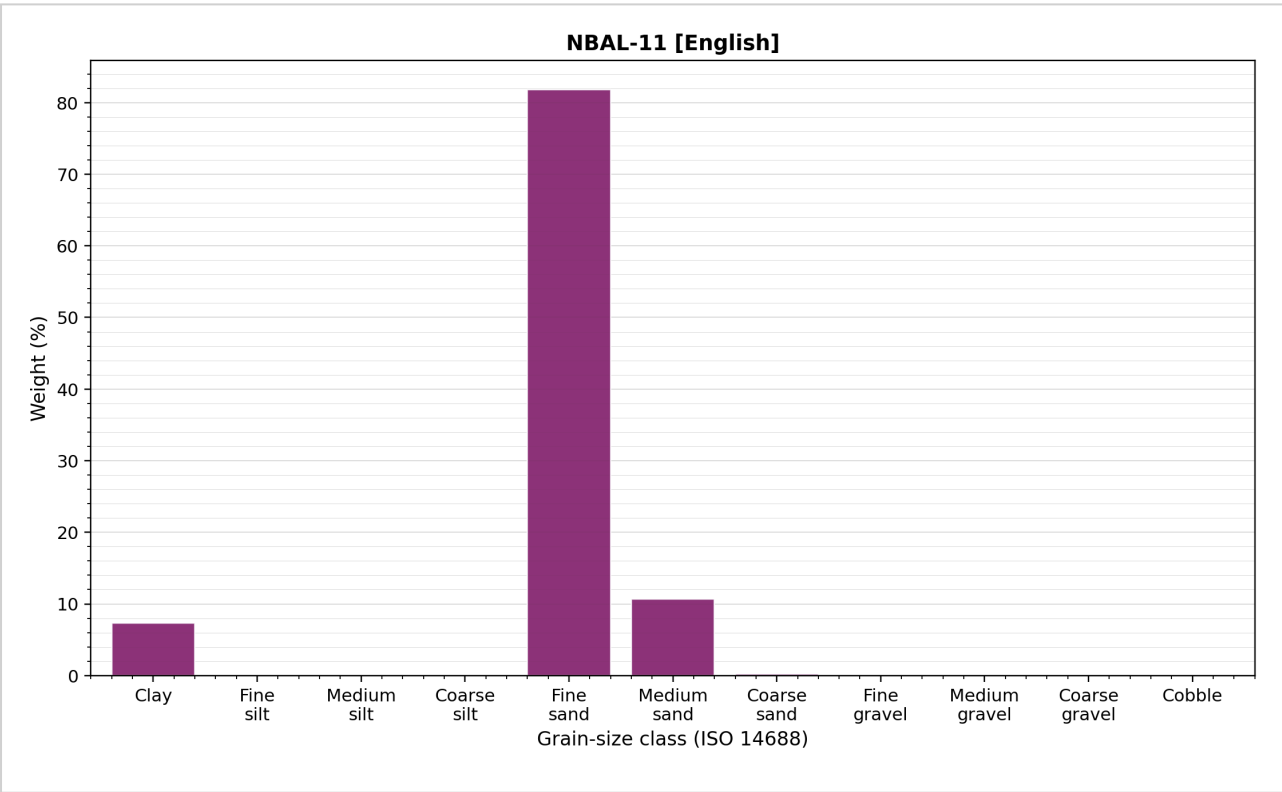
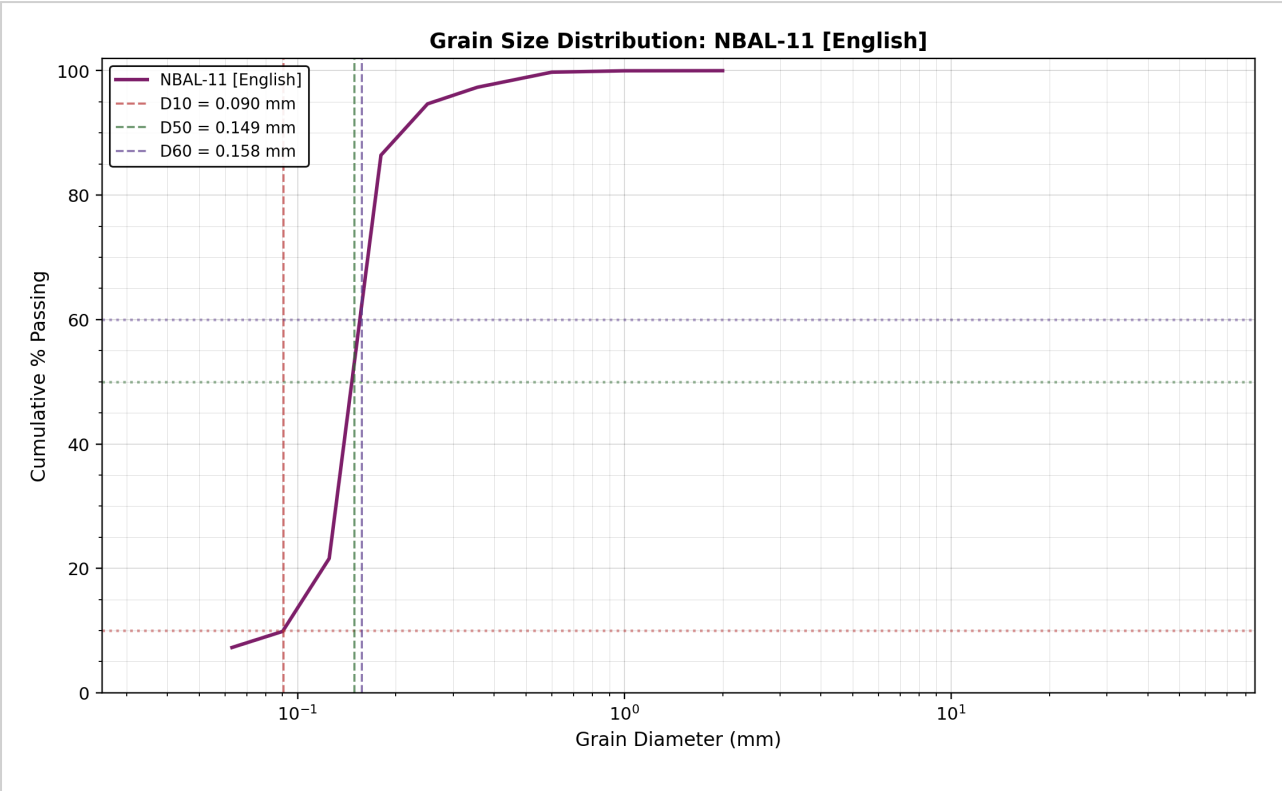


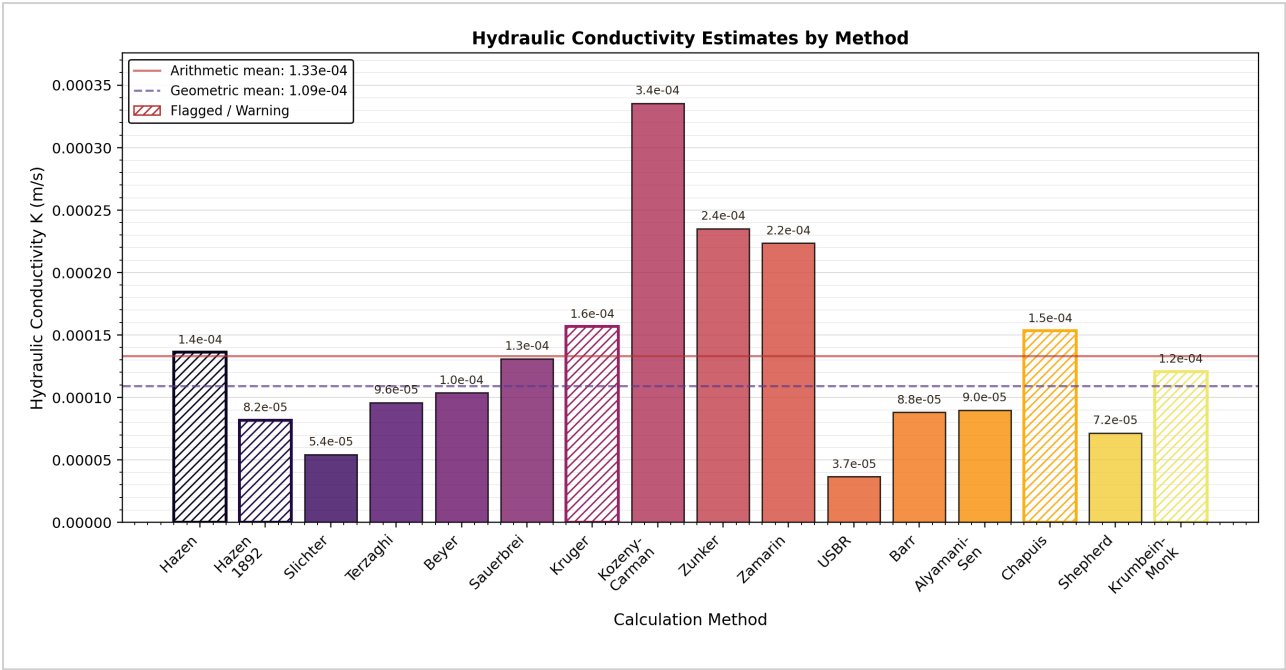
NBAL-10 [English]



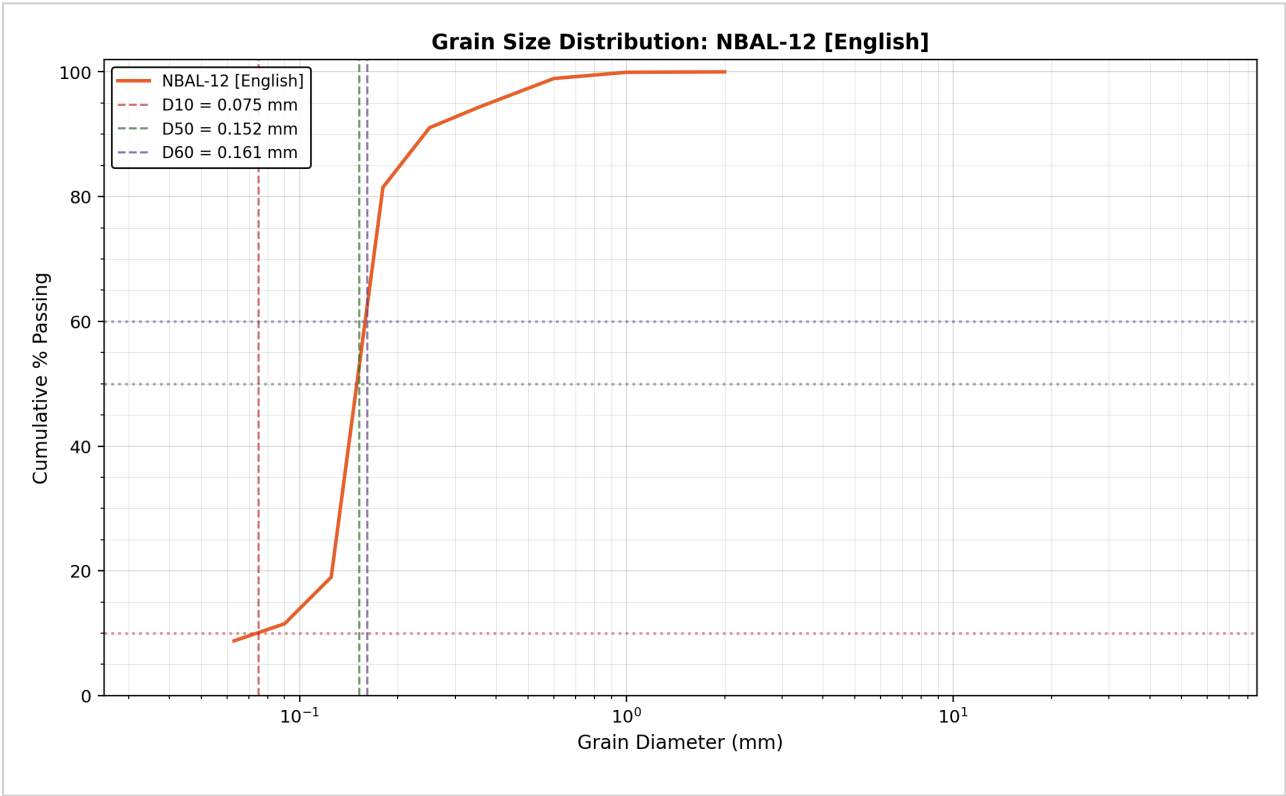


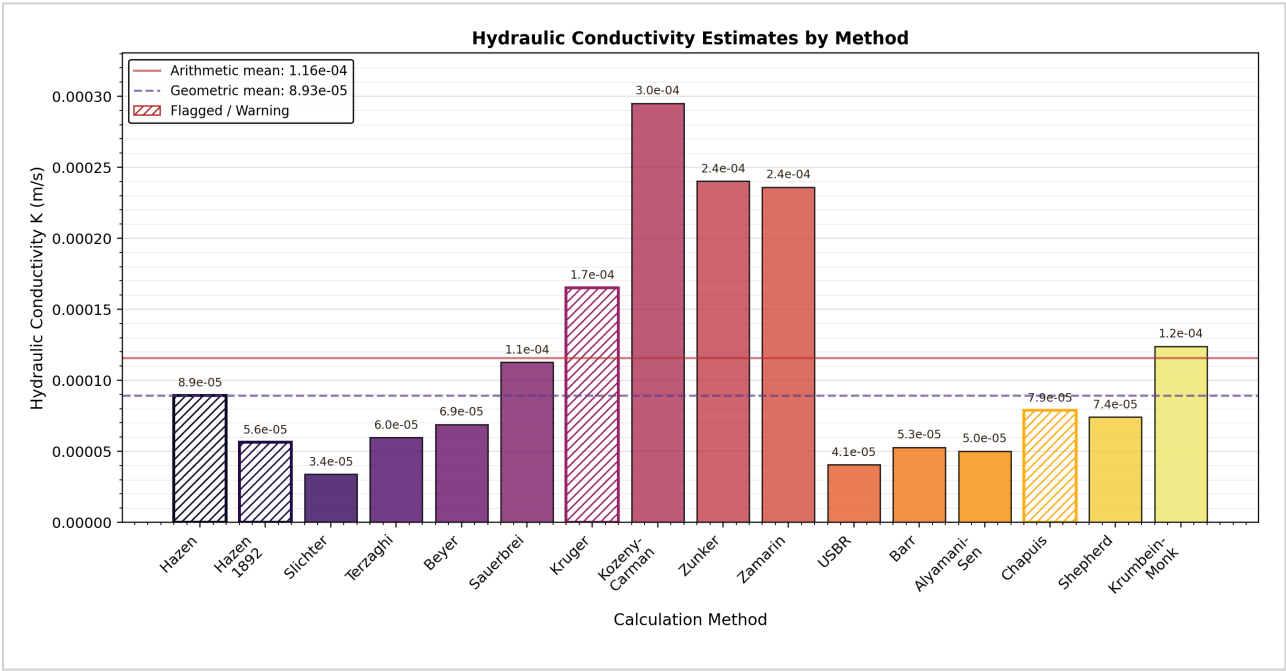
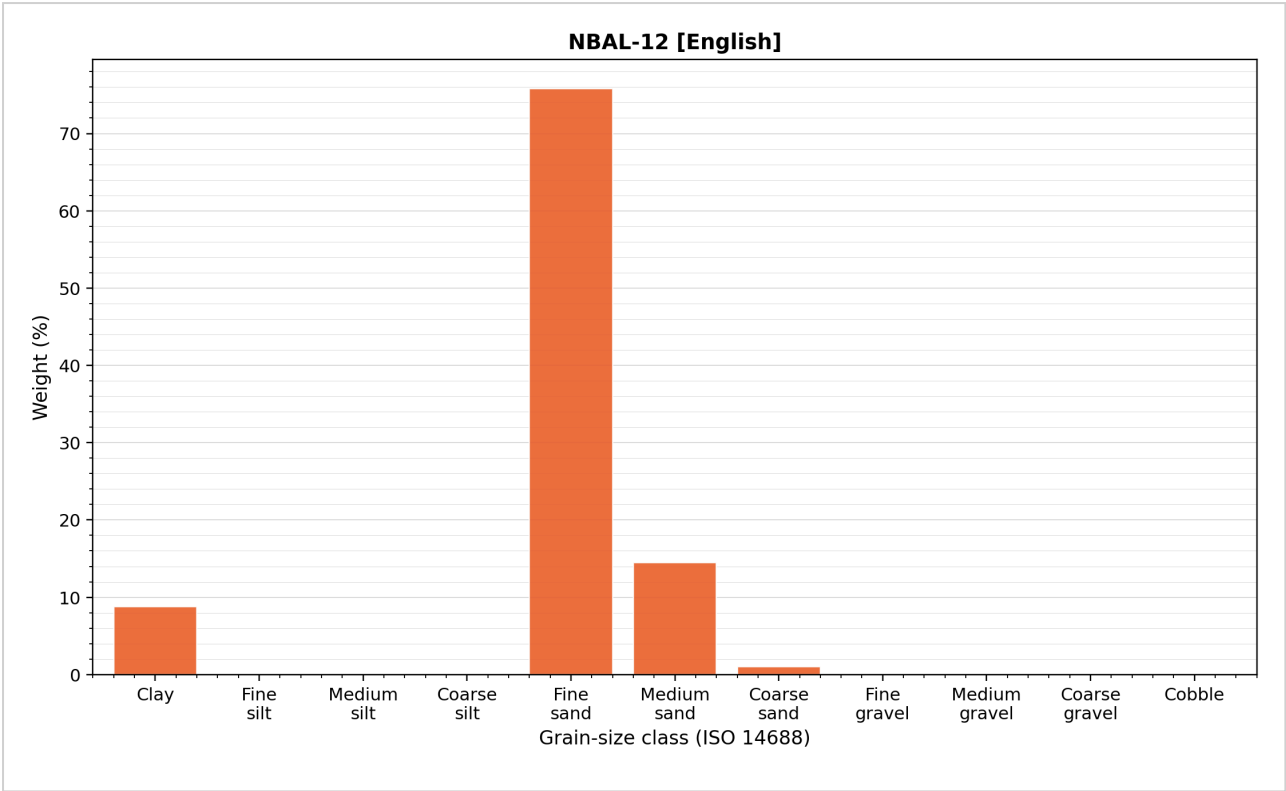
NBAL-11 [English]



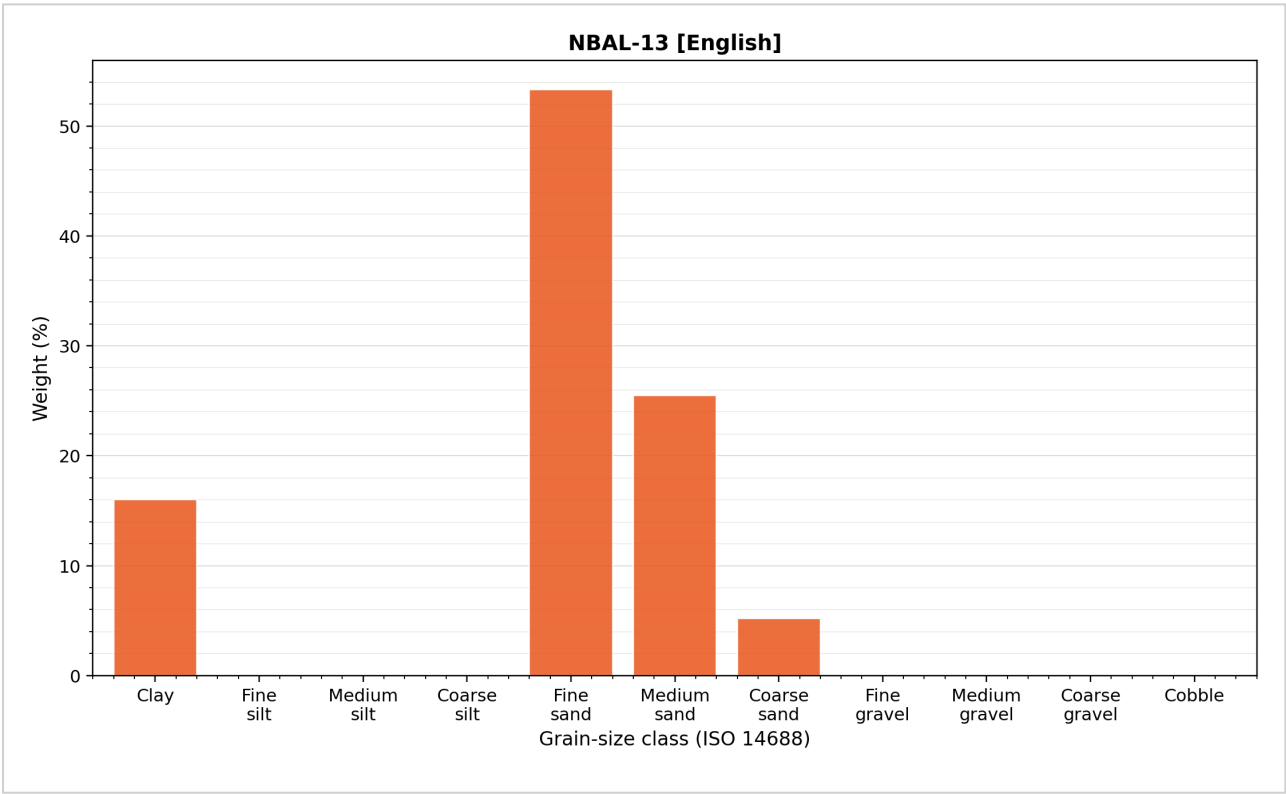
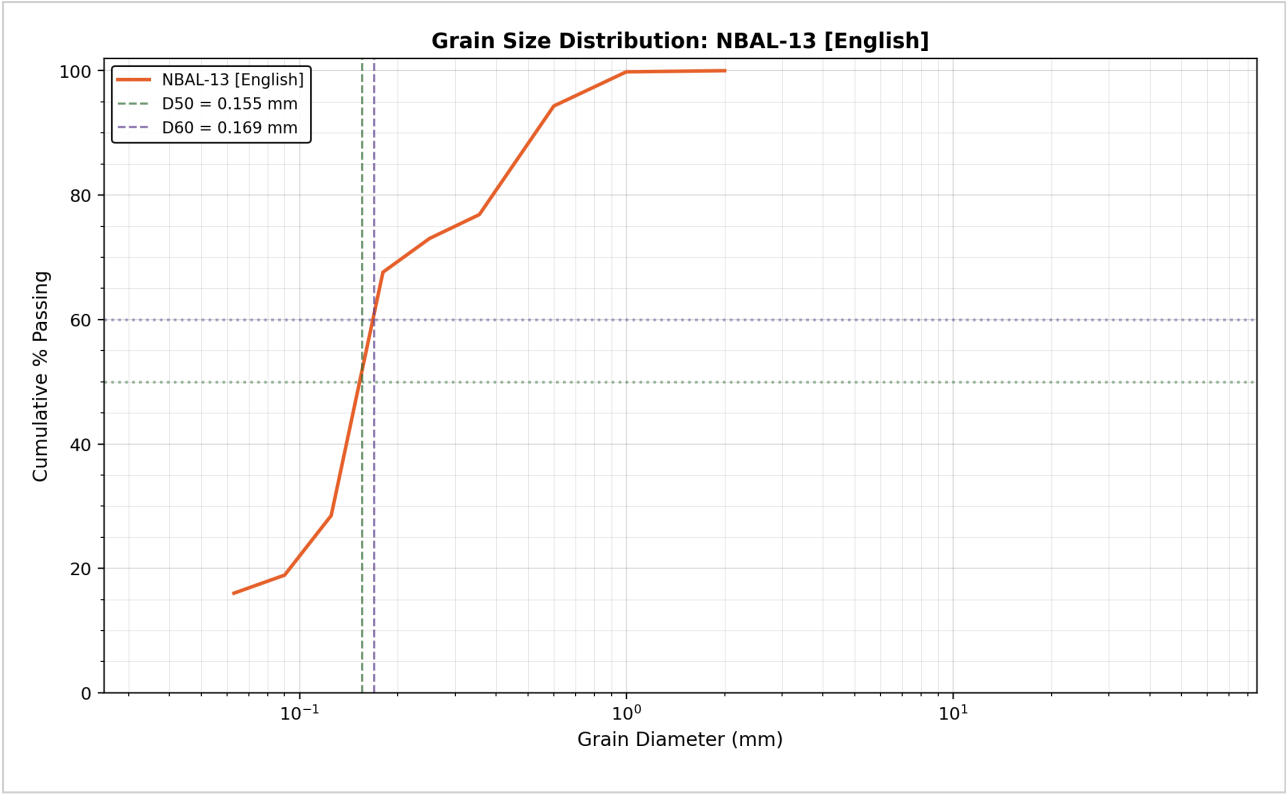


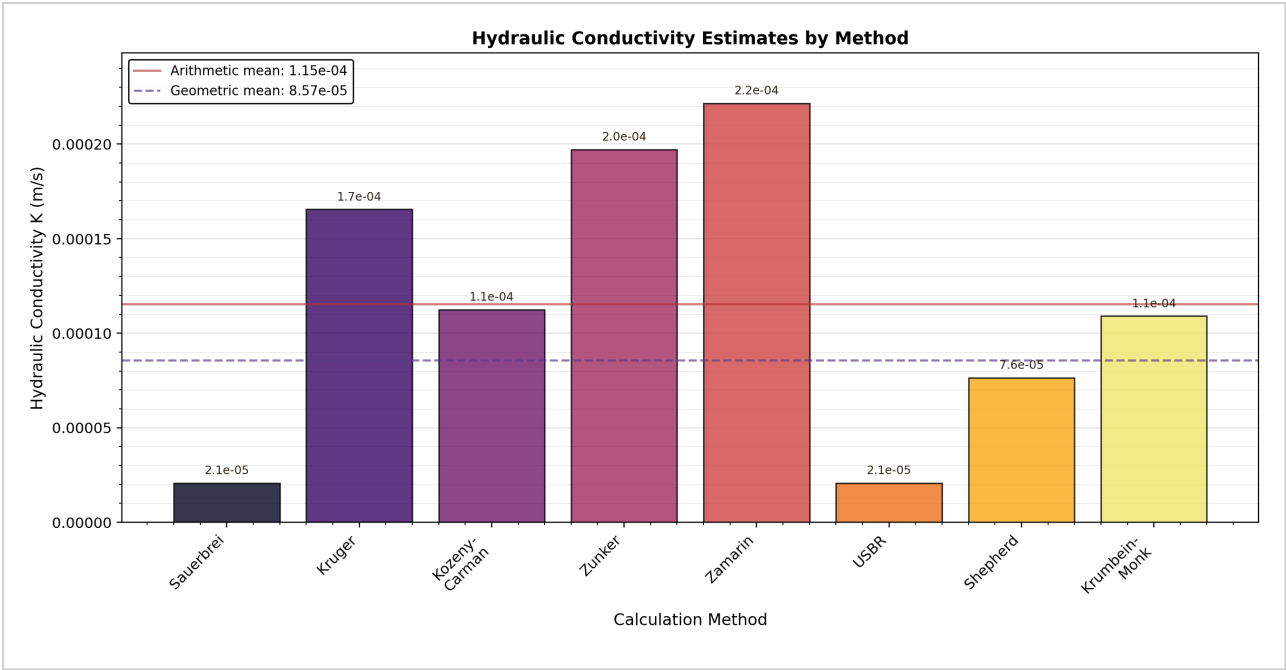
NBAL-12 [English]



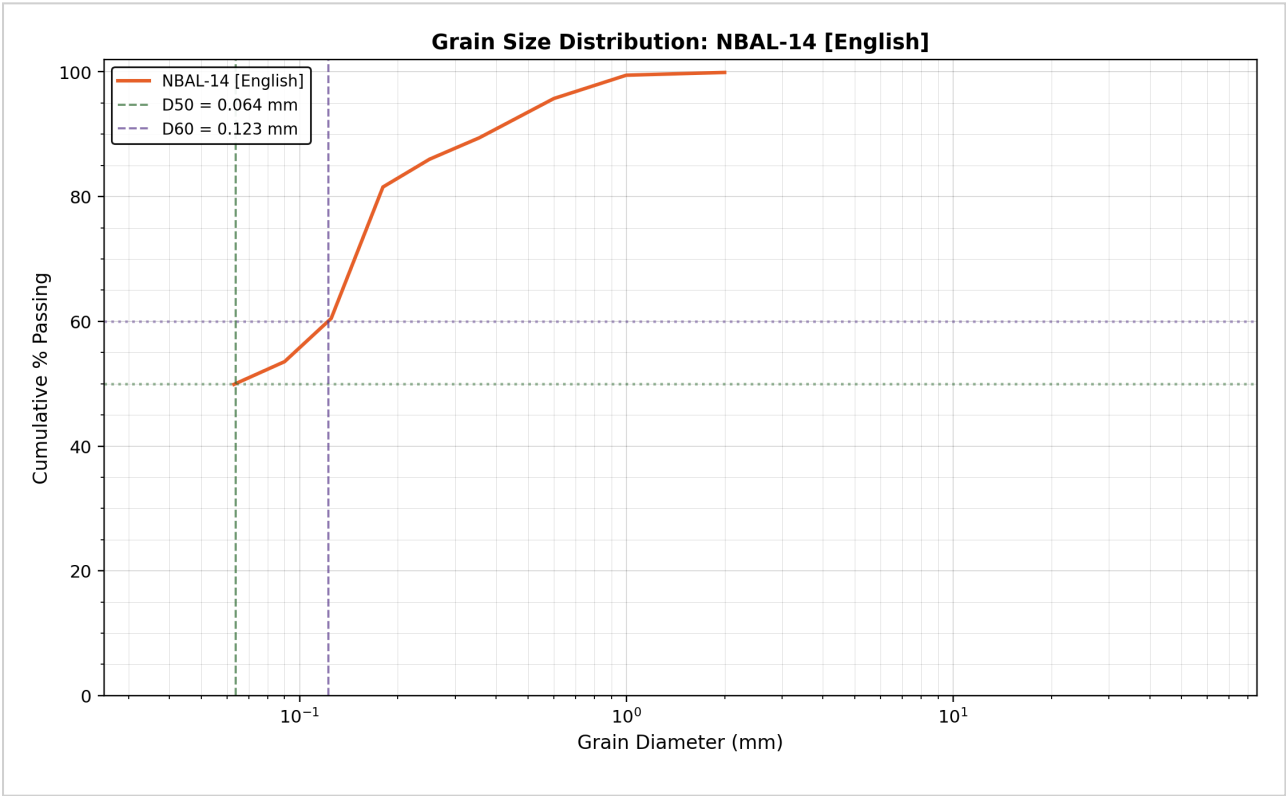


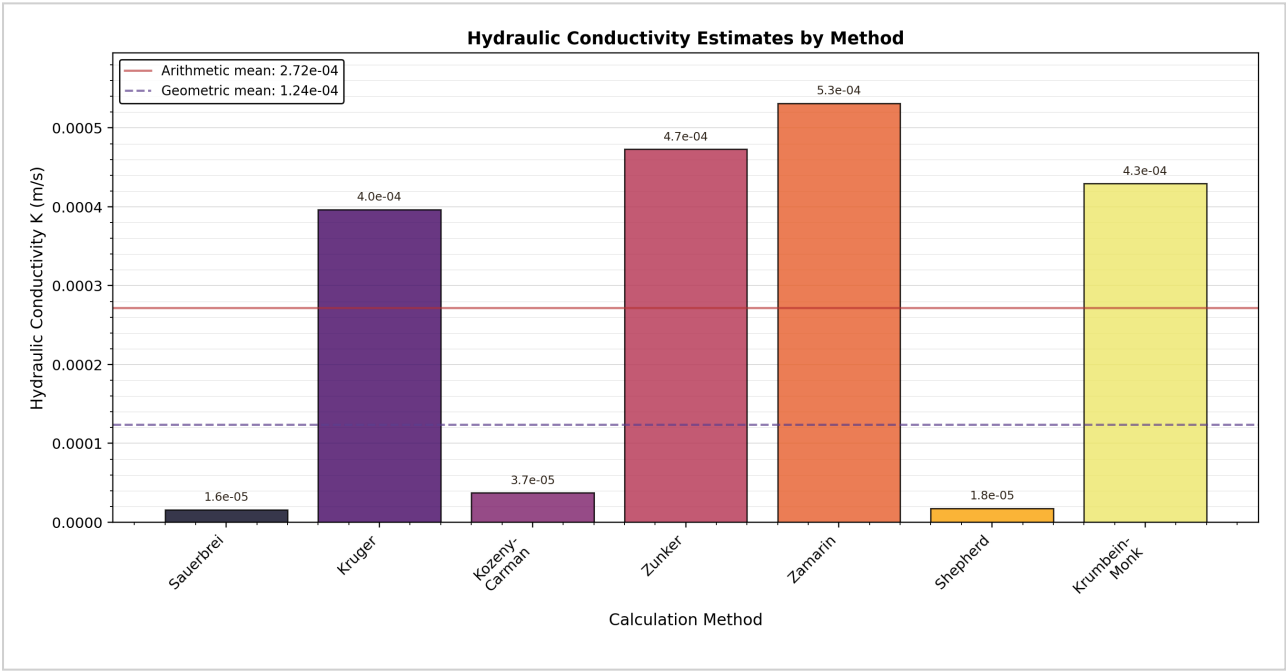
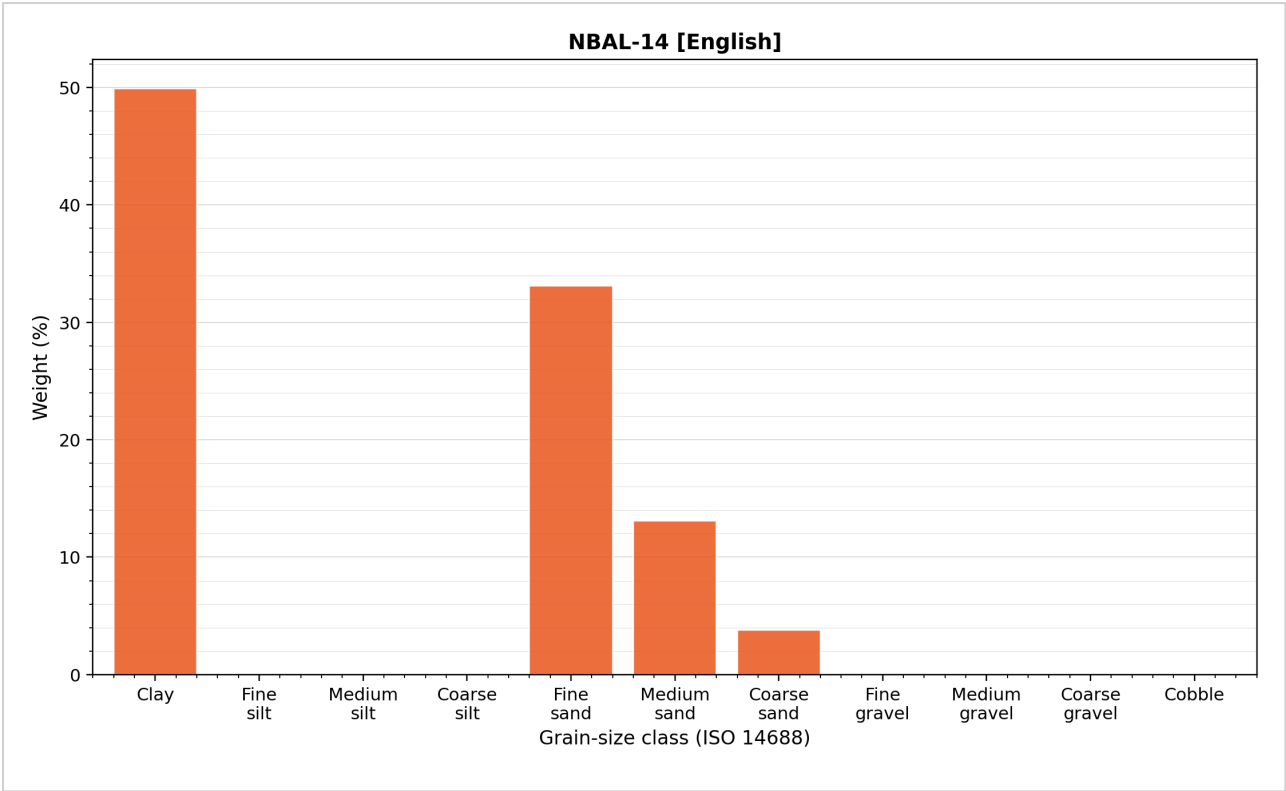
NBAL-13 [English]



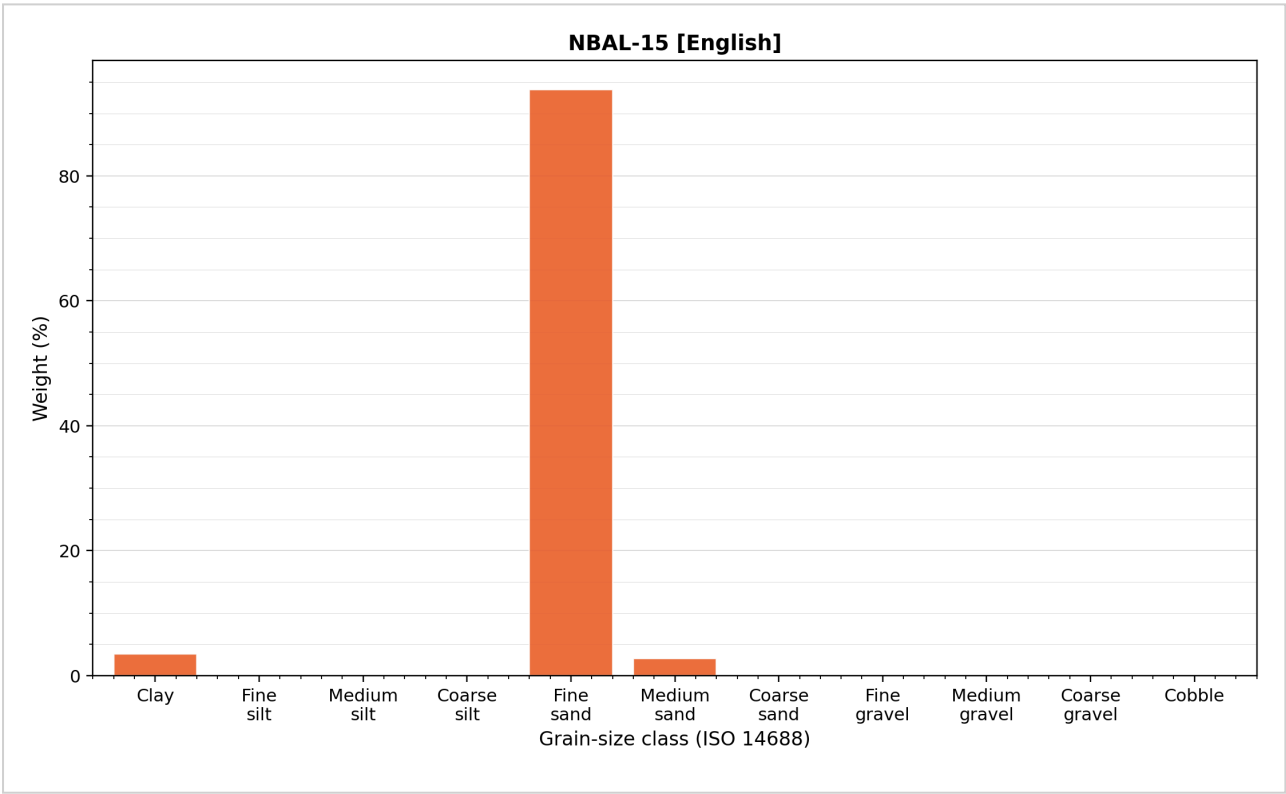
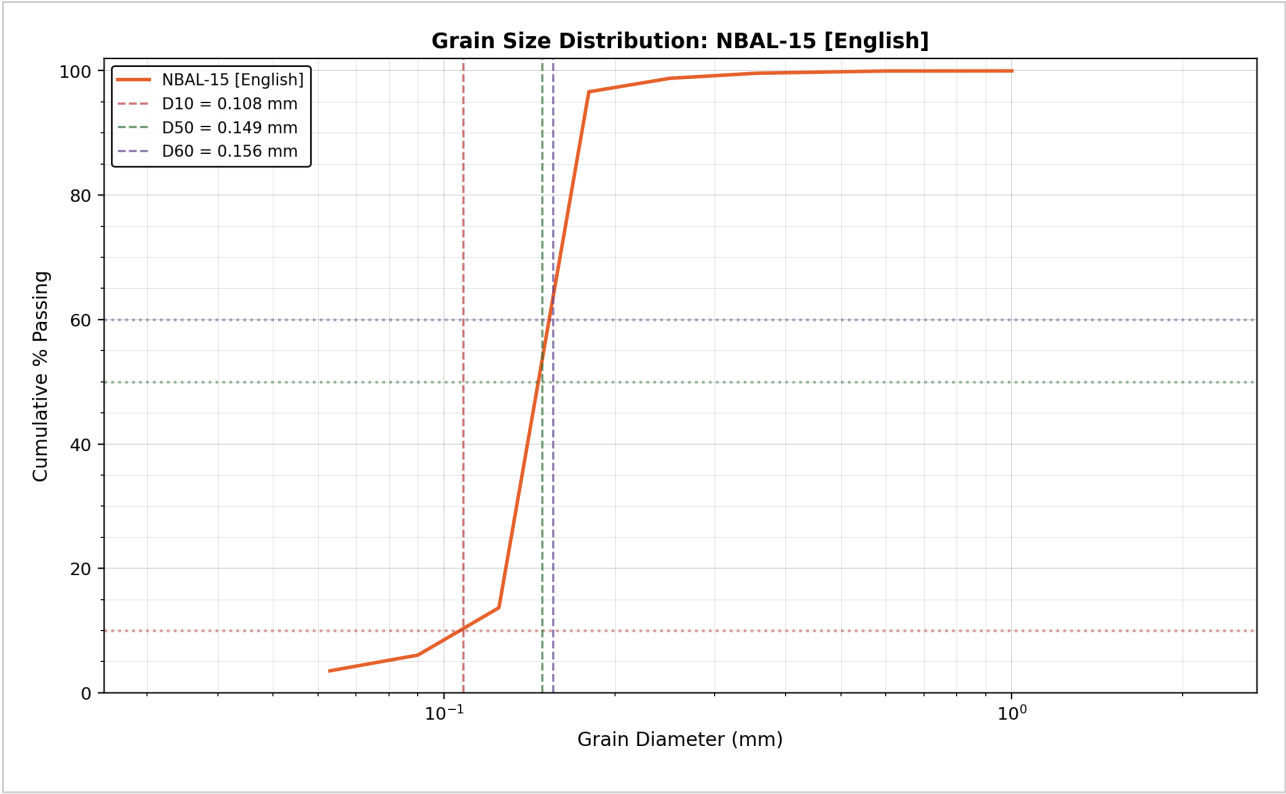


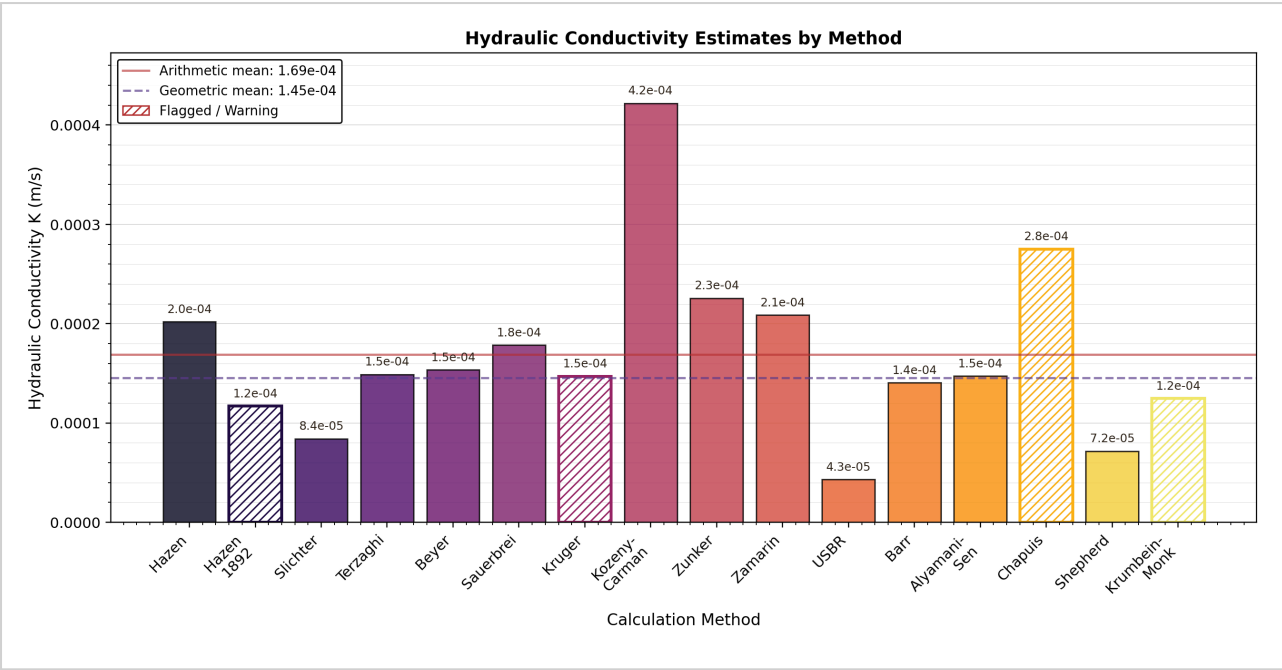
NBAL-14 [English]



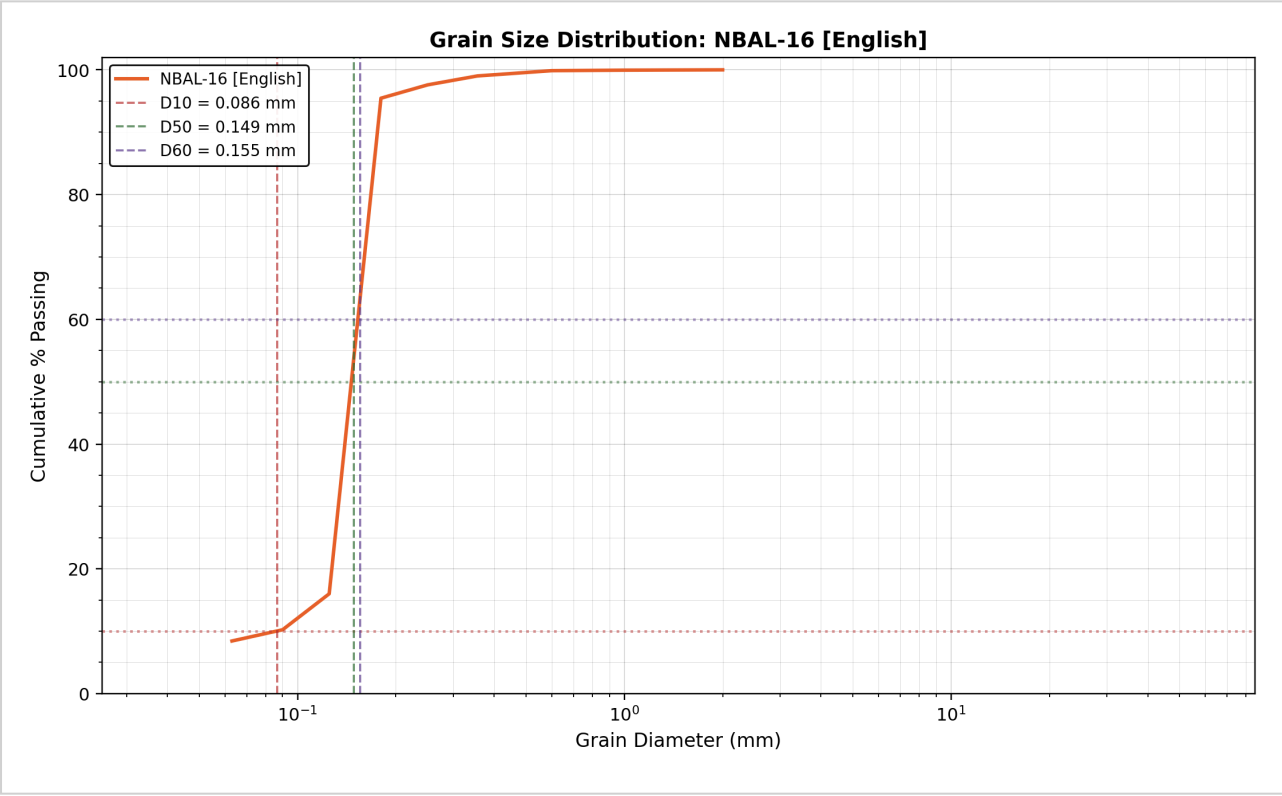


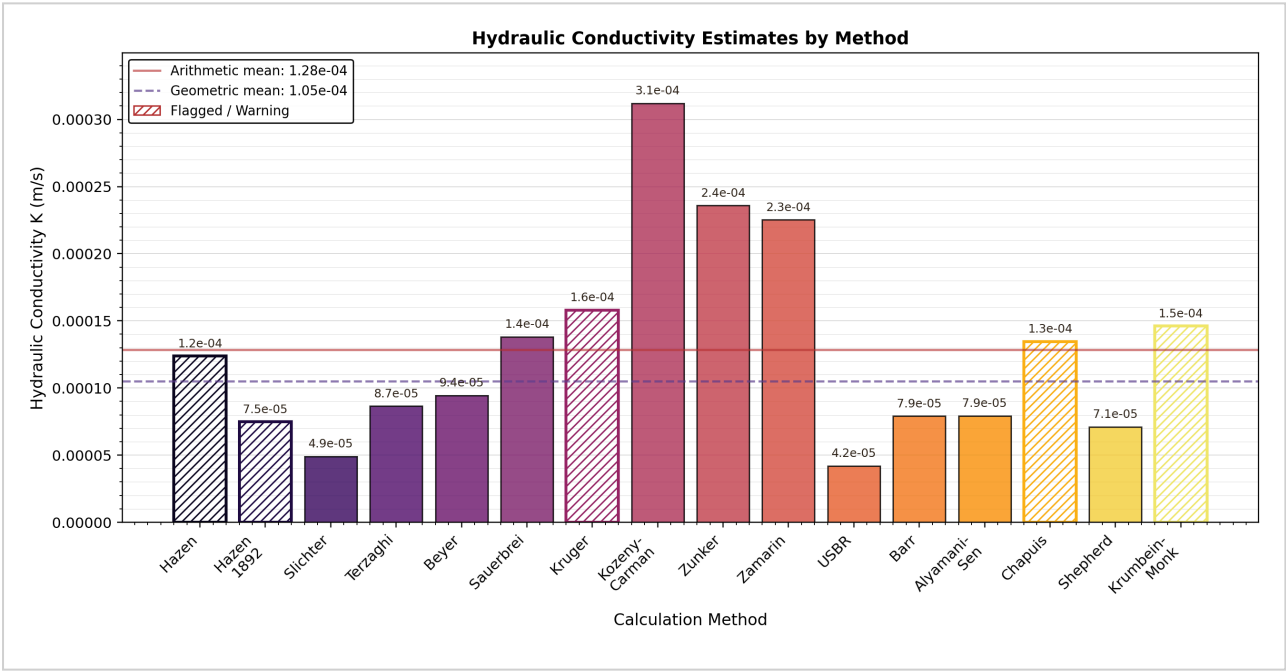
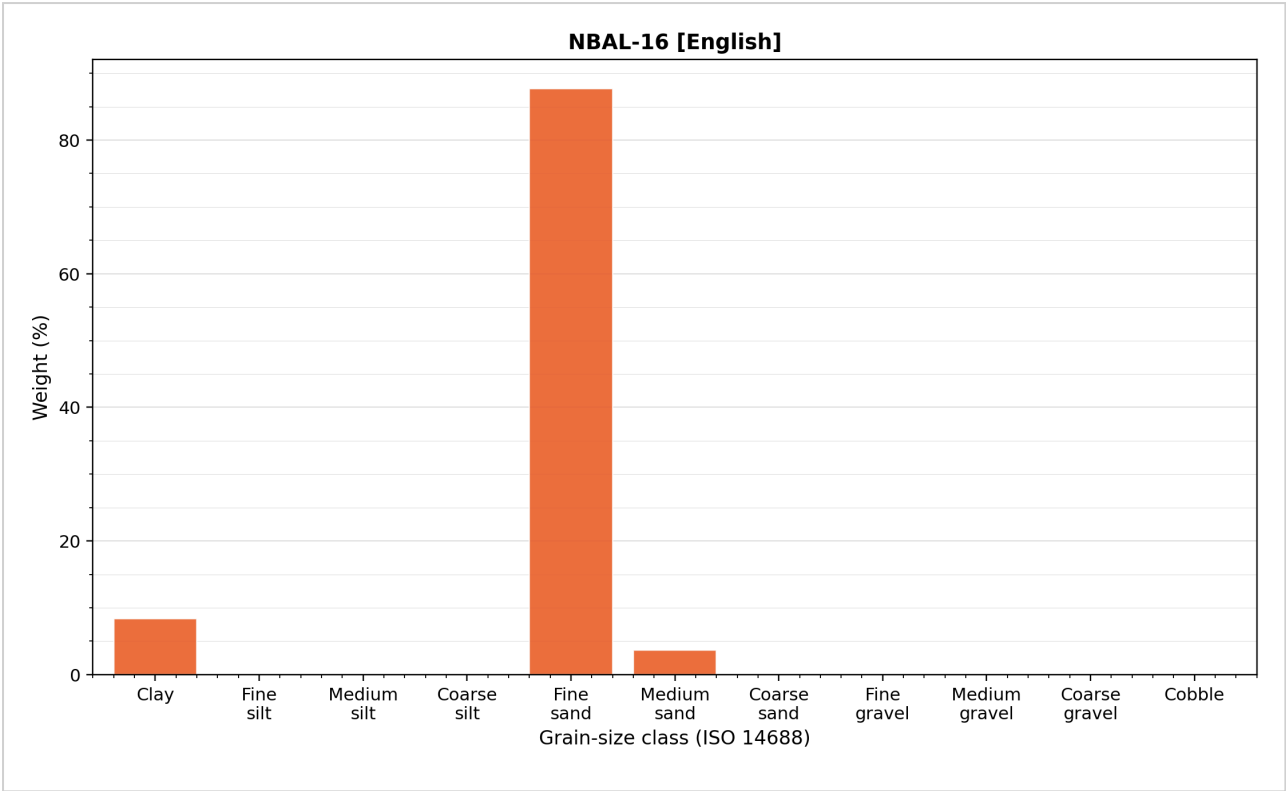
NBAL-15 [English]



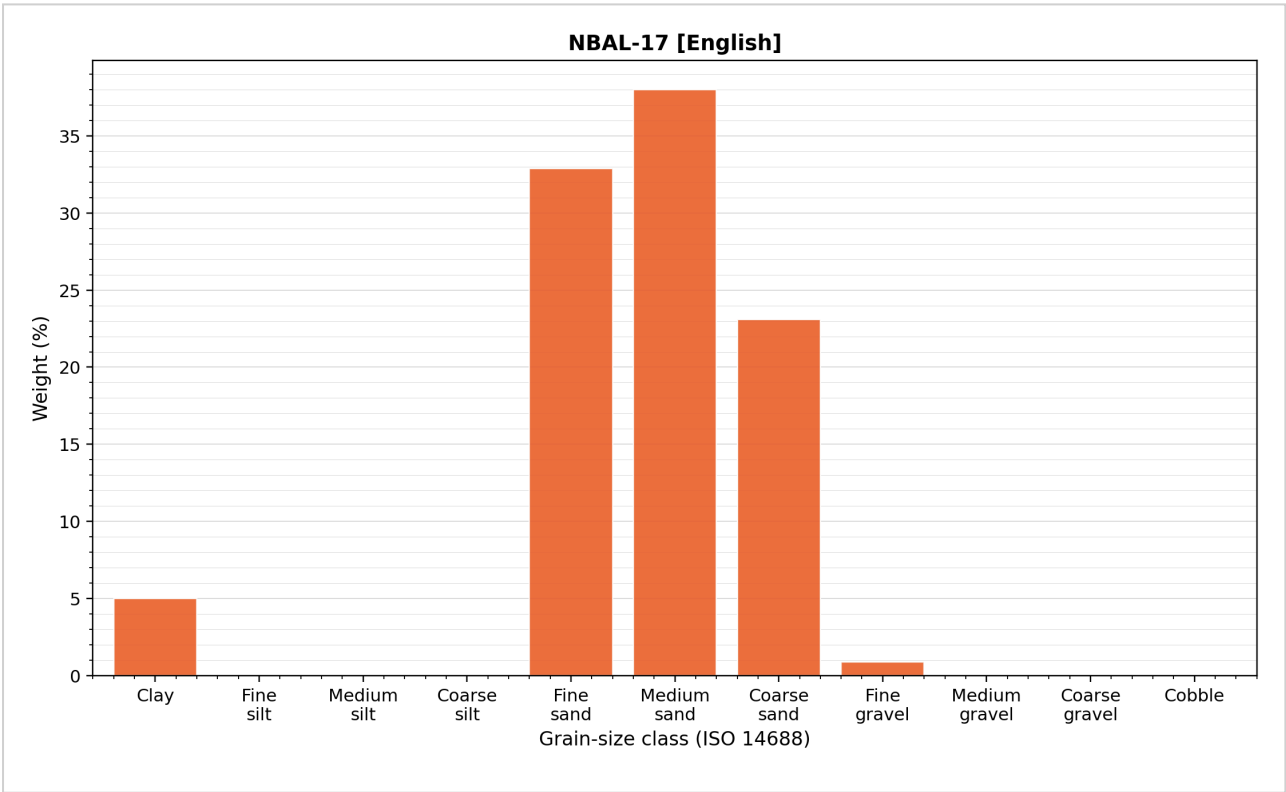
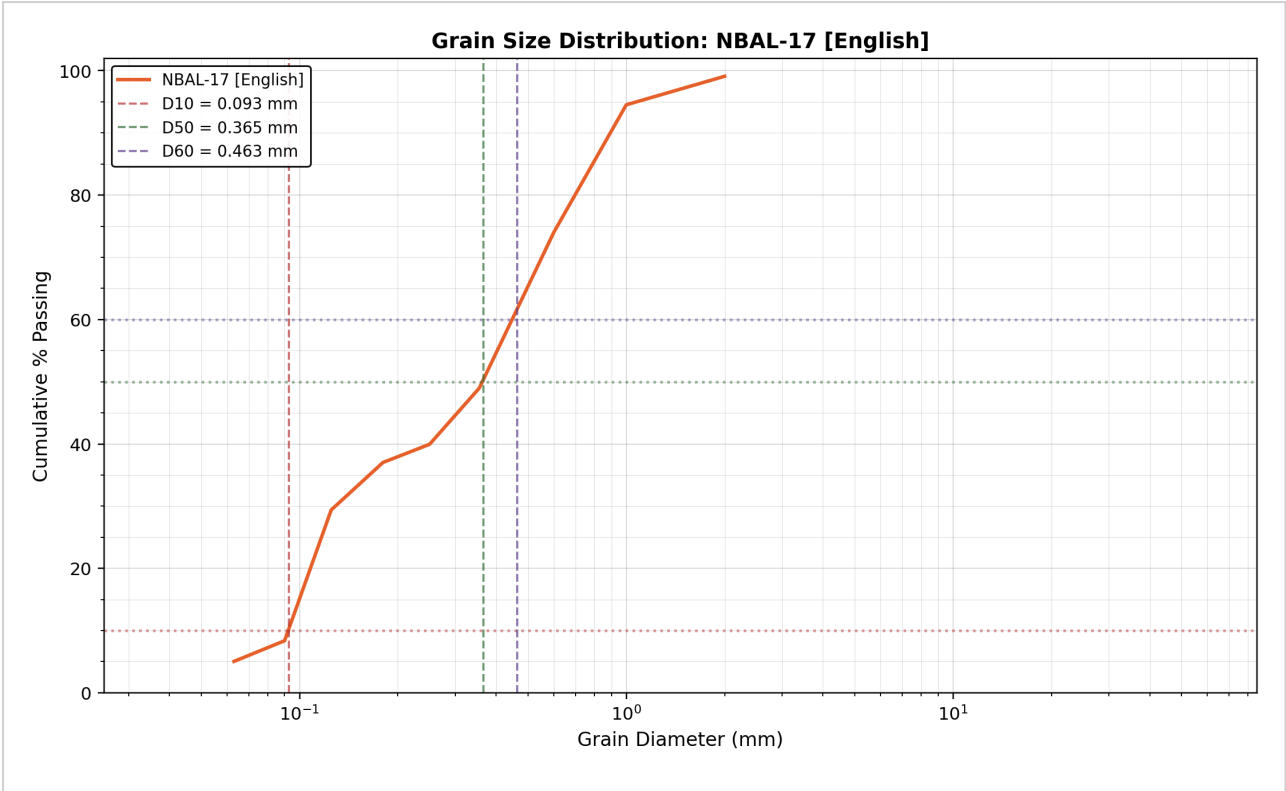


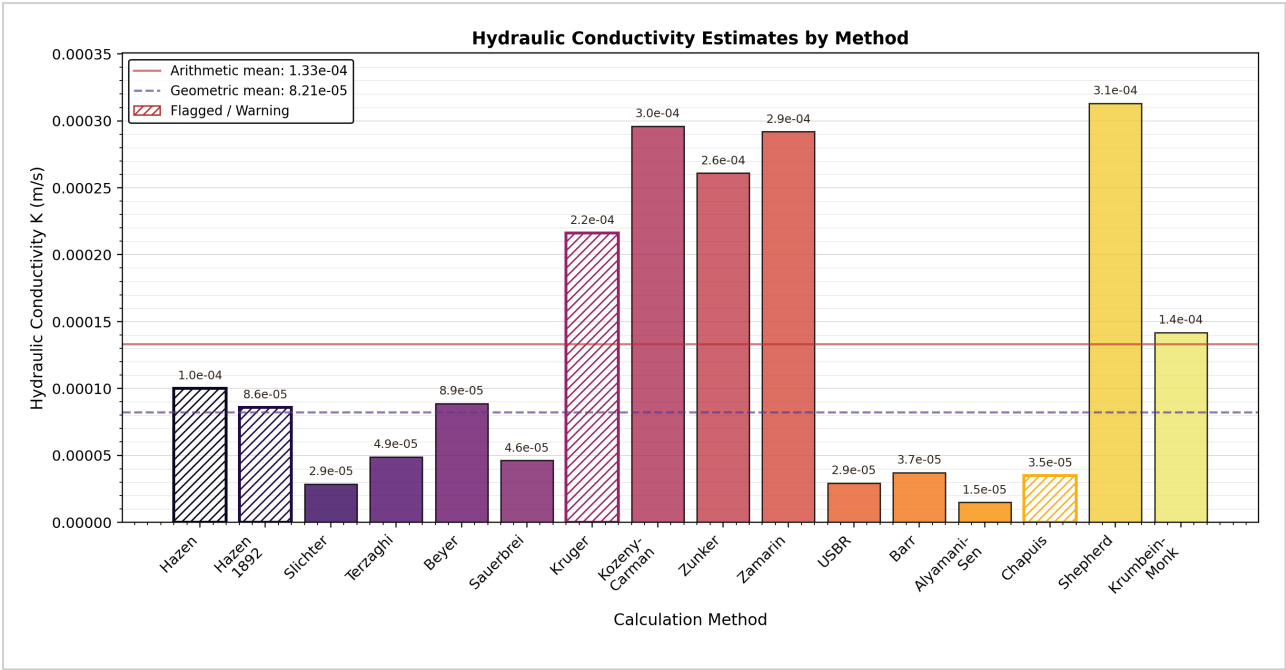
NBAL-16 [English]



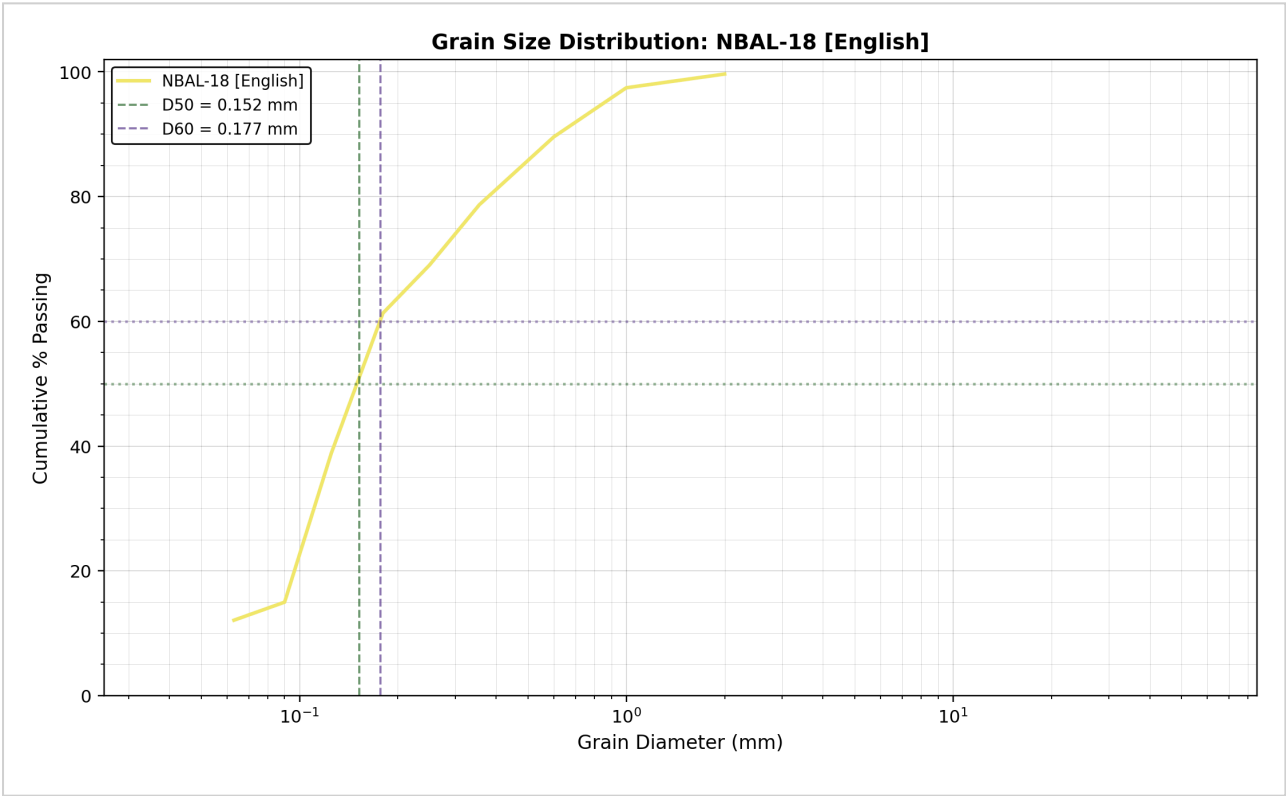


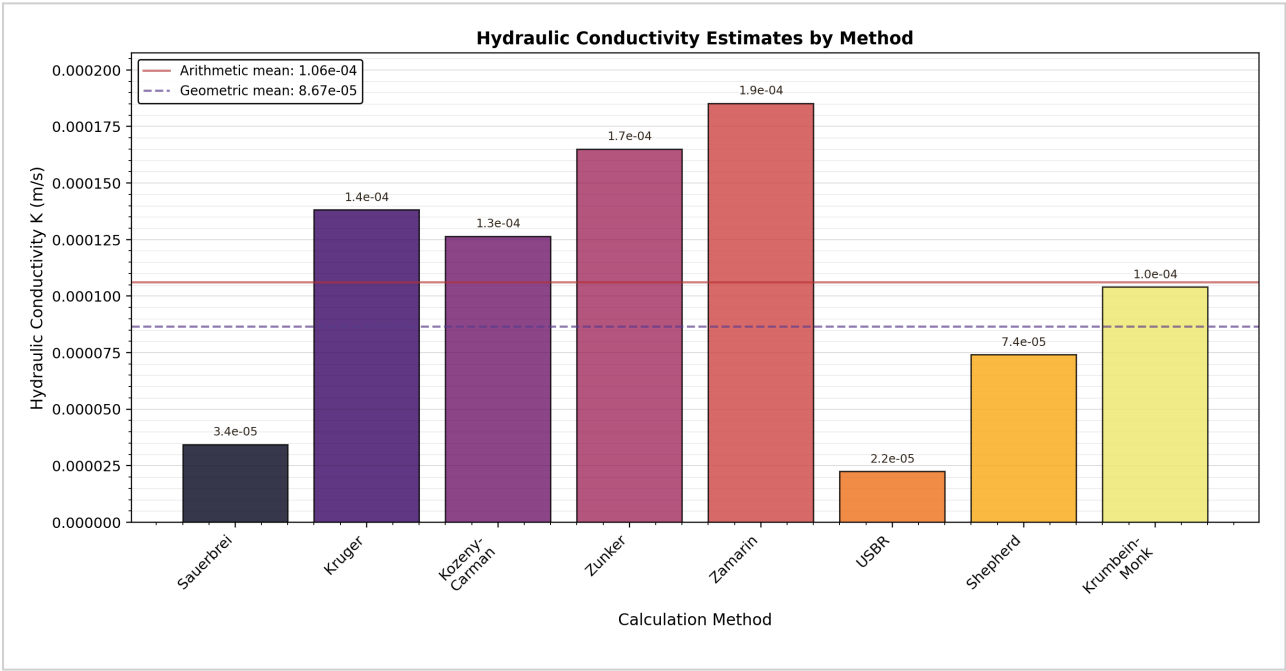
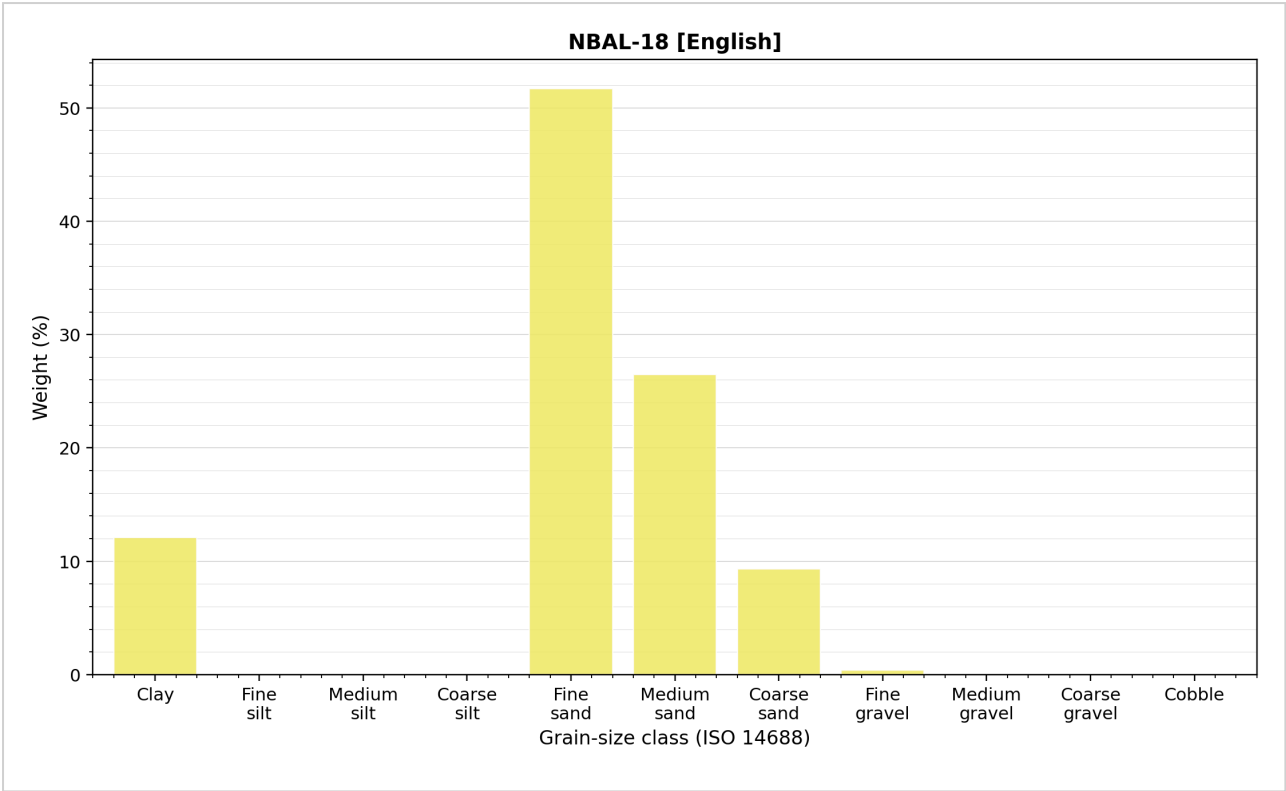
NBAL-17 [English]



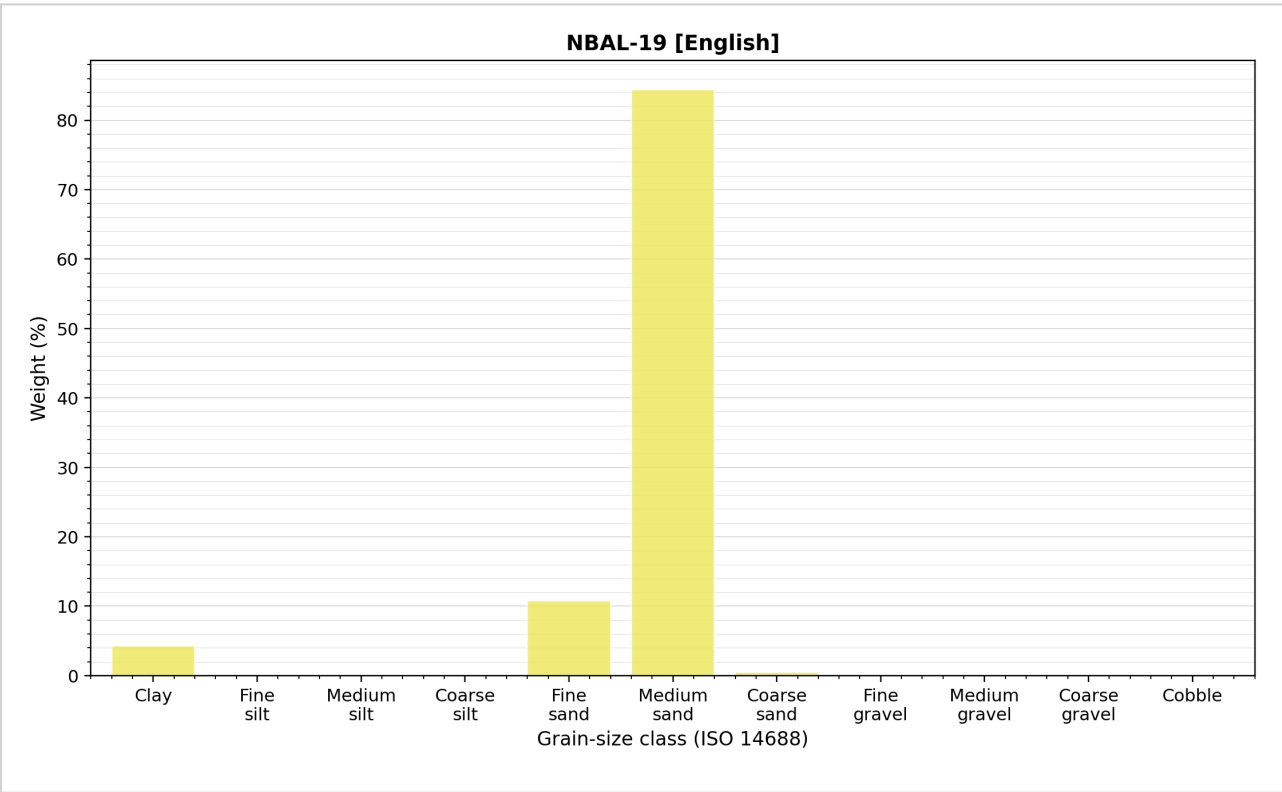
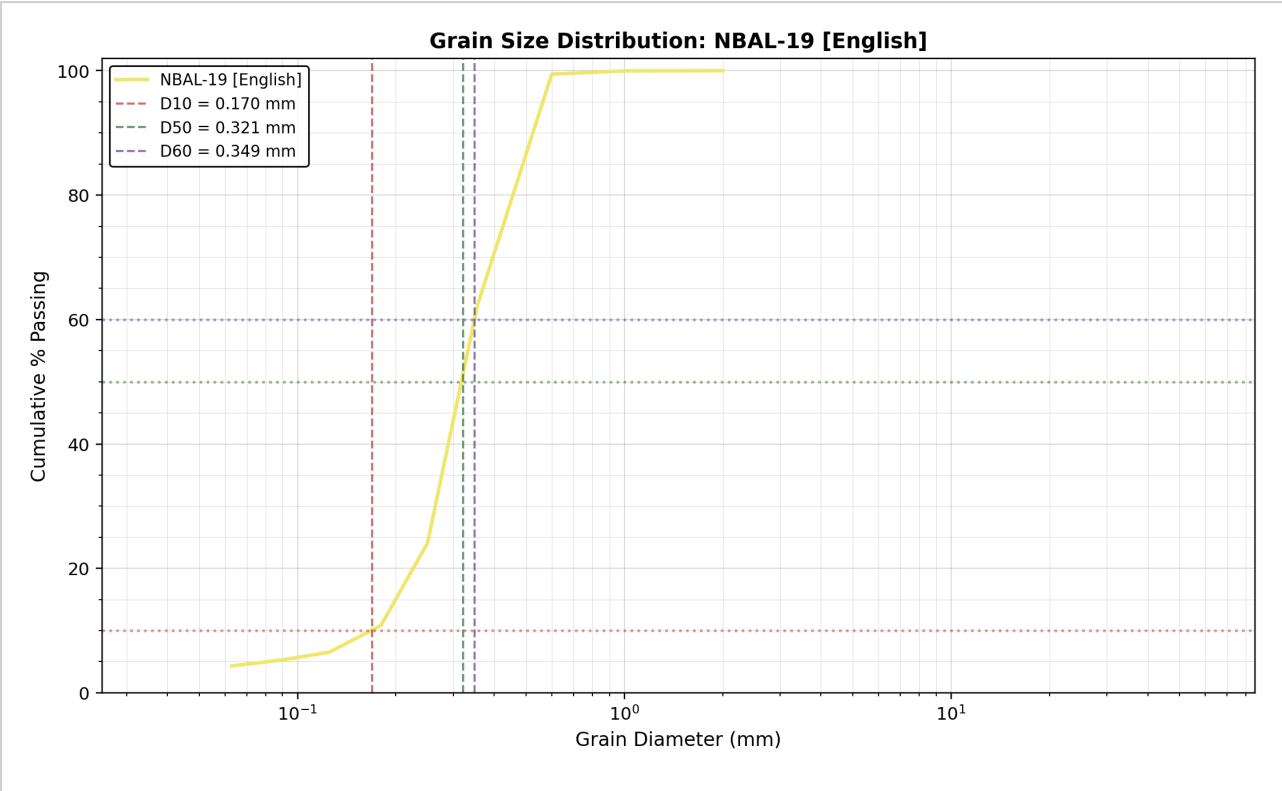


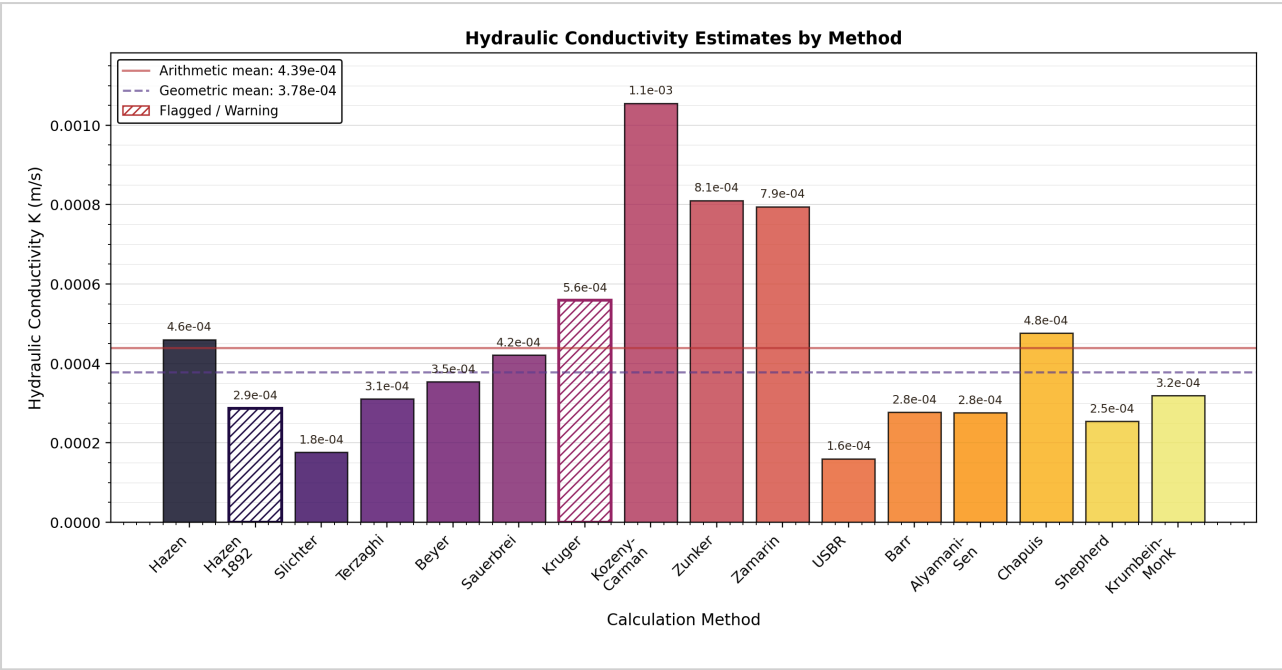
NBAL-18 [English]



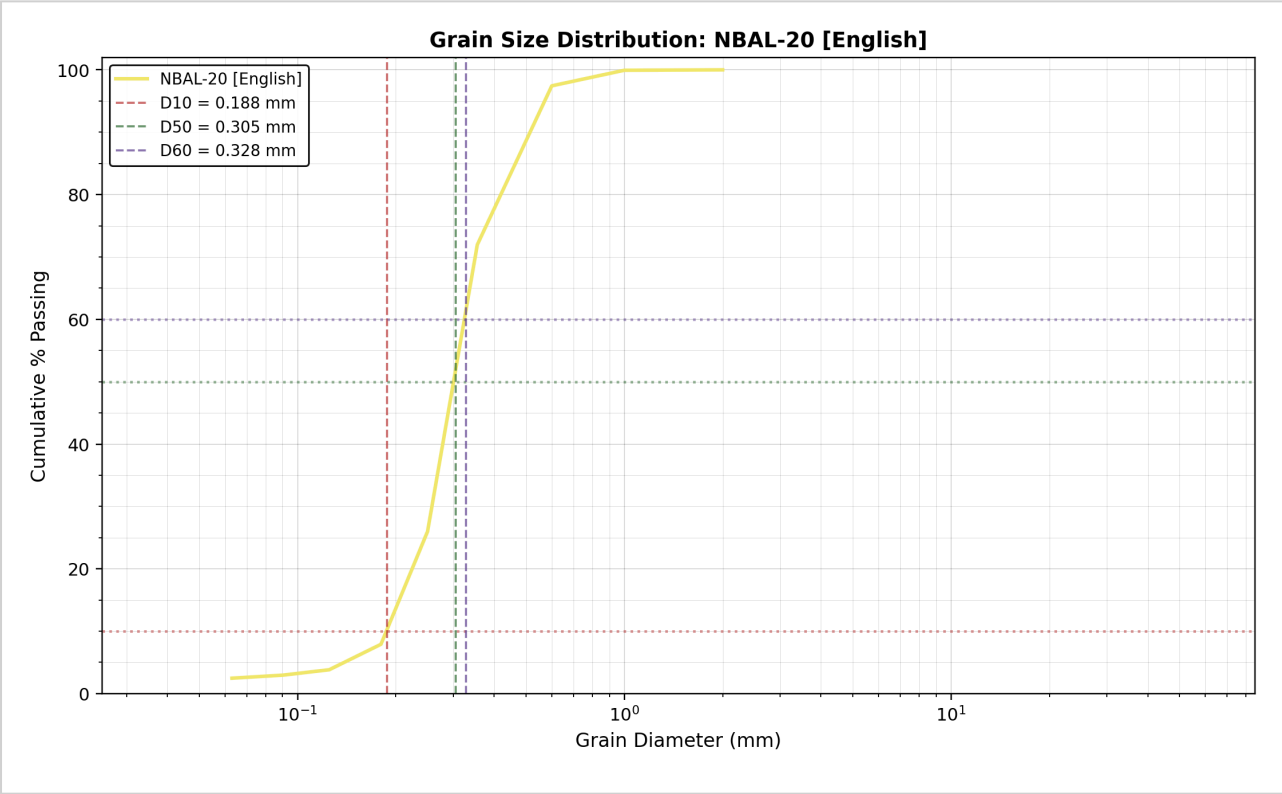


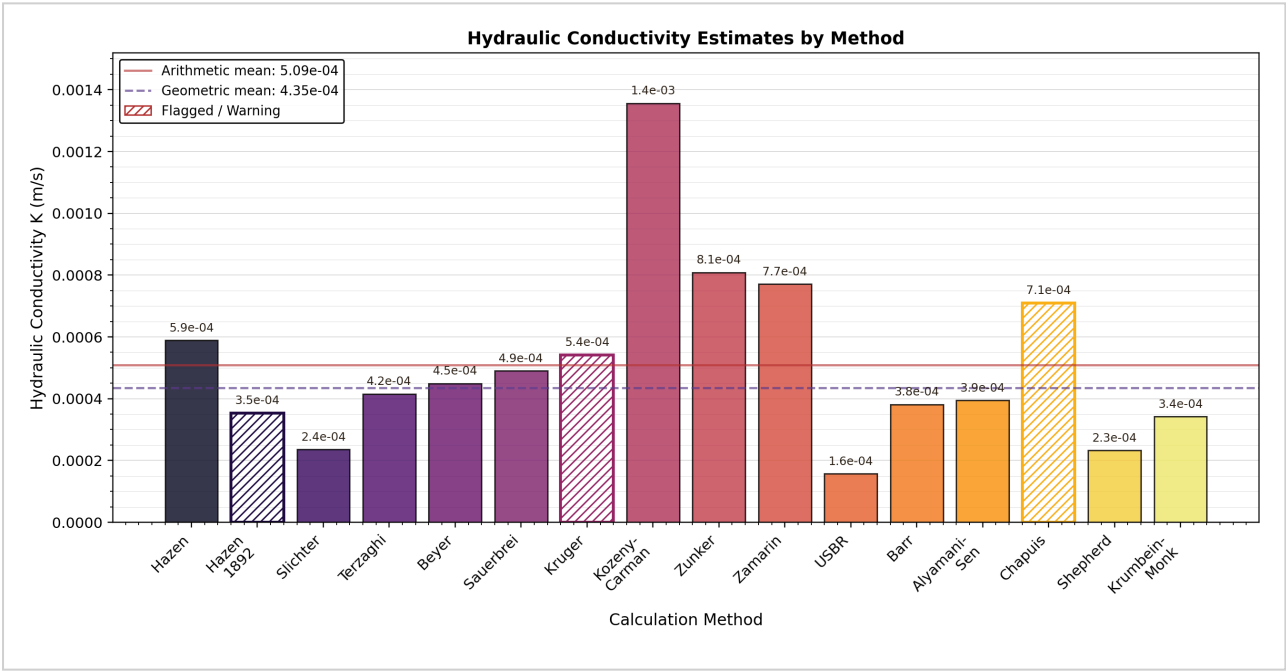
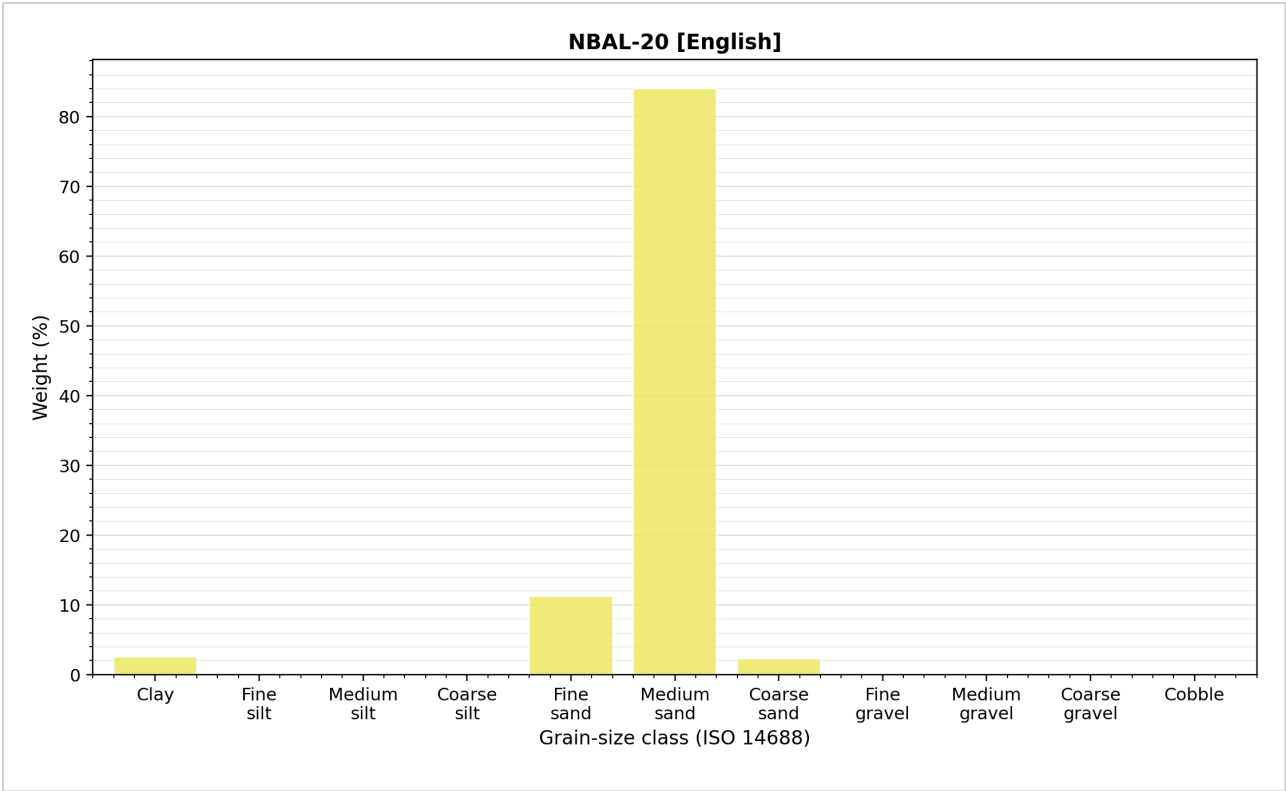
NBAL-19 [English]



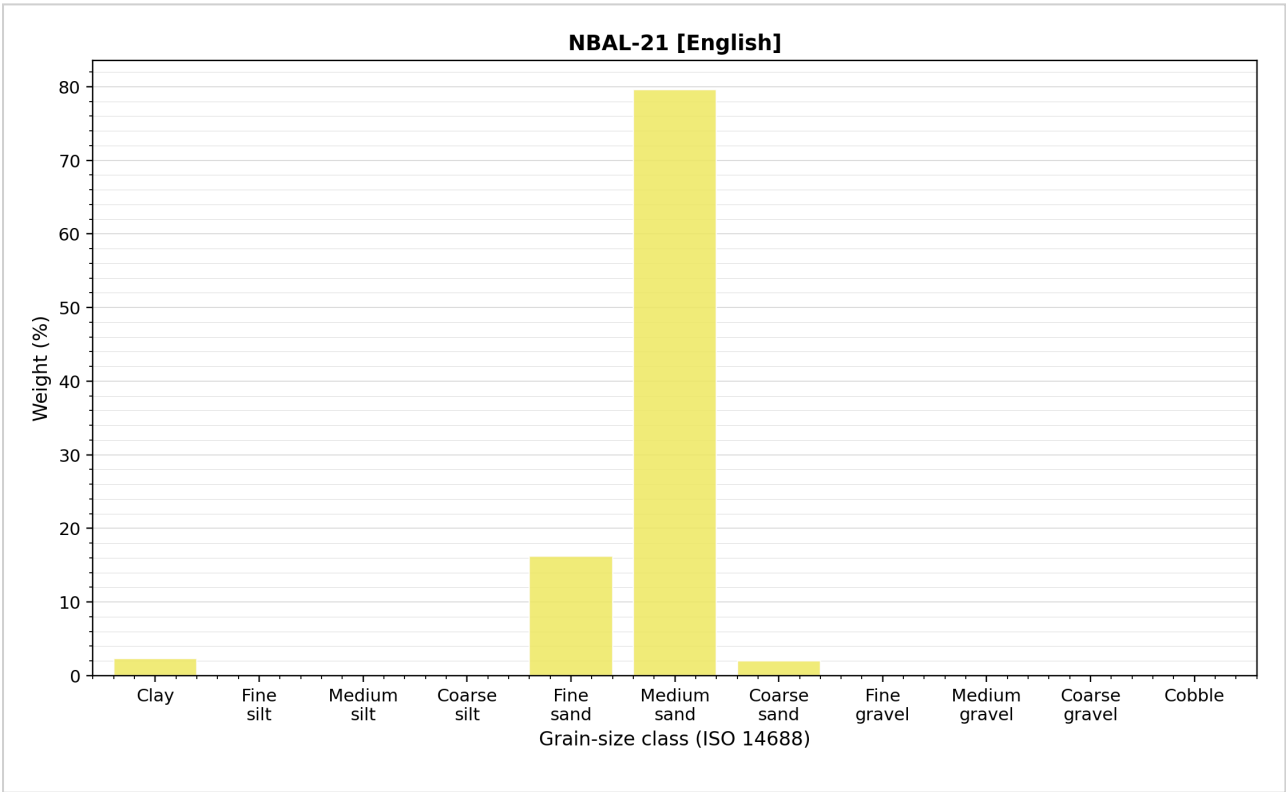
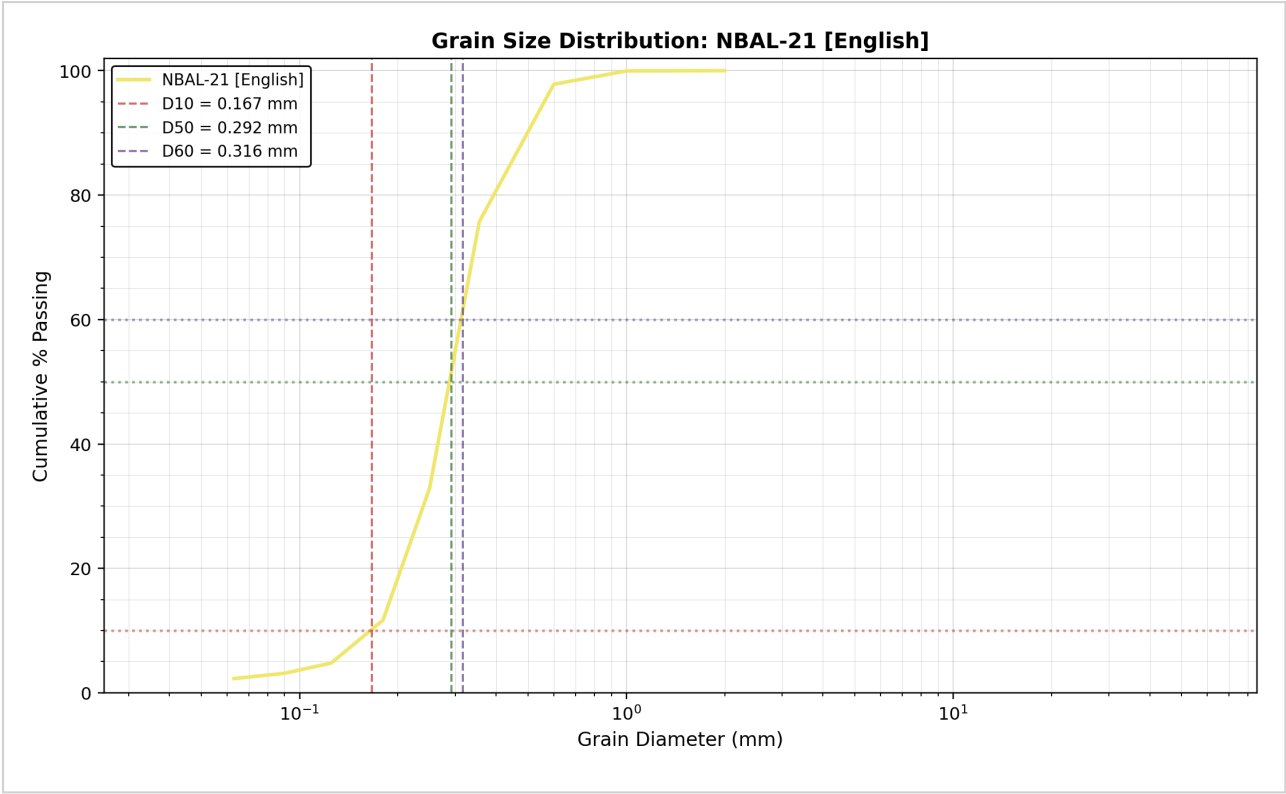


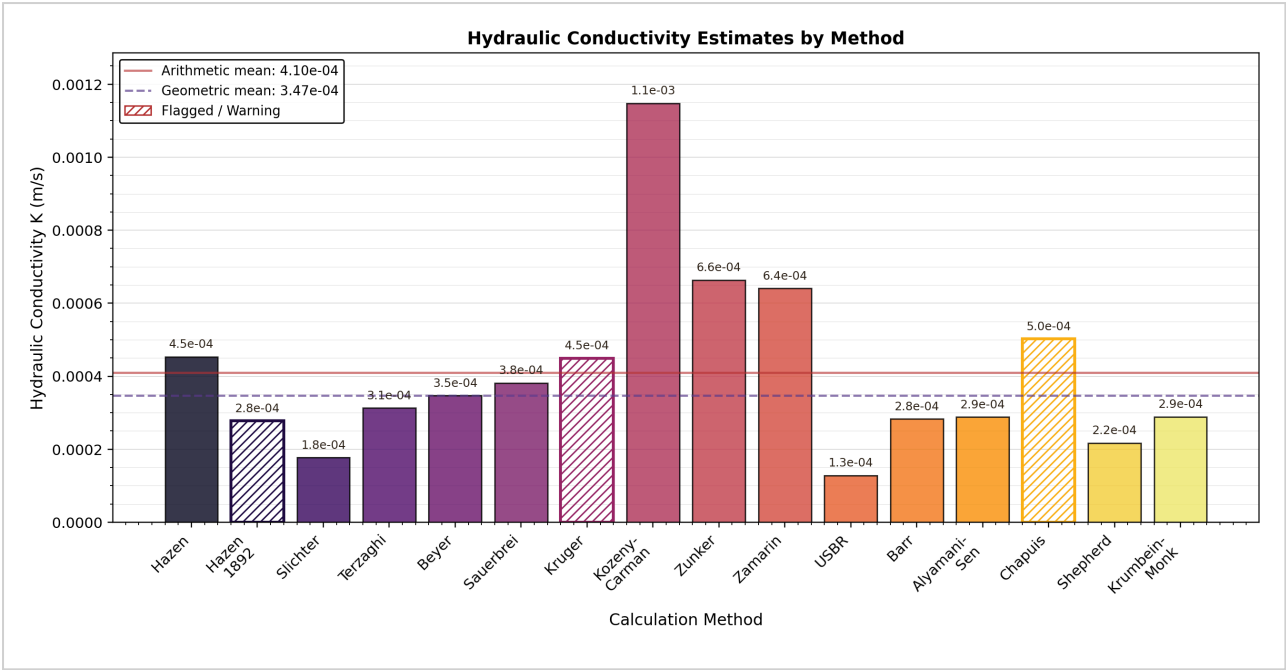
NBAL-20 [English]



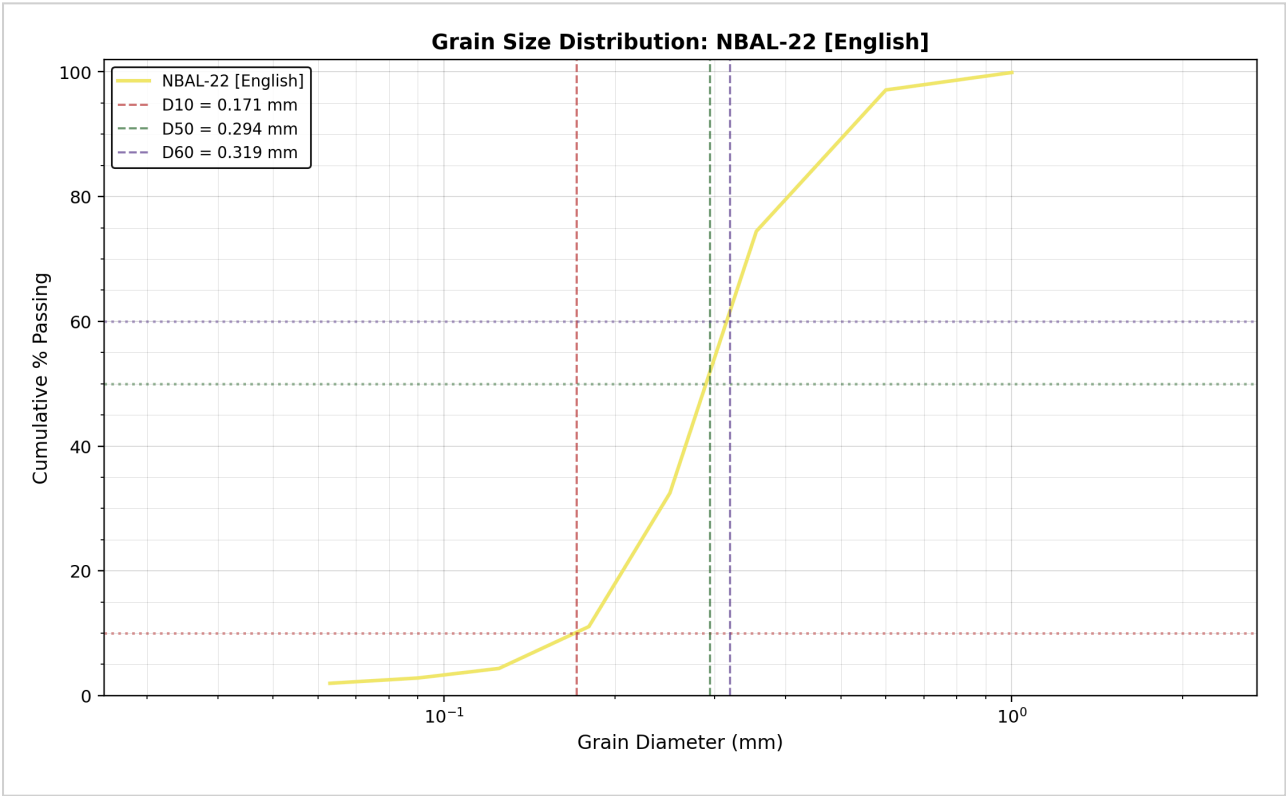


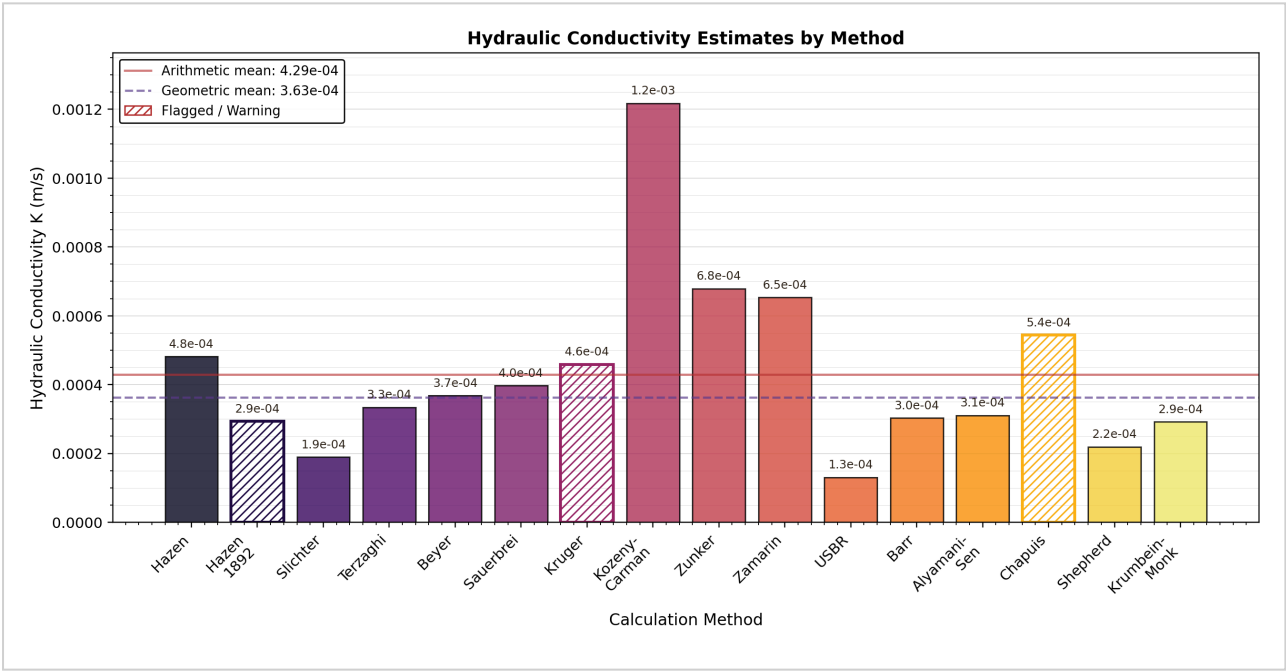
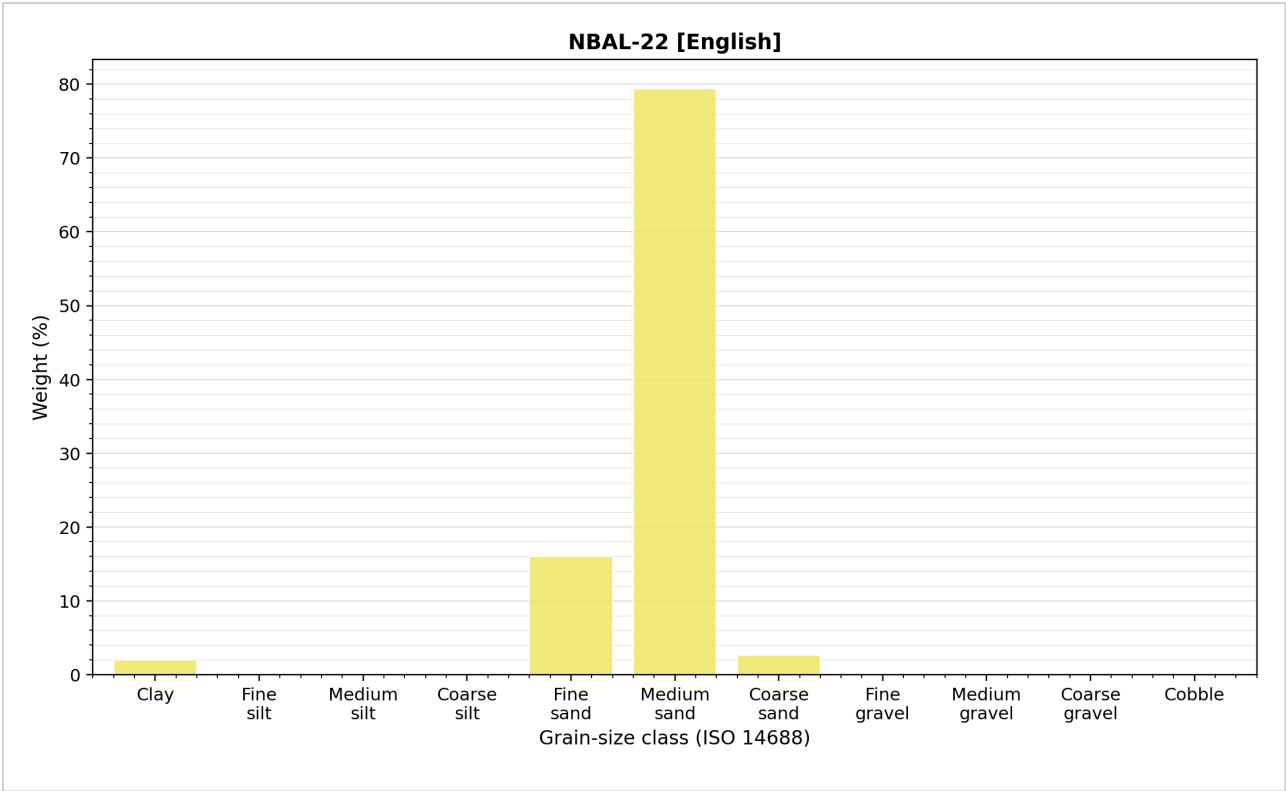
NBAL-21 [English]



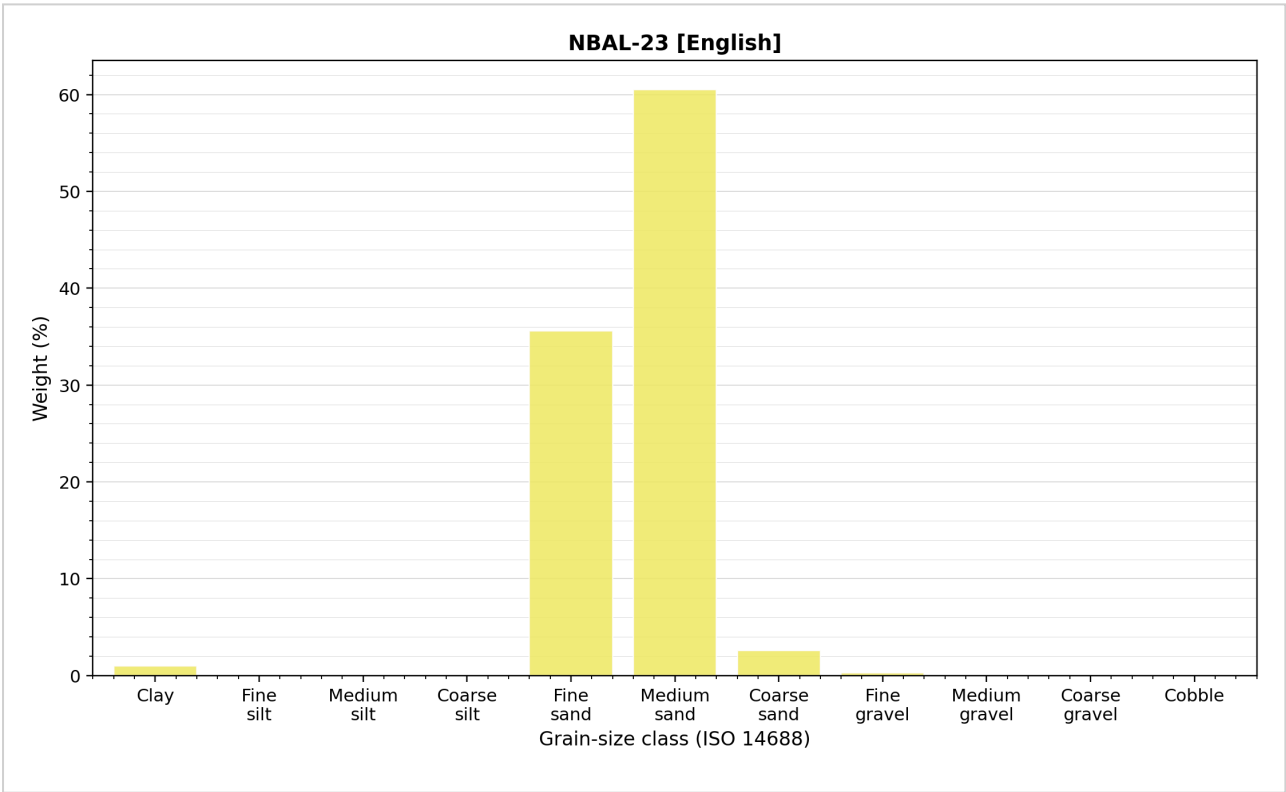
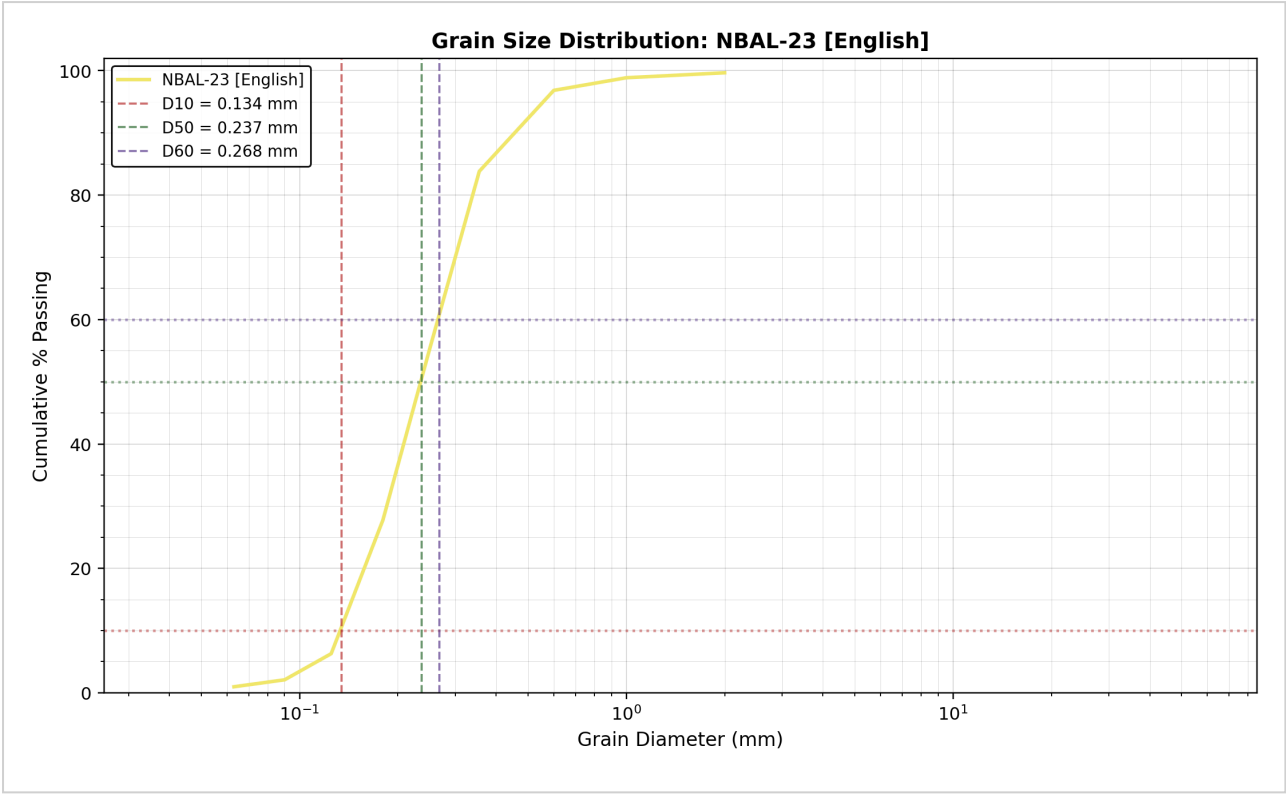


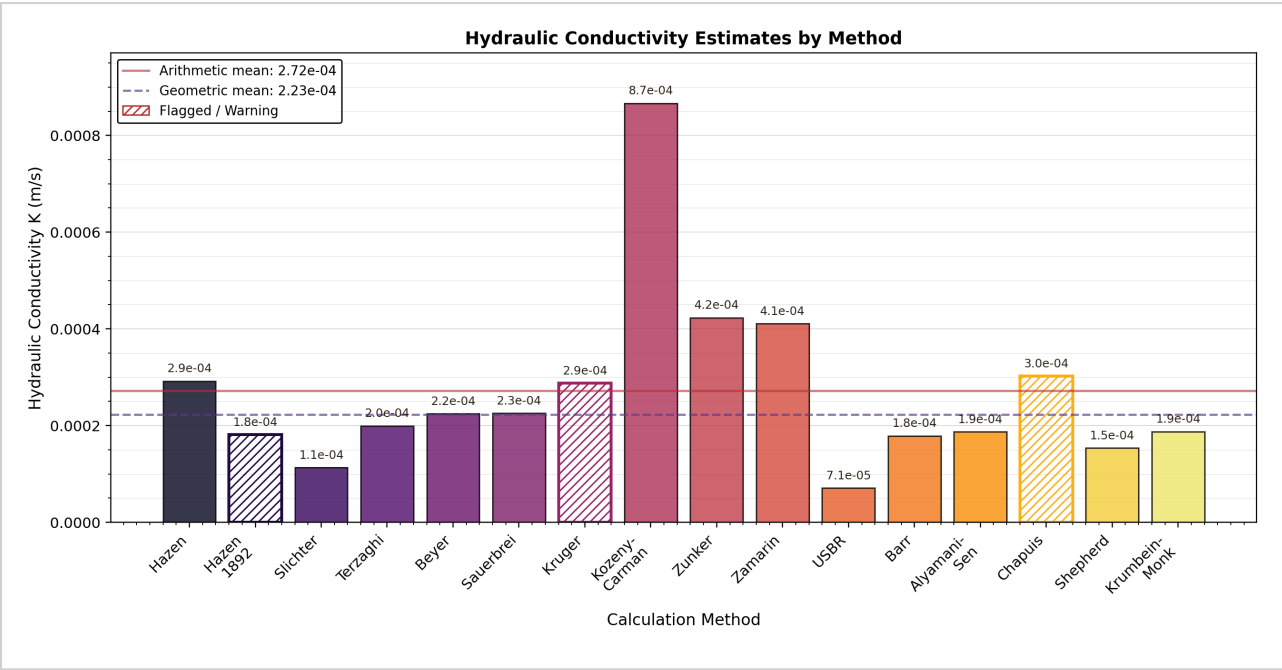
NBAL-22 [English]



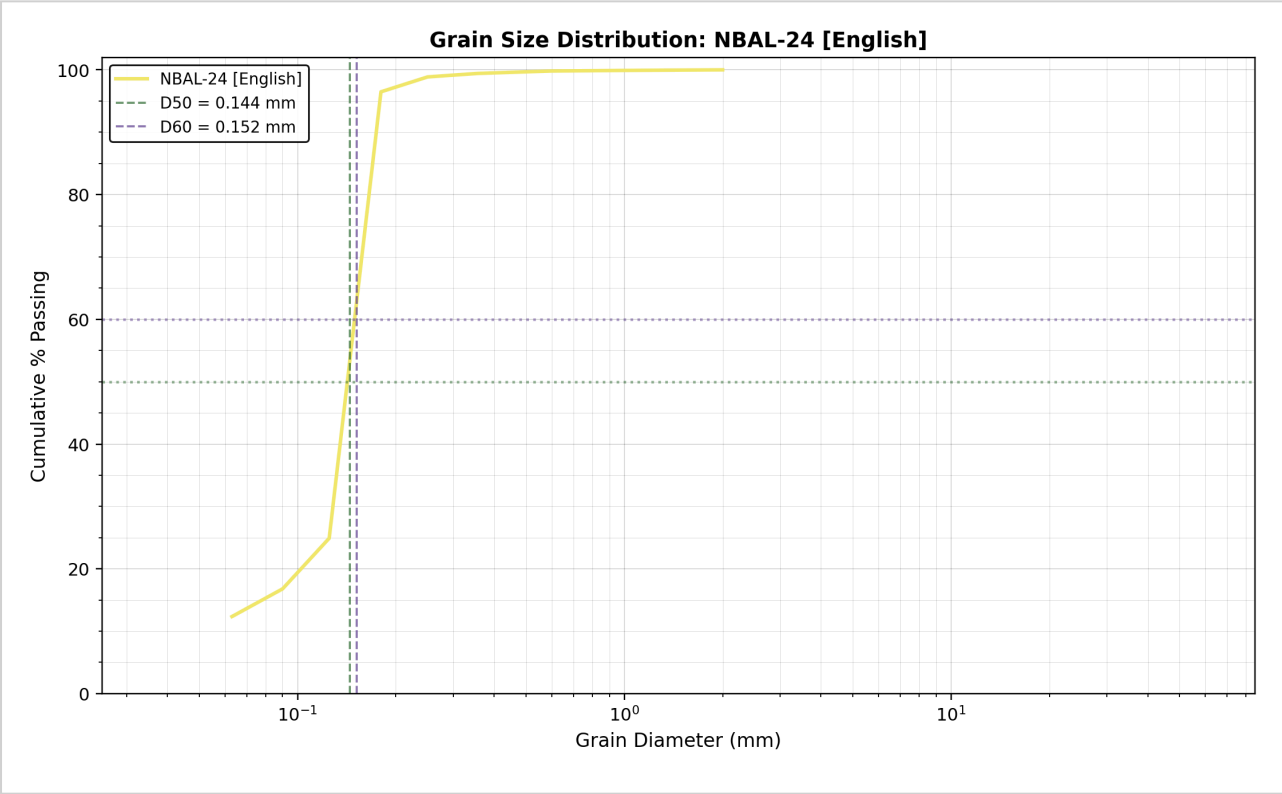


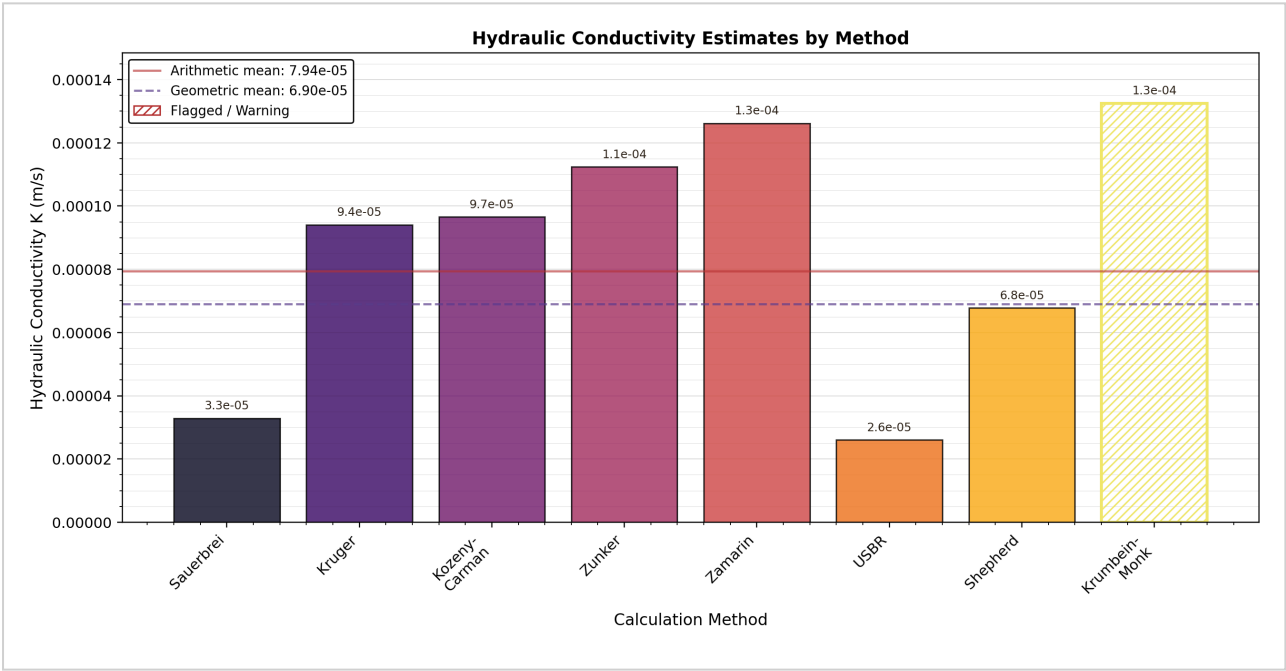
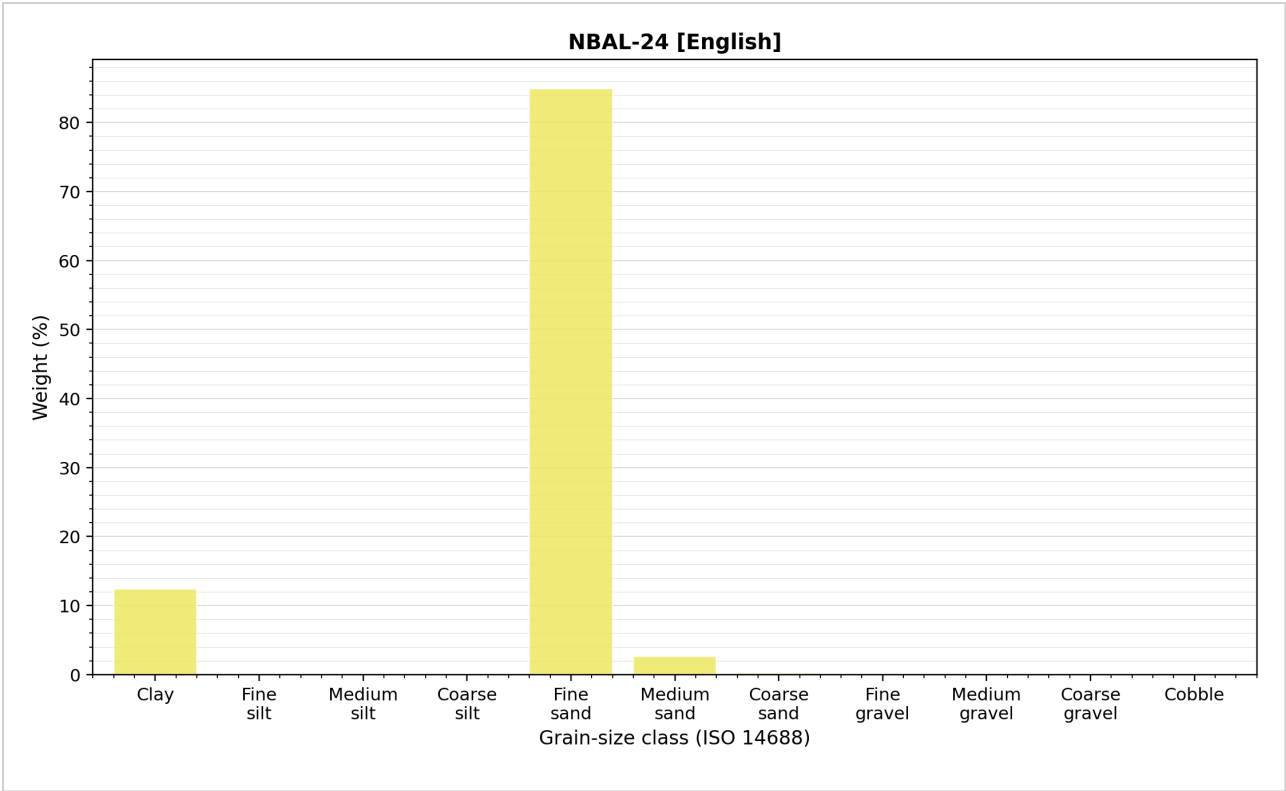
NBAL-23 [English]



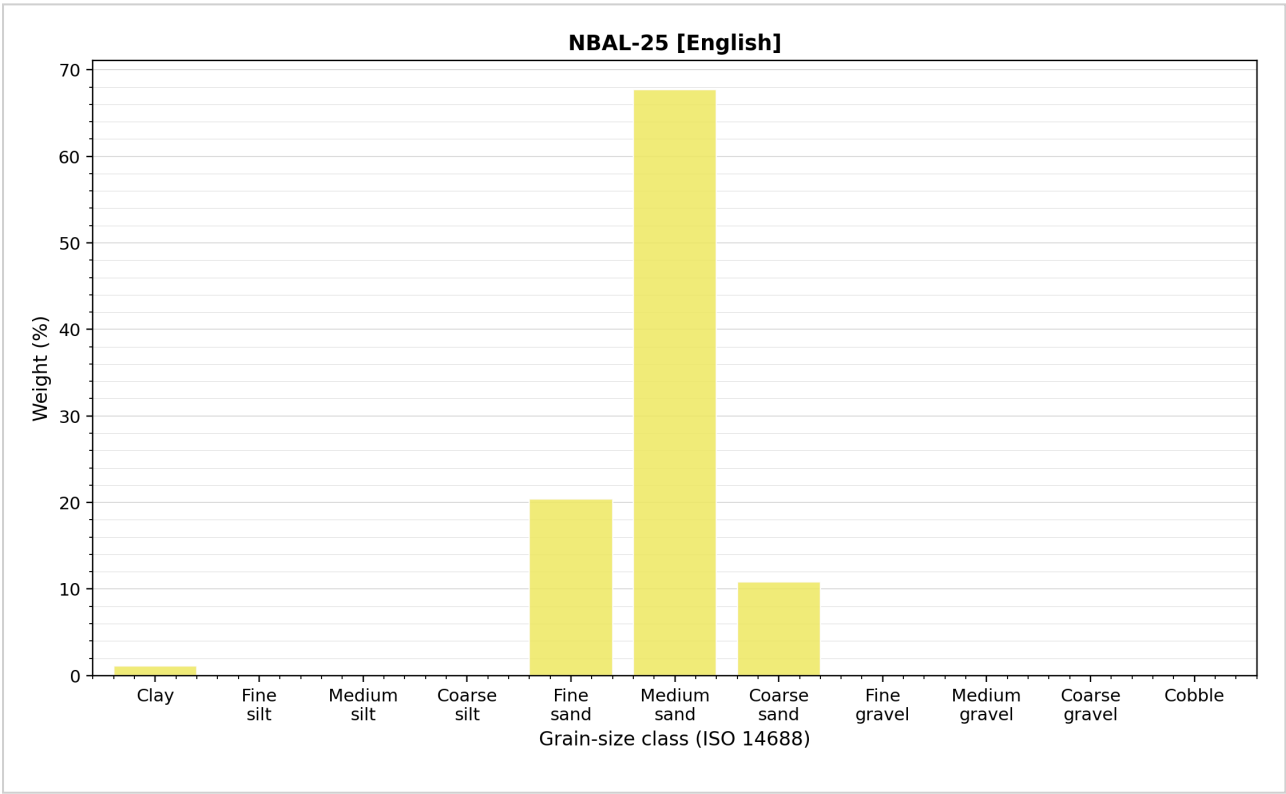
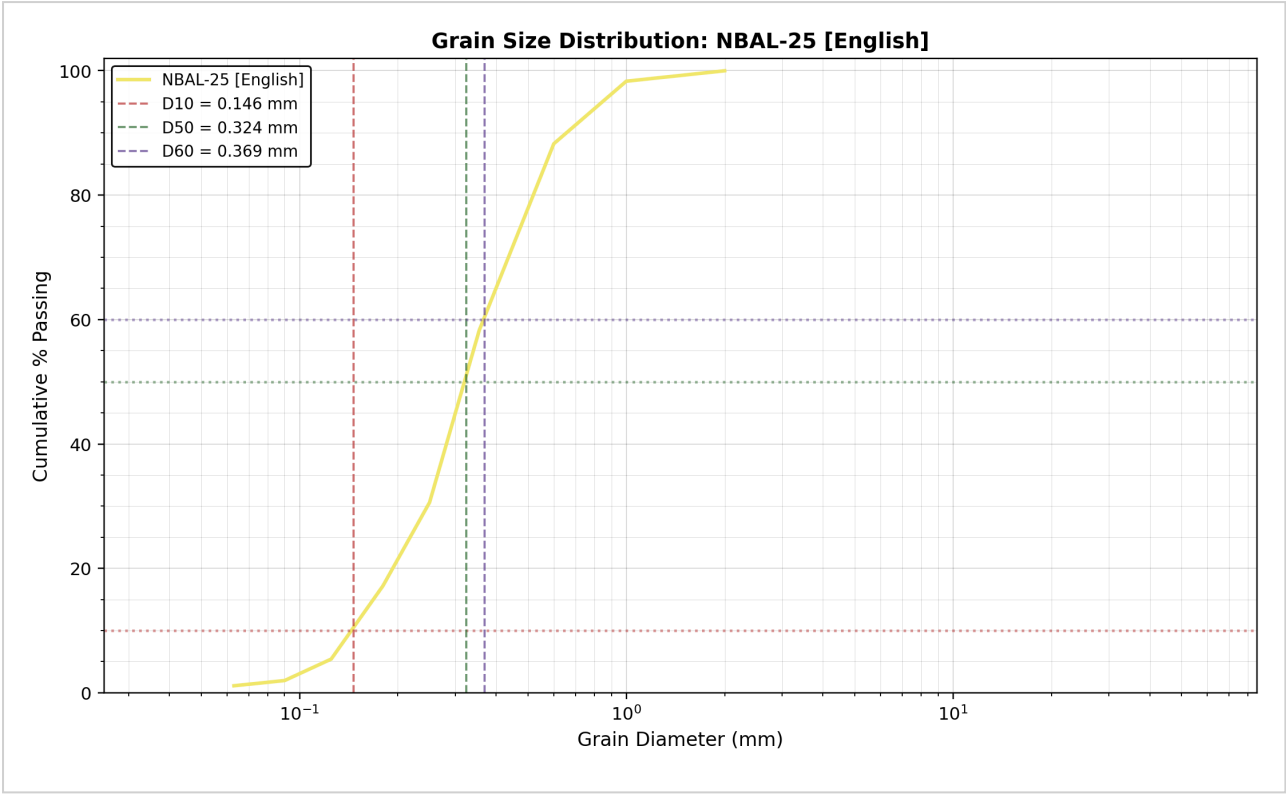


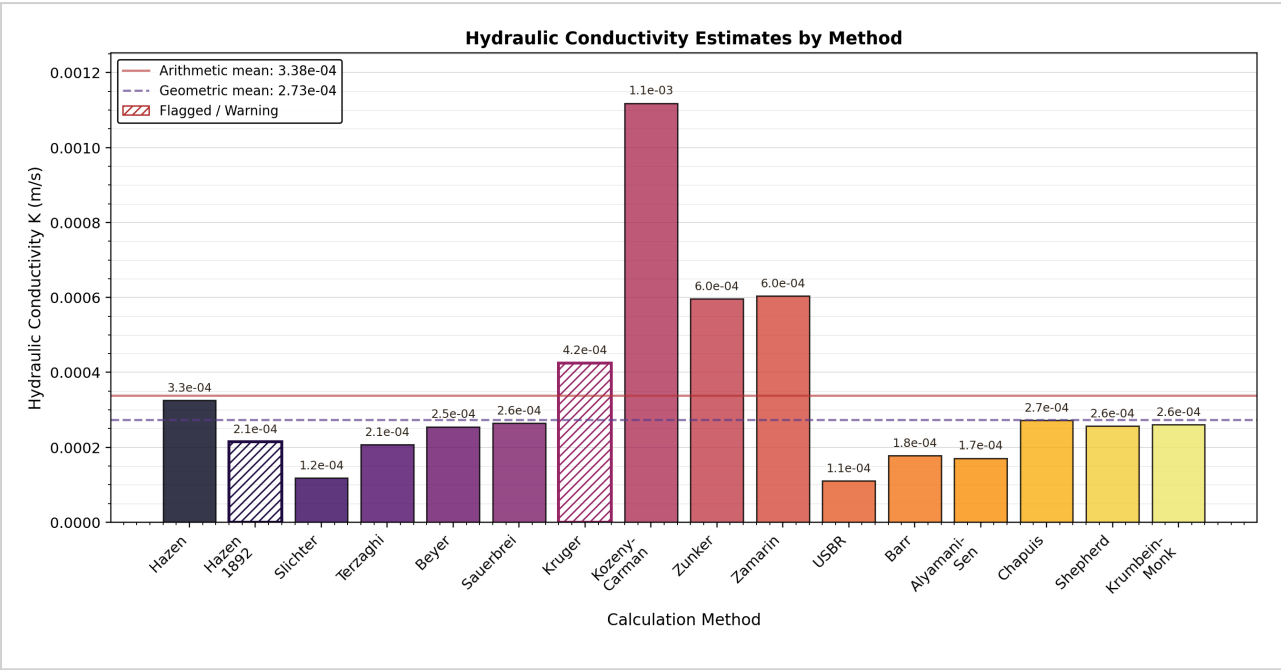
NBAL-24 [English]





NBAL-25 [English]





Interpretation & Discussion

Comparative Analysis

Permeability Characteristics:

- The highest permeability sample is NBAL-06 [English] with $K = 8.28 \times 10^{-4}$ m/s, classified as High (Clean Sand).
- The lowest permeability sample is NBAL-24 [English] with $K = 6.90 \times 10^{-5}$ m/s, classified as Moderate (Fine Sand).
- The 12.0-fold difference in permeability reflects the variability in grain size distribution among the samples.

Statistical Overview:

- Average sample K geometric mean: 2.72×10^{-4} m/s
- Standard deviation: 2.07×10^{-4} m/s
- Coefficient of variation: 76.1%